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(AUTONOMOUS)

SION(W), MUMBAI-22

DEPARTMENT OF INFORMATION TECHNOLOGY

MSc(IT),SEMESTER IV

Practical Journal

For the Subject

Data Visualization

Submitted by

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(Autonomous)

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Department of Information Technology

CERTIFICATE

This is to certify that **Neerajkumar Satyaprakash Gupta,** Seat No. **SMSC2425224** has successfully completed the necessary course of experiments in the subject of **Data Visualization** with the subject code **SIPITELP621** during the academic year 2024 – 2025 complying with the requirements of University of Mumbai, for the course of **MSc.in Information Technology Part 2 [Semester-4]**

_____ _____

Faculty-in-charge **Course Co-ordinator**

Anita Gupta **Sudha Bhagavatheeswaran**

_____

External Examiner

Date: **College Seal**

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PRACTICAL 1

Aim: Installation and overview of data visualization tool (Tableau Public)

WHAT IS TABLEAU?

Tableau is an easy-to-use business intelligence software. It makes data visualization, data analytics, and reporting as easy as dragging and dropping. Anyone can learn to use Tableau without having a prior programming experience. Tableau can combine data from various data sources such as spreadsheets, databases, cloud data, and even big data- all into one program to perform dynamic analysis.

WHY TABLEAU?

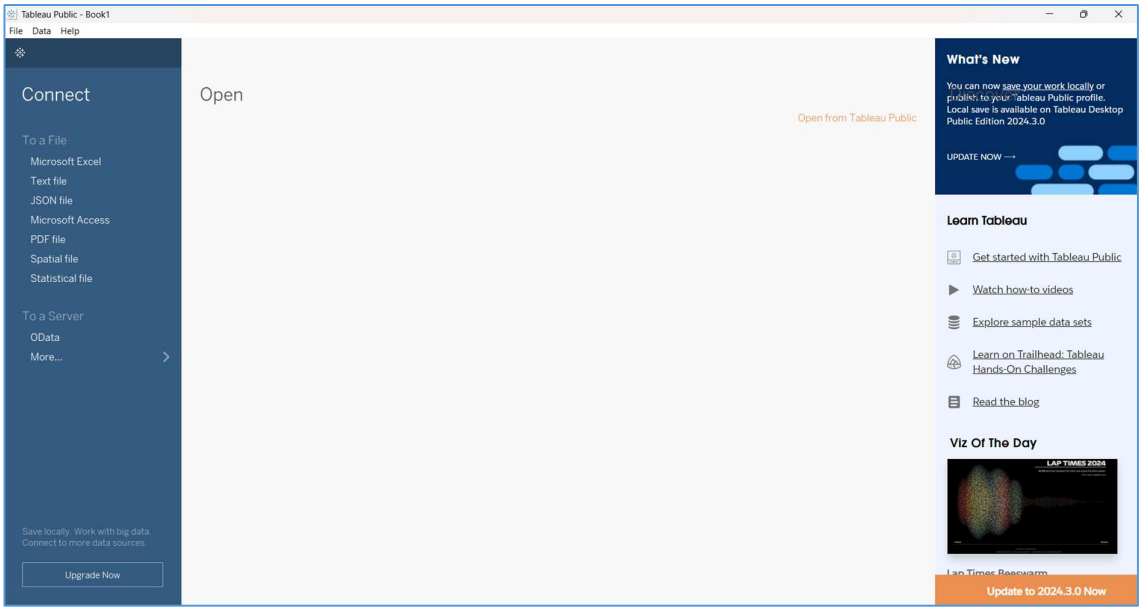
Whether it's small or large, profitable or non-profit, every organization needs to analyze their data for optimal decision making. Analysing data has never been easier with traditional business intelligence tools.

DOWNLOADING AND INSTALLING TABLEAU PUBLIC

Visit the website and sign up to download tableau public software for free and install it in your system.

Website: - https://public.tableau.com/s/.

After successful installation, Tableau Public App will look like this:



GETTING STARTED WITH TABLEAU

The first thing to do in Tableau is to connect to your data. There are mainly two types of connections- Connecting to local files or to a server.

CONNECTING TO LOCAL FILE

Tableau can connect to any local file or database such as-

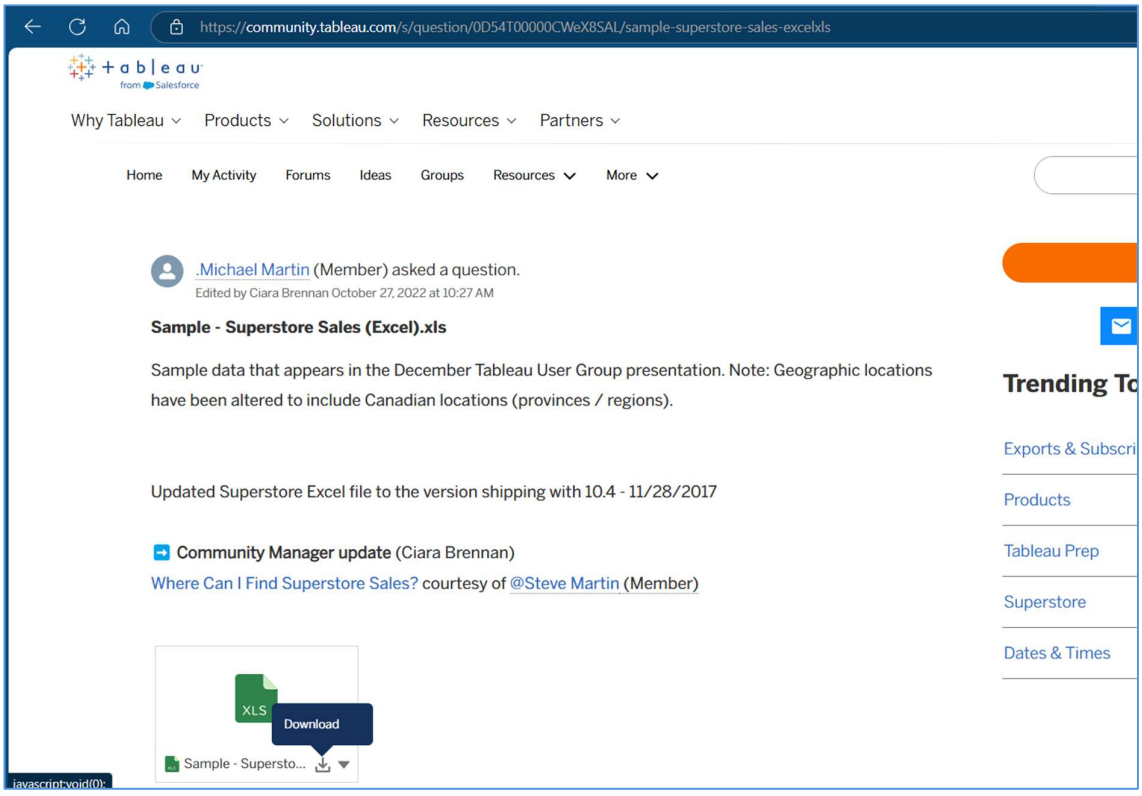
- Excel
- Text File
- Access
- Statistical File
- Other Database file

Local connection gives the maximum speed of data processing.

While working on Tableau, data can have Live Connection where any change in the source data will be automatically updated in Tableau. On the other hand, data can be Extracted to Tableau repository so that any change made here will not affect the original source data.

CONNECTING TO EXCEL FILE

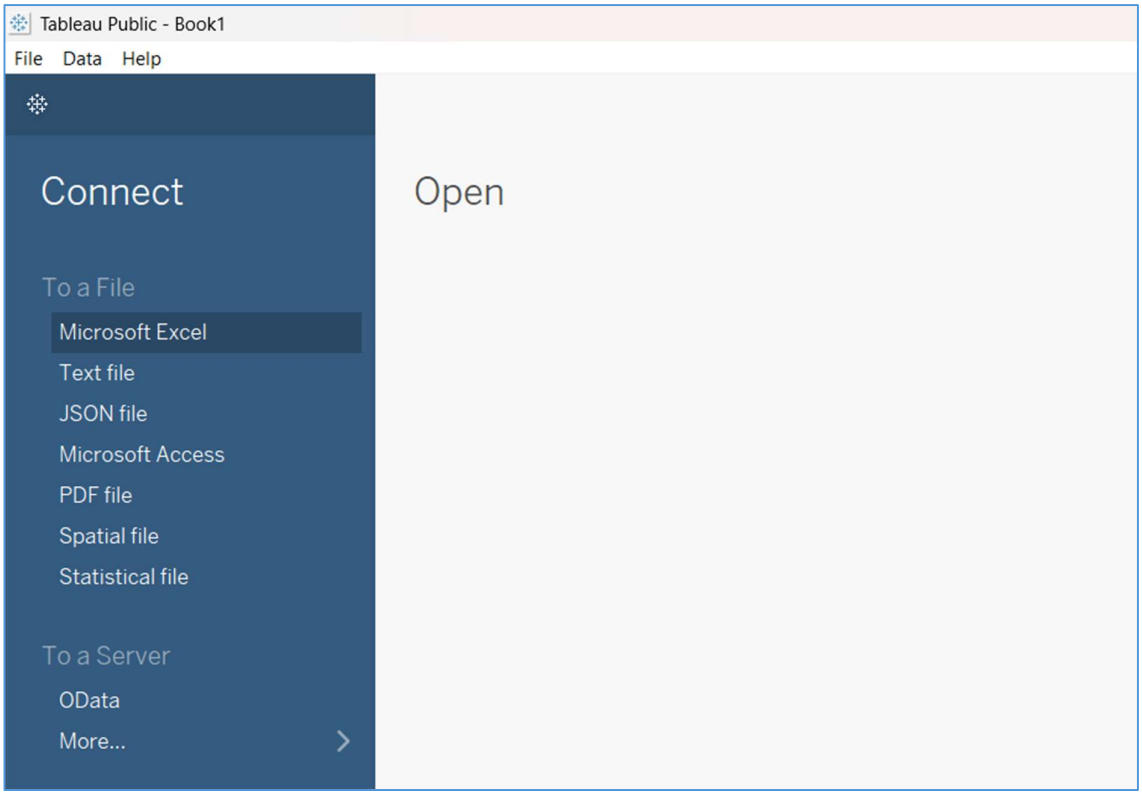
Before connecting to excel file, we need to download the sample superstore file from tableau community or from Kaggle which is freely available

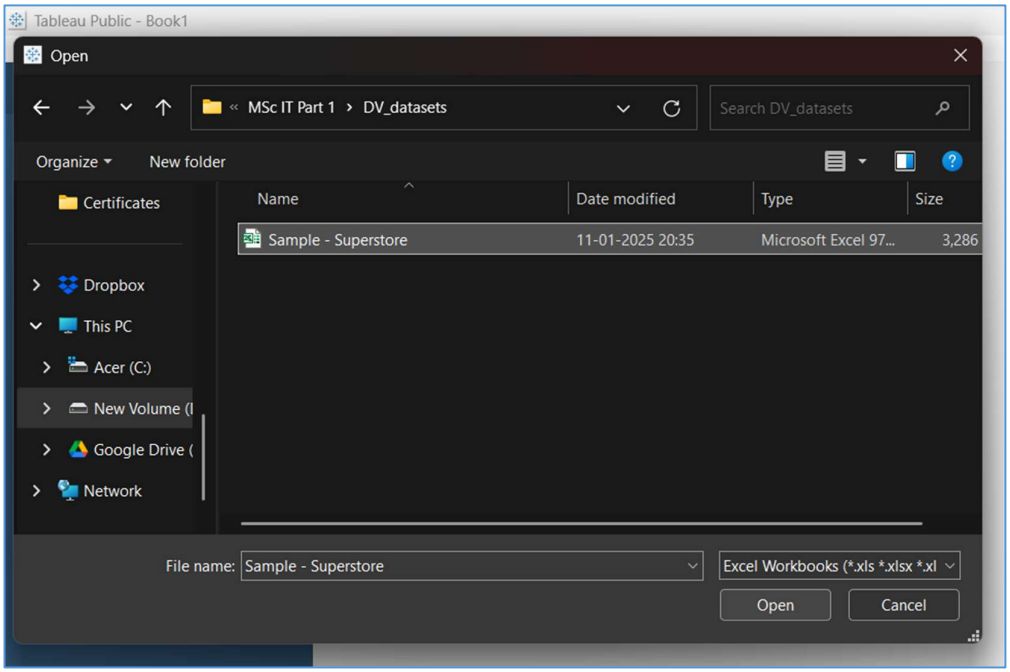


To connect to an excel file, click “Microsoft Excel” on the left-hand side. Navigate to the file on your computer and double click to open it. For this practical, we will use a sample

file that comes with the installation called “superstore”. You should open the appropriate file that you will be working with.

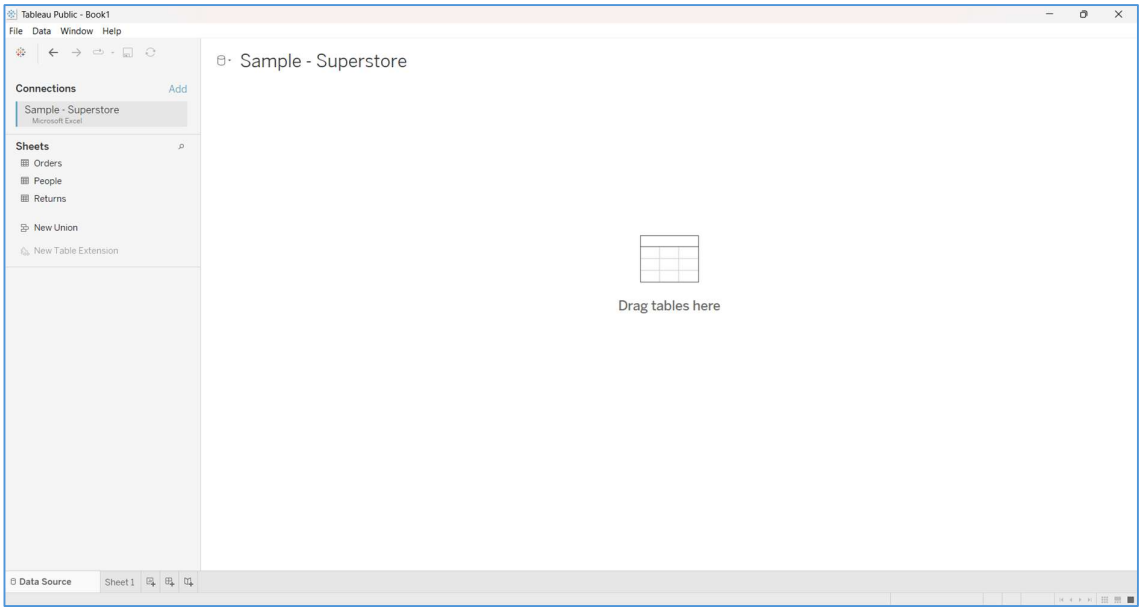
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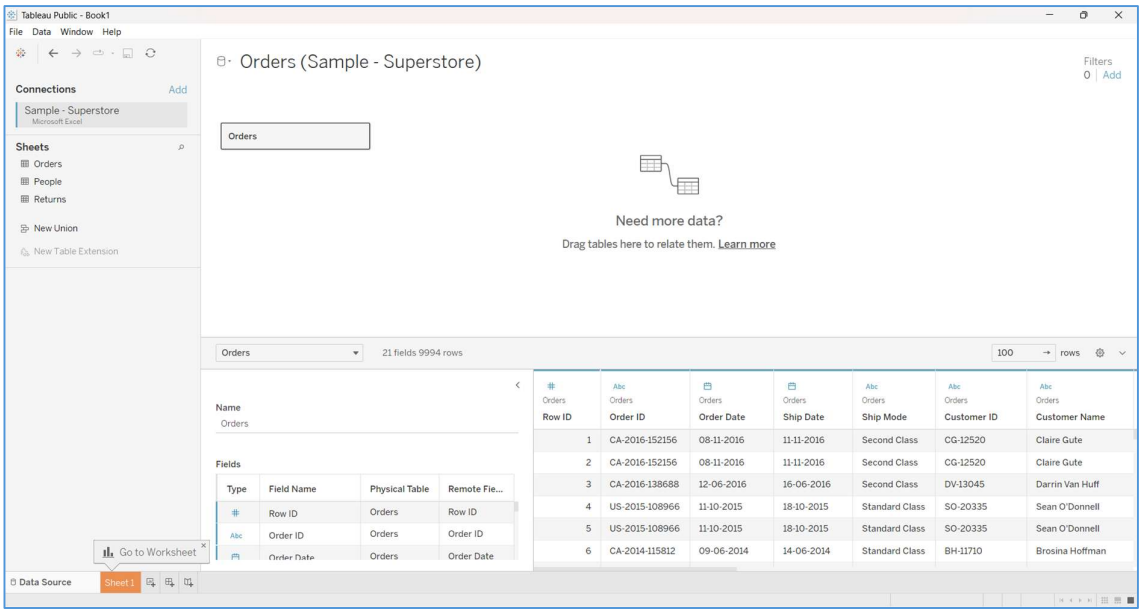




Now you are in the data connection window. It looks like the following: -

Notice that I have three sheets in this file- Orders, People, and Returns. I can simply drag the table I want (in these practical we will drag or use the Orders table). If I drag more than one table, Tableau automatically creates the join between the tables.





EXPLORING TABLEAU WORKSHEET

Creating charts based on the data we connected is easy. At the bottom of the page, click on a sheet (sheet 1) and we will see the following screen:

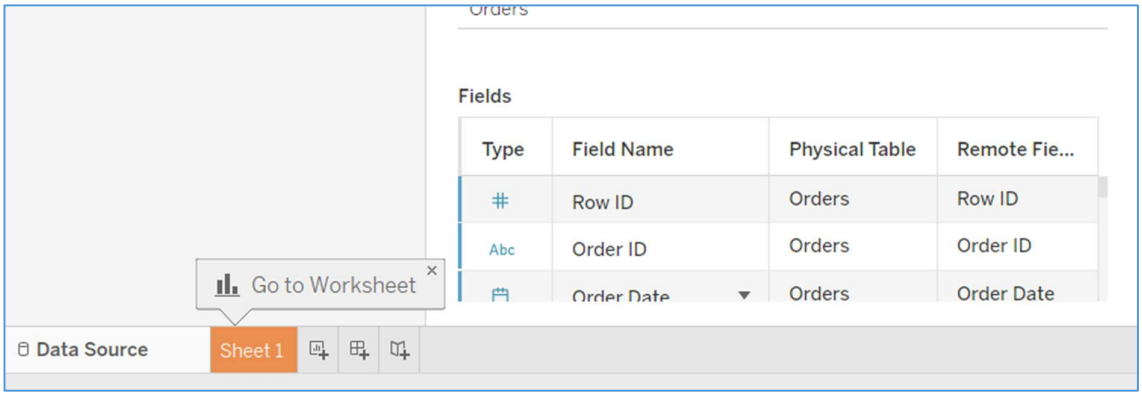
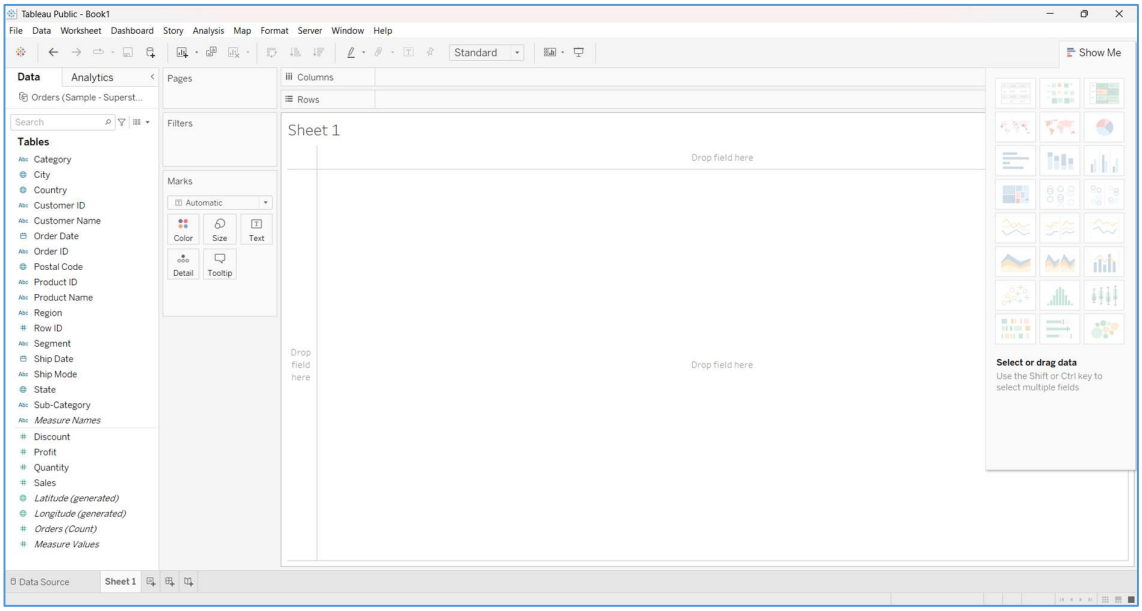


Tableau automatically separates the data into Dimensions and Measures. Dimensions are the categorical fields. These fields will create labels in the chart. Measures are the quantitative fields. These are the numbers we want to analyse. They create axis in the chart.



By looking at the above worksheet we can see all the basic components of tableau public as follows: -

1. Workspace Area

This is the central area where your visualization takes shape. Here, you'll drag and drop fields to create visual representations of your data.

2. Columns and Rows Shelves

These shelves are key to structuring your visualization:

- Columns Shelf: Place fields here to define columns in your chart.

- Rows Shelf: Place fields here to define rows in your chart.

3. Data Pane

Located on the left of the screen, it contains:

- Dimensions: Qualitative fields (e.g., categories, names, dates).

- Measures: Quantitative fields (e.g., sales, profit).

4. Filters Shelf

Drag dimensions or measures here to filter the data displayed in your visualization.

5. Marks Card

This card lets you customize the appearance of your visualization. You can adjust:

- Color: Change the color of marks.

- Size: Adjust the size of marks.

- Label: Display labels.

- Detail: Add details to your marks.

- Tooltip: Customize tooltips that appear when hovering over marks.

6. Pages Shelf

Use this shelf to create a series of pages for your visualization, which can show changes over time or different categories.

7. Analytics Pane

Located next to the Data Pane, the Analytics Pane provides tools for adding trend lines, reference lines, and other analytical features to your visualization.

8. Show Me Panel

This panel suggests different types of visualizations based on the fields you've selected. It helps in choosing the best visualization type for your data.

9. Sheet Tabs

Tabs at the bottom of the interface allow you to switch between different sheets, dashboards, and stories within the same workbook.

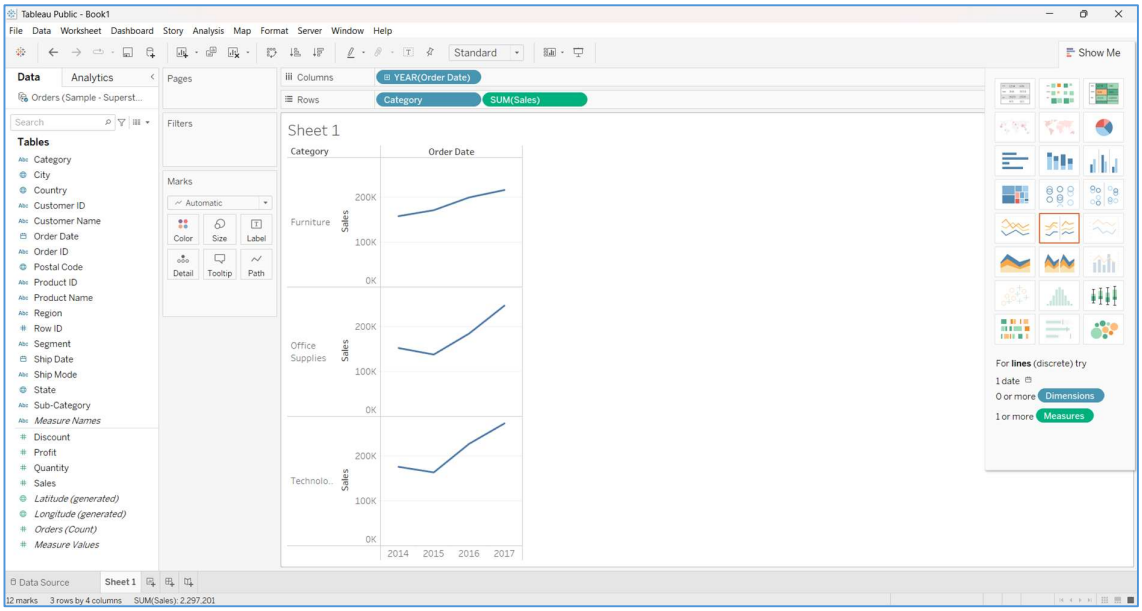
10. Toolbar

The toolbar at the top includes various options for saving, undo/redo actions, creating new worksheets, and other quick access tools.

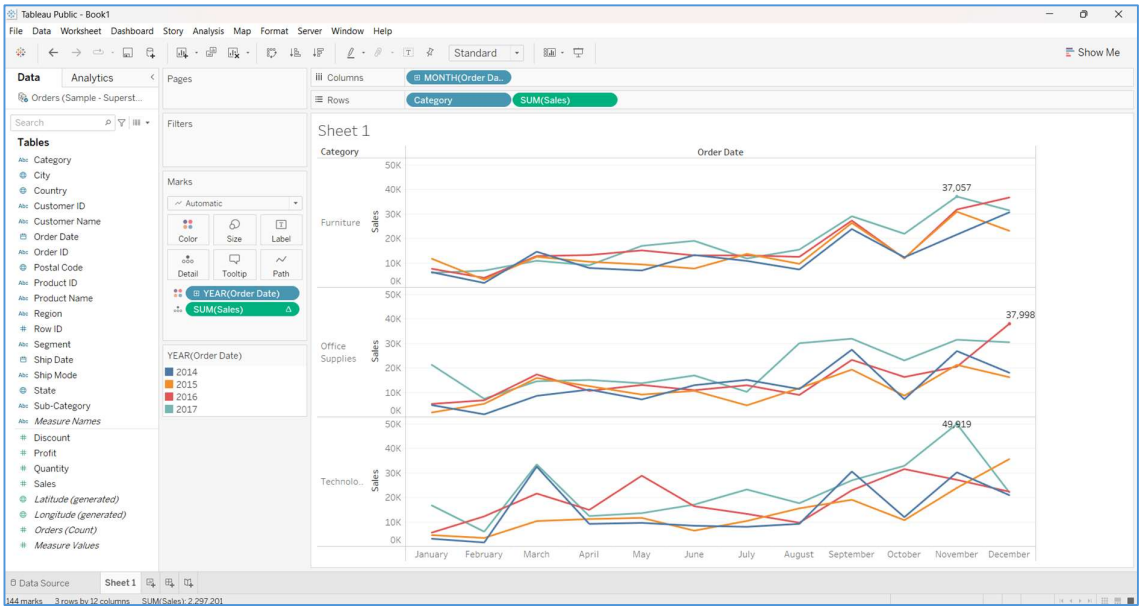
CREATING CHARTS

Now we will perform basic data visualization operations to create charts you can drag any tables, Dimensions: Qualitative fields (e.g., categories, names, dates) into columns and any measures: Quantitative fields (e.g., sales, profit) into rows.

After adding Order Date into columns and Category, Sales into rows the chart looks like this:



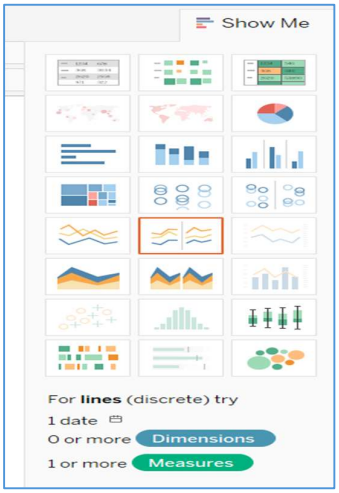
You can change the duration of time from year to month and month to days by clicking on the plus in the year (Order Date) which is in the columns and also you can right click on it and click on continuous to show the chart in continuous manner which is in discrete form by default.



Here I have months in the x-axis, and category and sales are in y-axis. Different colour represents different years.

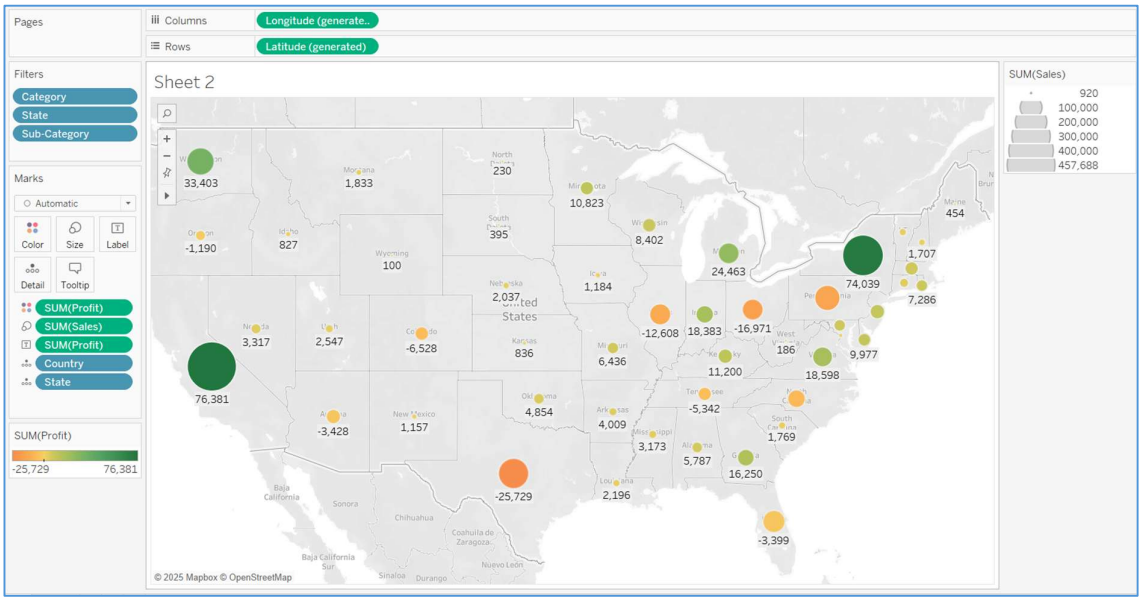
How do I know what type of chart to use?

Sometimes, it might be confusing what type of chart should be used for a specific data. Tableau has an interesting feature called “Show me”. “Show me” is the list of the possible charts that can be created using different combination of data.



Below are two more charts created from the same source of data:

This chart shows global sales and profit filtered by categories:

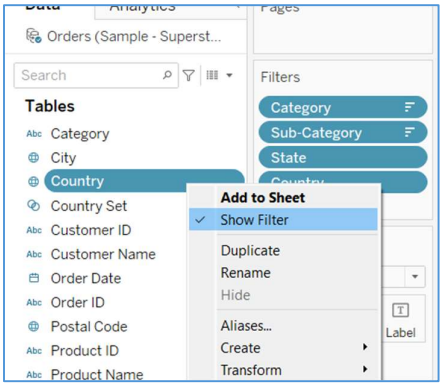


This chart shows sales by category:



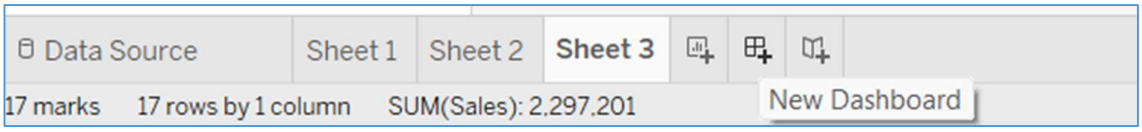
Adding Filters to the charts:

Adding filters to the charts enable the end user to easily drill down to obtain desired information. To add a filter, simply right click on the dimension that will be used for filter and choose “Show filter”.

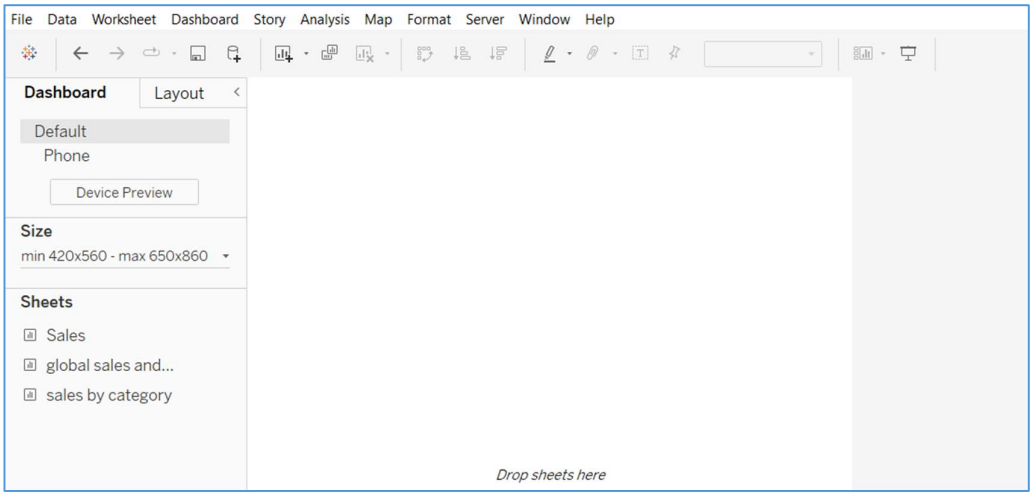


CREATING DASHBOARD

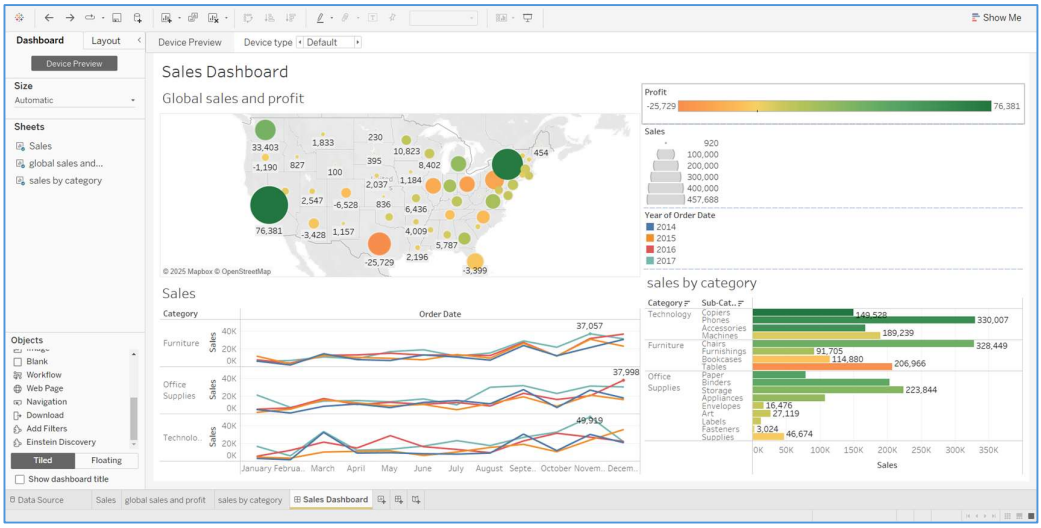
Tableau Dashboard contains all the related features intuitively interconnected to provide interactive and real-time dashboard experience for non-technical users. To create a dashboard, click “New Dashboard” icon at the bottom the page.



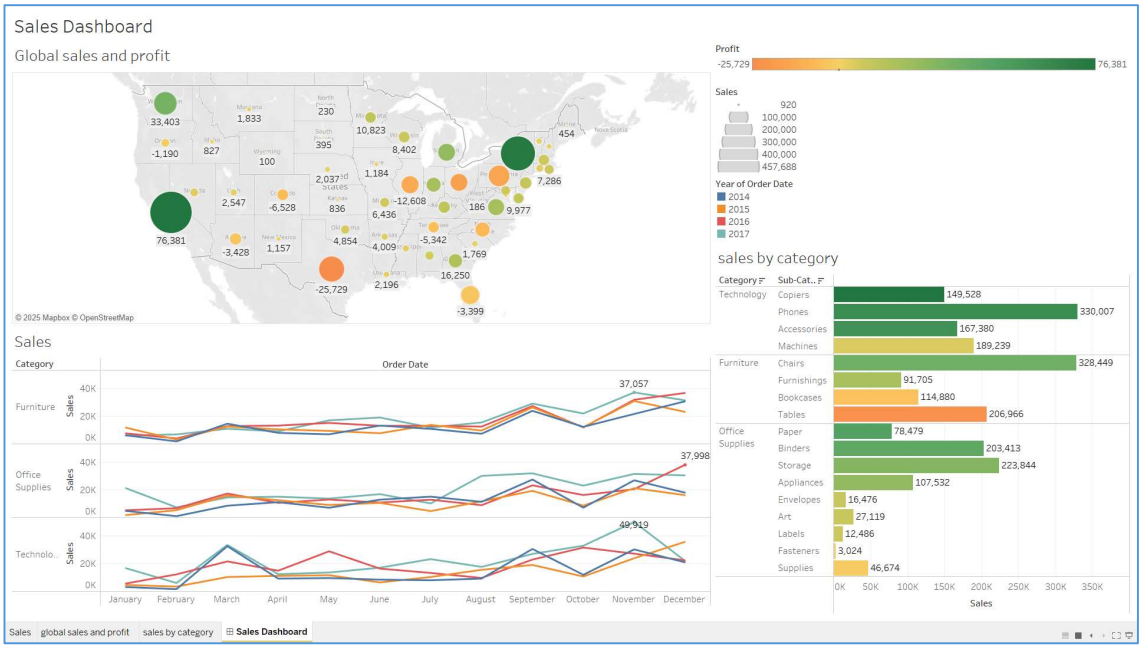
On the top left corner, dashboard pane will display the available sheets. In my case, I have sales (sheet1), global sales and profit (sheet2), and sales by category (sheet3). To add those sheets on the dashboard, simply drag and drop them on the empty space to the right.



This is how the final dashboard looks like after adding three sheets:



Sales Dashboard:



PRACTICAL 2

Aim: Demonstrate Time Series using visualization tool

WHAT IS TIME SERIES?

Time series forecasting is a critical requirement for many organizations. The starting point of forecasting is a time series visualization, which provides the flexibility to reflect on historical data and analyse trends and seasonal components. It also helps to compare multiple dimensions over time, spot trends, and identify seasonal patterns in the data. A few examples include stock market analysis, population trend analysis using a census, or sales and profit trends over time.

Time series analysis is a statistical technique used to record and analyse data points over a period of time, such as daily, monthly, yearly, etc. A time series chart is the graphical representation of the time series data across the interval period.

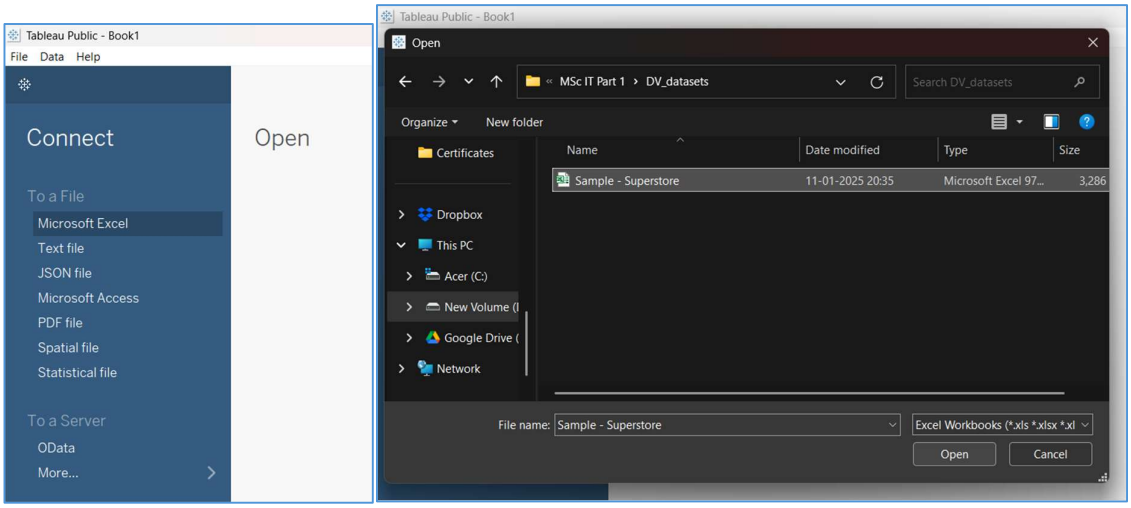
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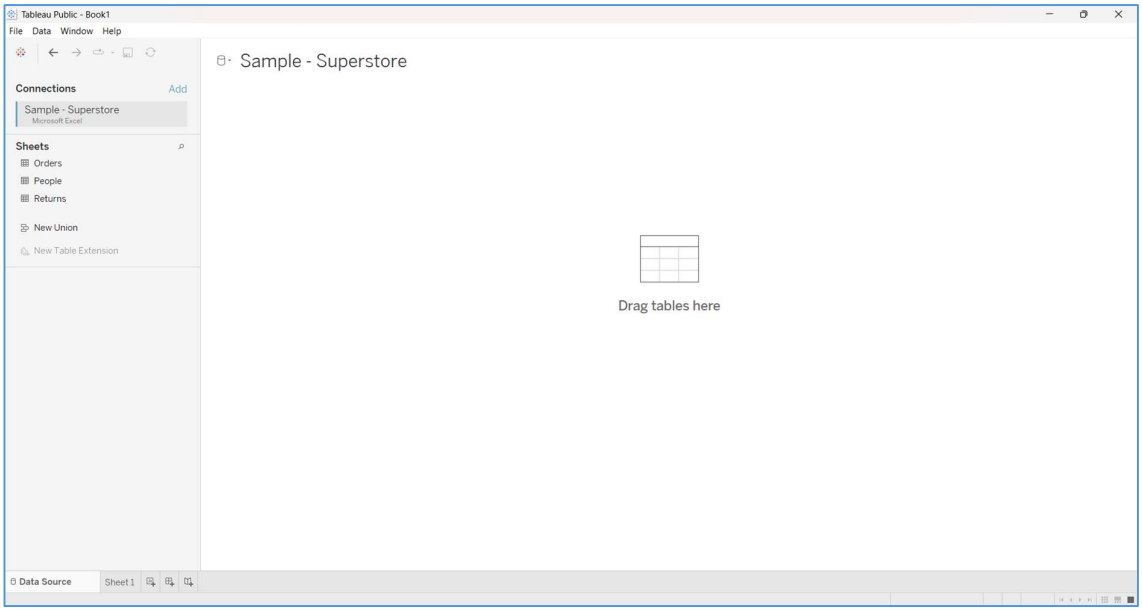
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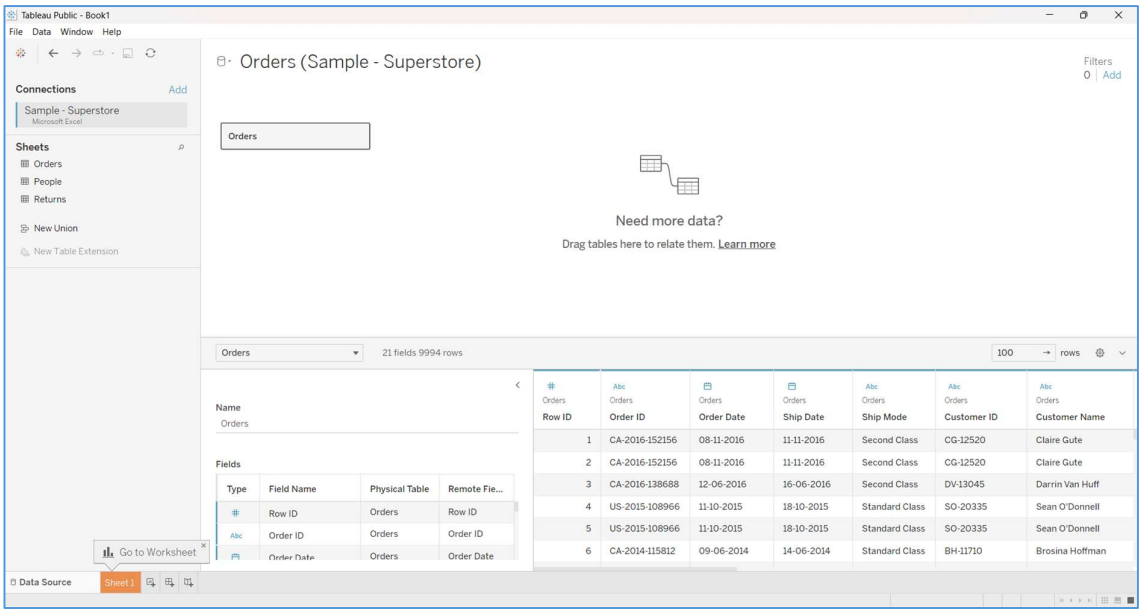
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OPEN WORKSHEET

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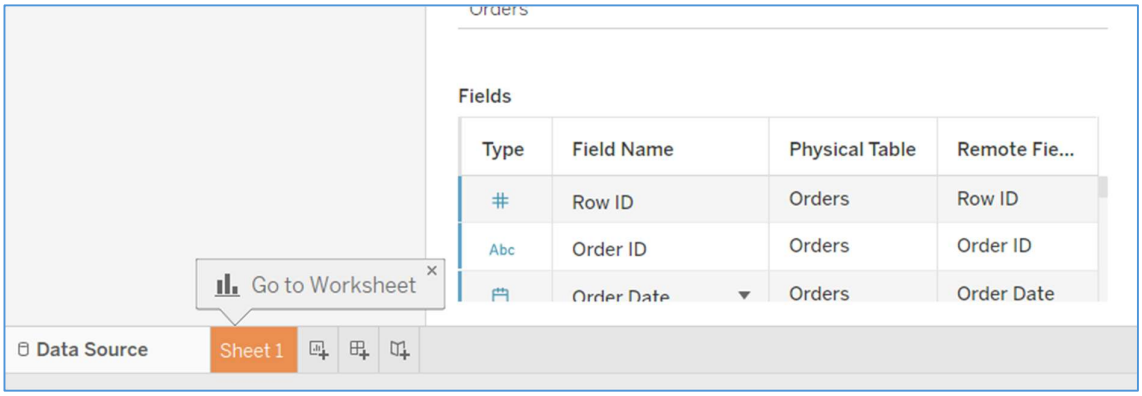
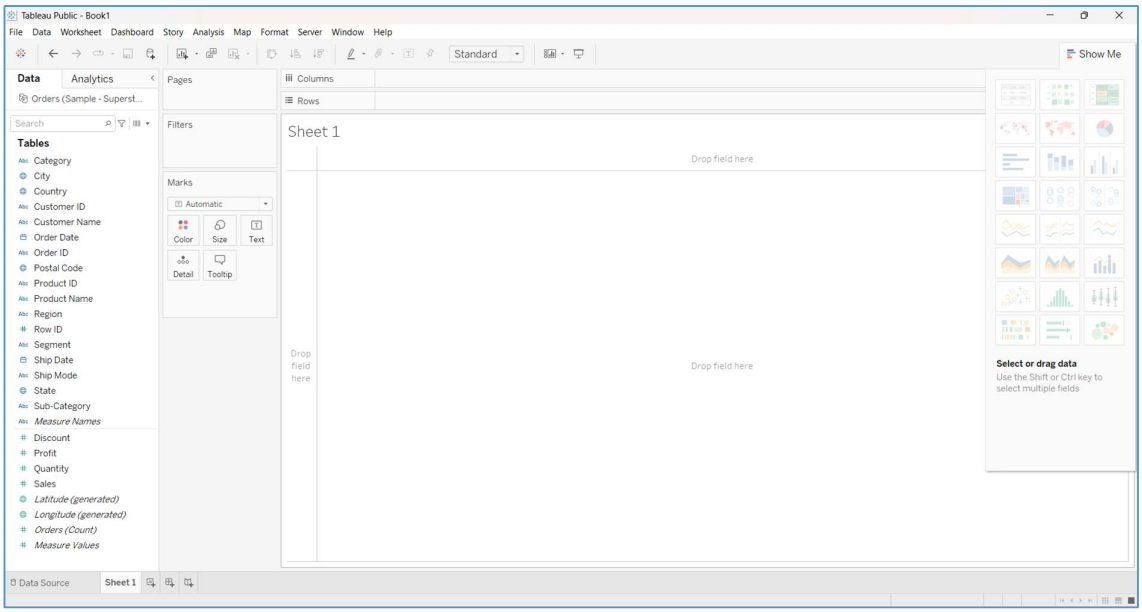
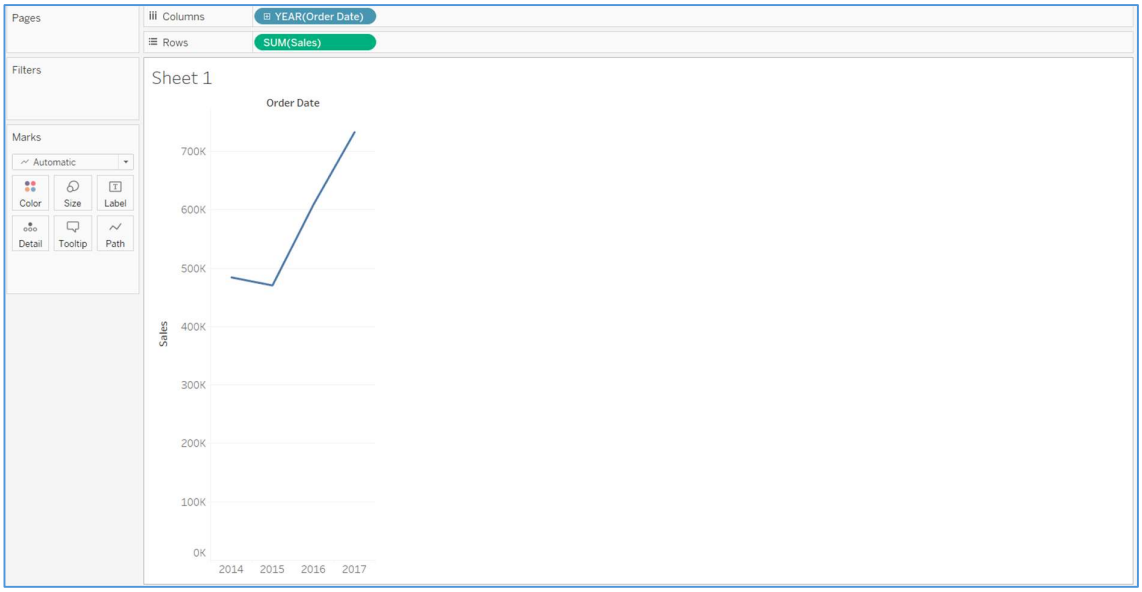


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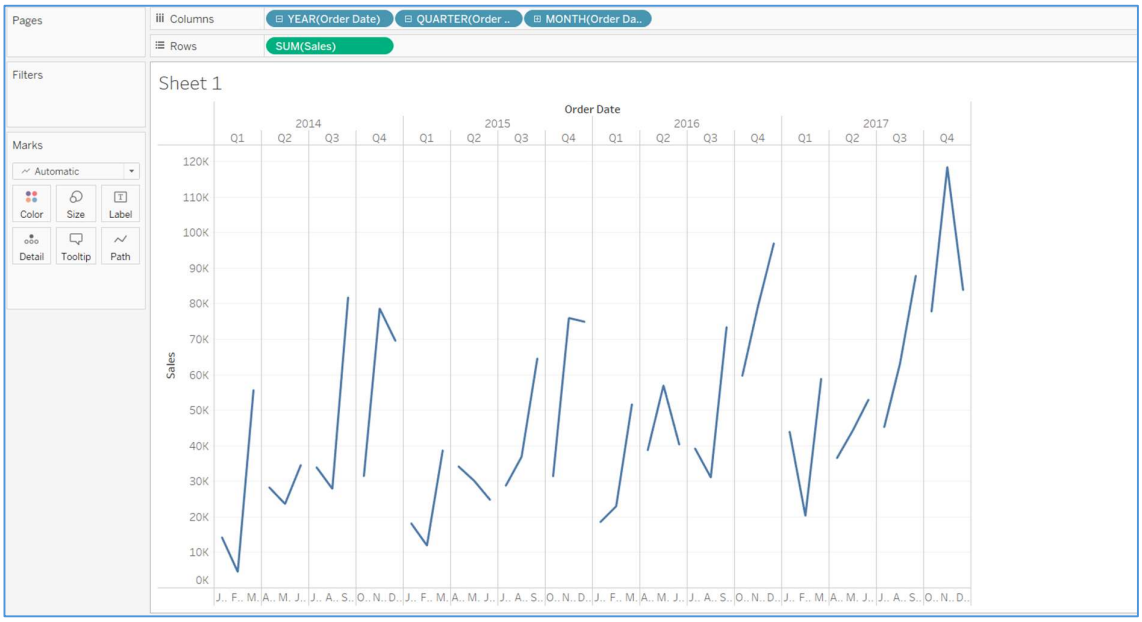


STEPS FOR CREATING TIME SERIES CHARTS

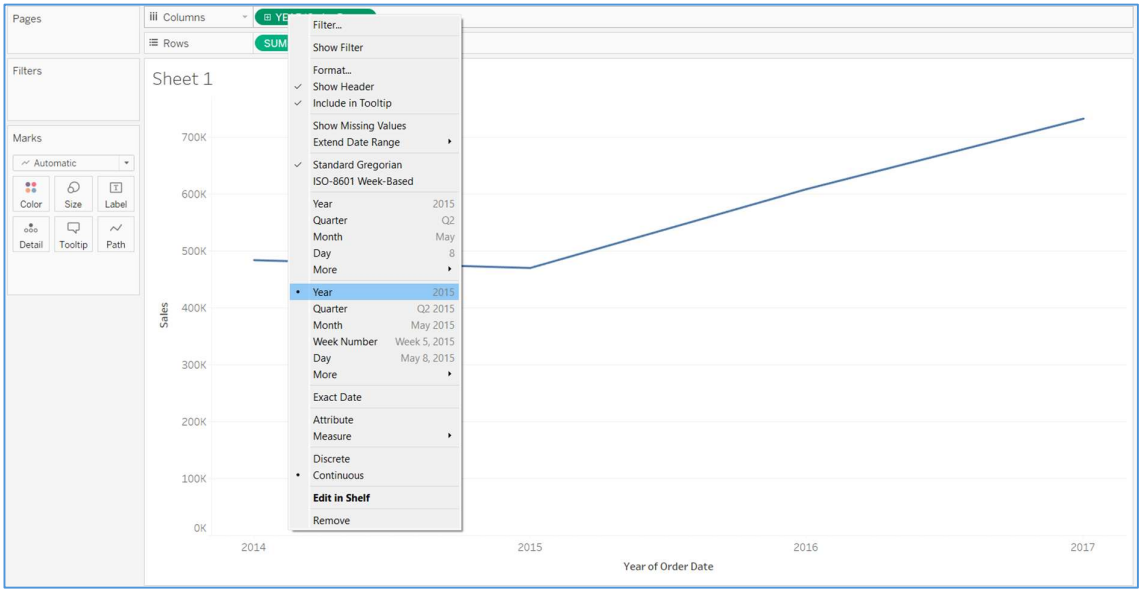
Step 1: Drag the **Order Date** field to the Columns shelf and the **Sales** variable to the Rows shelf. The default chart will give us a yearly trend line chart. The Marks shelf automatically selects a line graph for the chart.



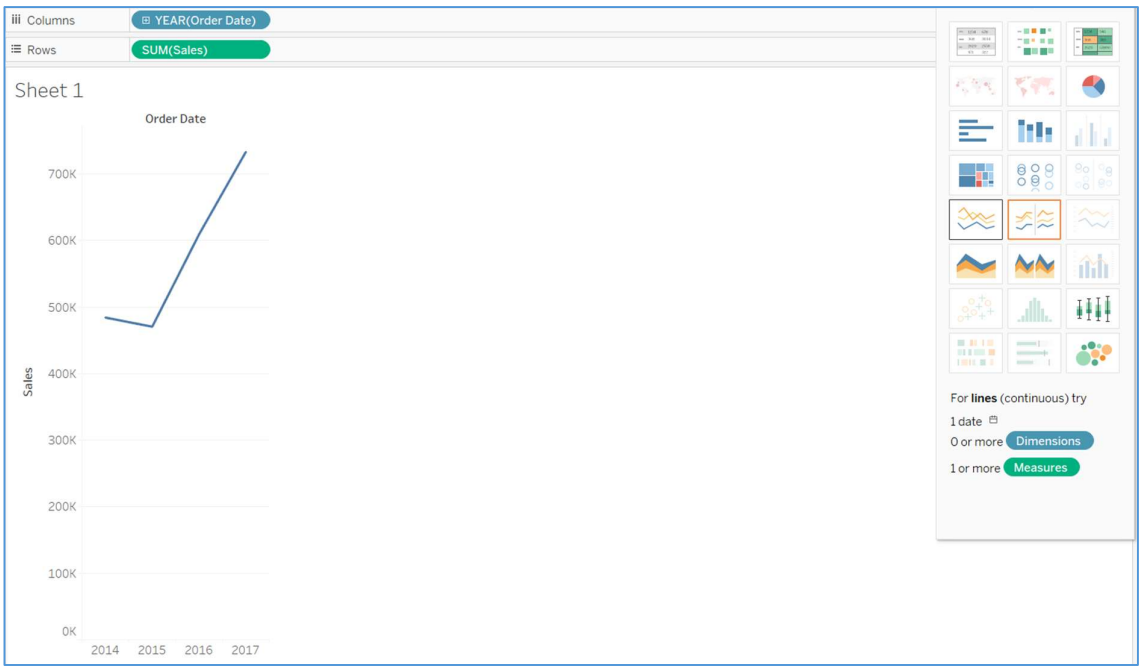
Step 2: In the chart above, we see that the display is in years. To further drill down to quarter and month levels, we can simply click on the **plus icon** on the order date in the **Columns** shelf. This will generate the following output, which now displays the data broken down to the month and quarter level.



Step 3: The above chart is useful, but it is displayed in a discrete format. It will be more beneficial if the data is displayed in continuous form. To convert the chart into a continuous format time series chart, the first step is to roll up the **YEAR (Order Date)** back to year level, and then the second step is to right-click on it and select the **Year** and **Continuous** options. This is illustrated in the chart below.

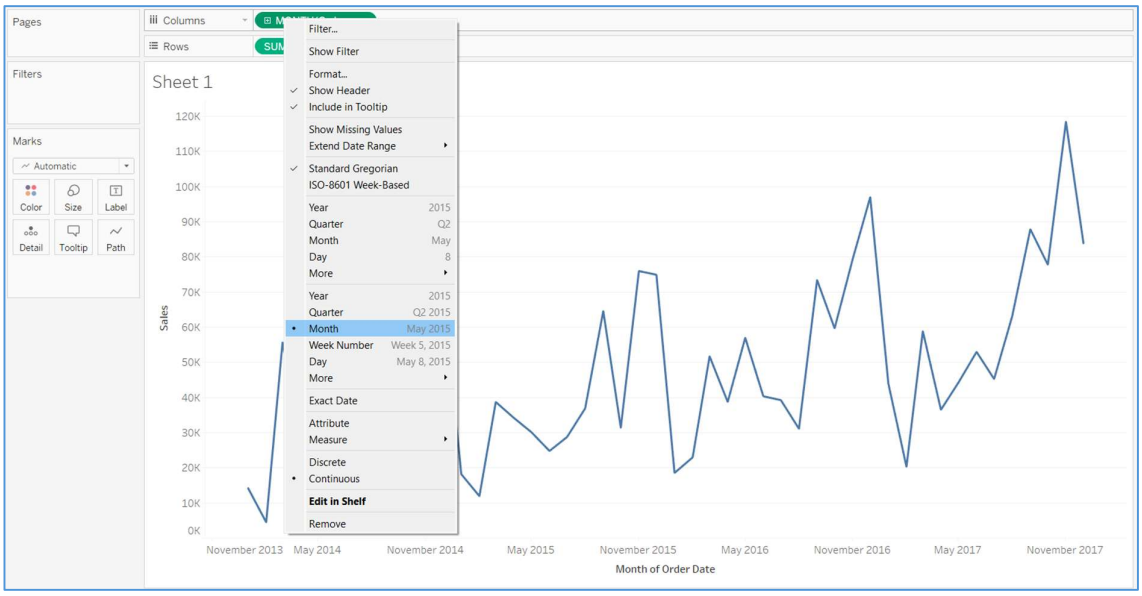


Another option in Tableau to build the continuous chart is to directly select the line chart type in the **Show Me** card, as shown in the chart below.



The above chart shows the trend of annual sales during the period 2014 through 2017. There is a continuous trend of increase in sales volume. However, it is better to analyse the time series data by breaking it down to a monthly level.

Step 4: It is easy to change the chart breakdown from annual to monthly. This can be done by simply changing the **Columns** shelf from **YEAR (Order Date)** to **MONTH (Order Date)**. This will generate a monthly time series chart. From an analytics perspective, this chart is more insightful as it allows us to see the sales fluctuations across months and years. This is also useful for decomposing the seasonality and trend components of the time series data.



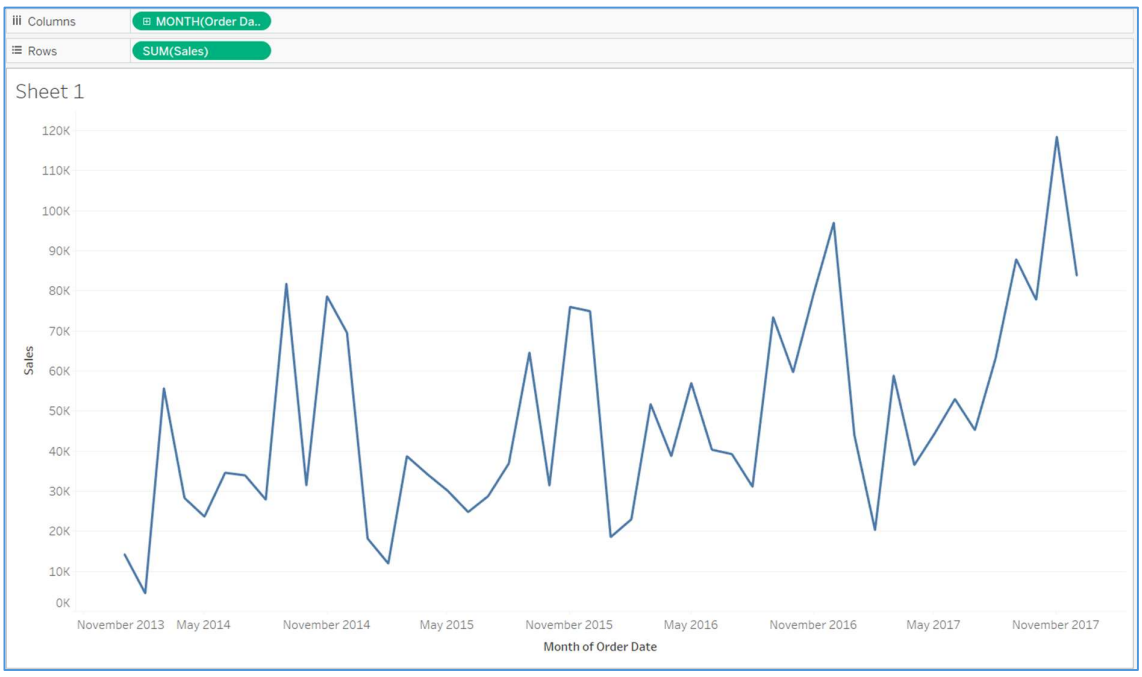
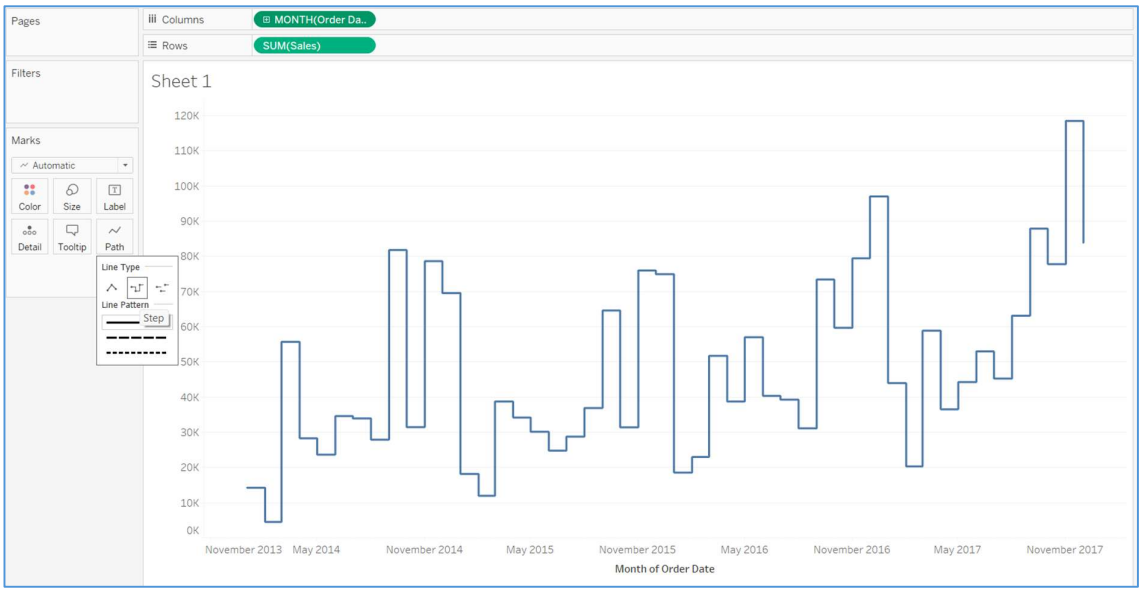


Tableau also provides the ability to change the path property as well as the chart type.

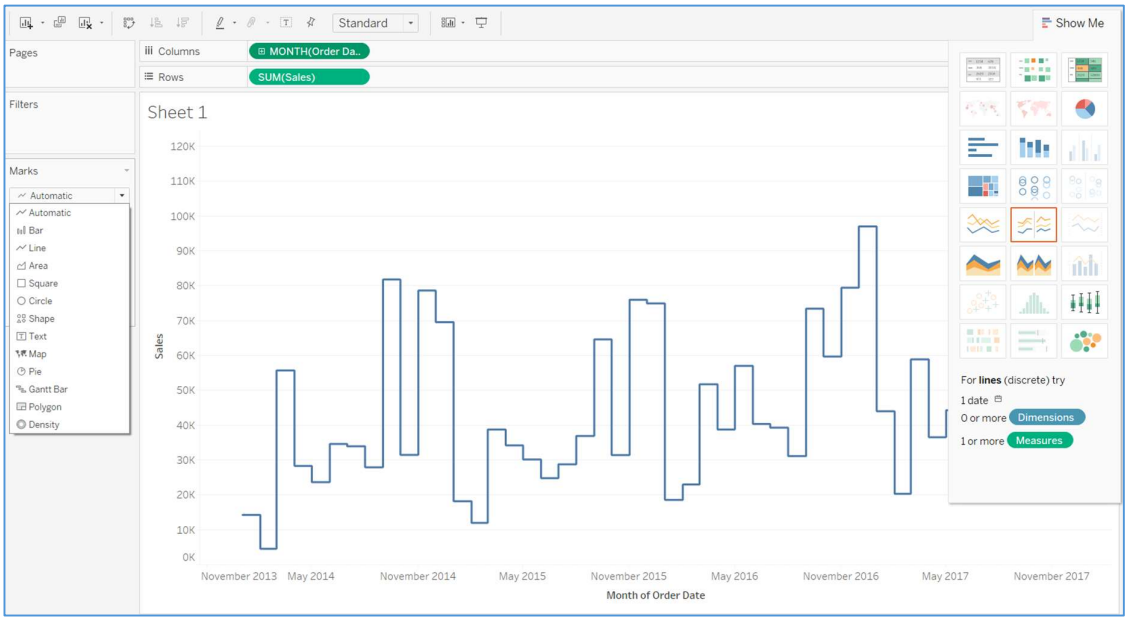
Step 5: Change the Path Property

We can change the path property by going into the **Marks** shelf and clicking on the **Path** option. There are three options for the type of line graph for the view, and selecting the second option will produce the following chart. The output is like the previous chart, but the trend shifts are more pronounced now.



Step 6: Change the Chart Type

We can change the chart type to options such as a bar or an area chart, either from the **Marks** shelf or from the **Show Me** option. However, for time series data, this is not suggested as the line chart is the best option.

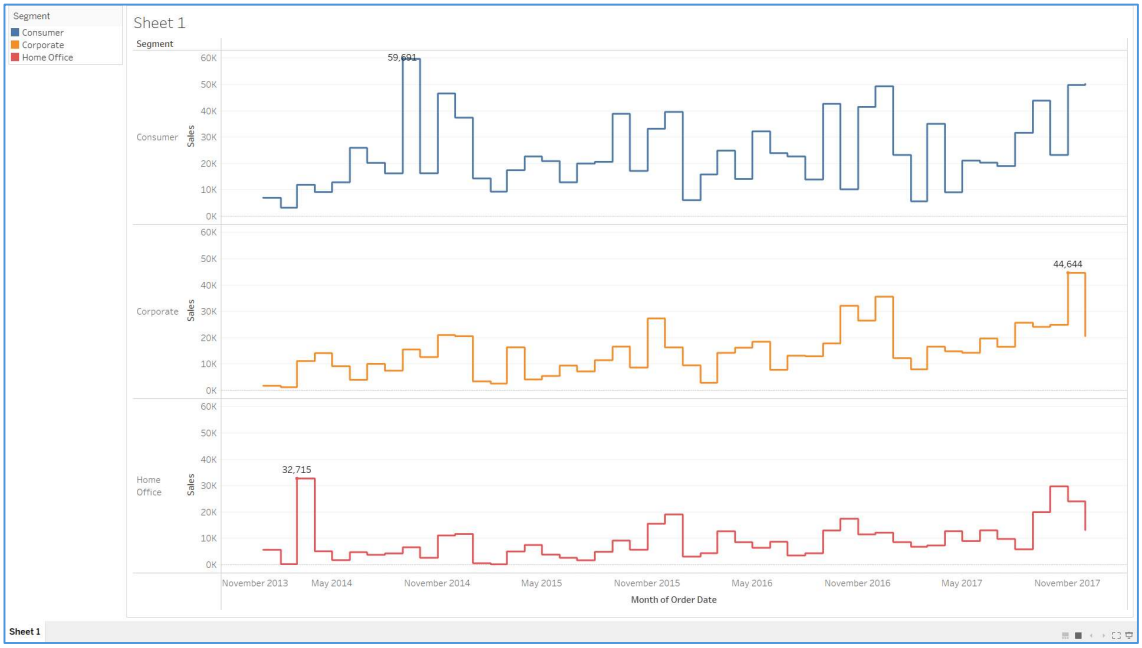


Step 7: Adding Categories to Time Series

In the previous sections, we have learned how to build a time series chart with two variables, **sales** and **time**. However, sometimes it is important to add more variables to a chart to understand and analyze it better. For instance, it could be useful to visualize sales by segment across time. This can be done easily in two ways. First, simply drag the **Segment** field to the **Color** pane in the **Marks** shelf. The second method is to move the category to the **Rows** shelf to show it separately.



Time Series Chart:



PRACTICAL 3

Aim: Statistical data graphics using visualization tool

WHAT ARE STATISTICAL DATA GRAPHS?

Statistical data graphs are visual representations of data that help us understand and interpret numerical information more easily. They can reveal patterns, trends, and relationships in the data that might not be immediately obvious in raw numbers.

We can create various statistical data graph types in Tableau to visualize and analyze our sales, profit data. This section will help us to understand some available statistical graph types and their use cases.

We'll go over the following statistical graph types:

1. Pie Chart
2. Bar Chart
3. Area Chart

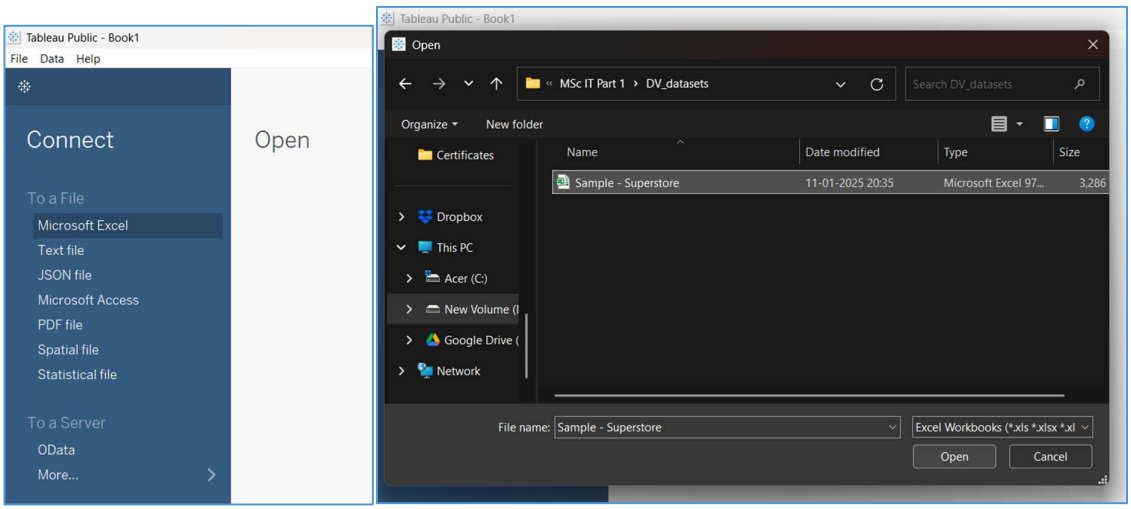
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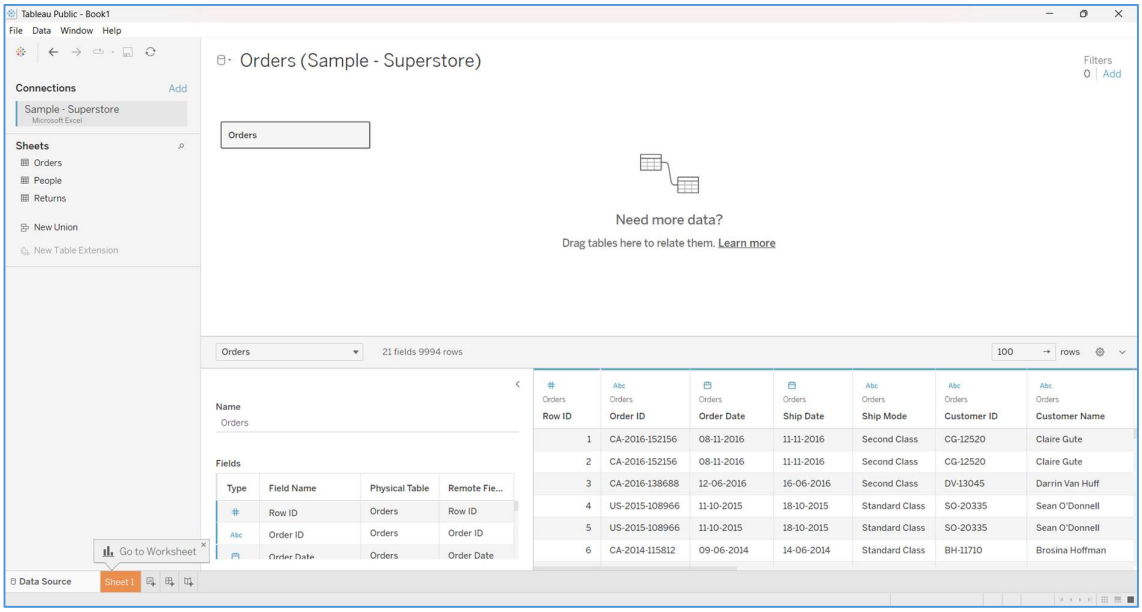
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OPEN WORKSHEET

Creating charts based on the data we connected is easy. At the bottom of the page, click on a sheet (sheet 1) and we will see the following screen:

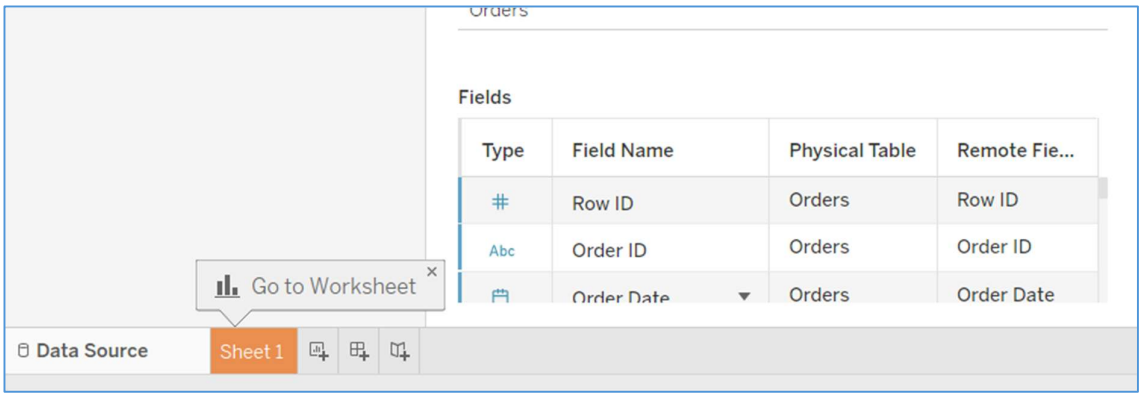
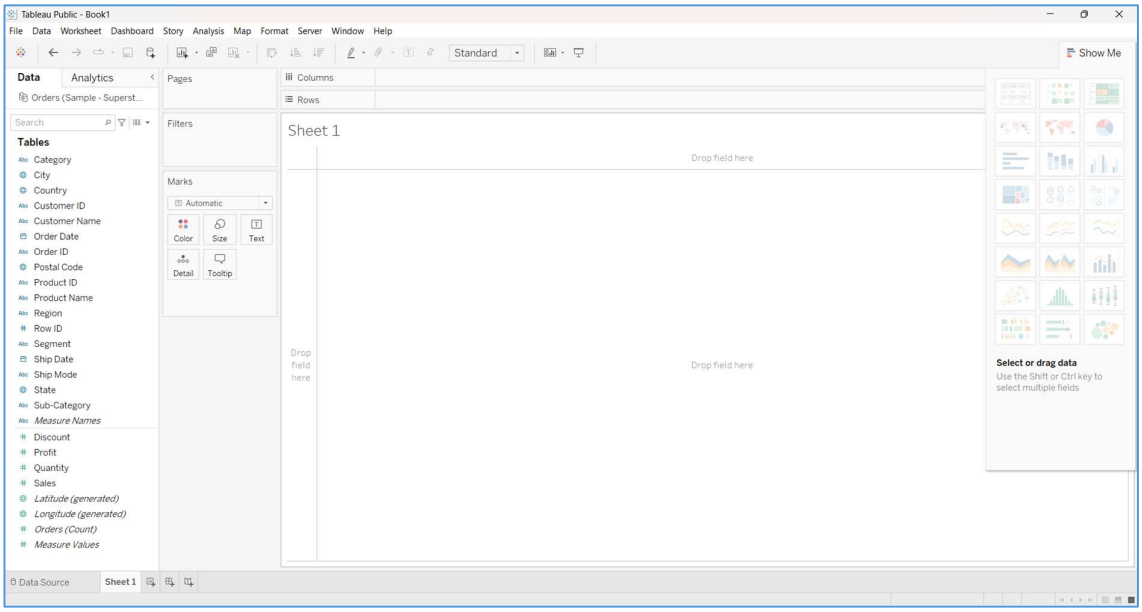


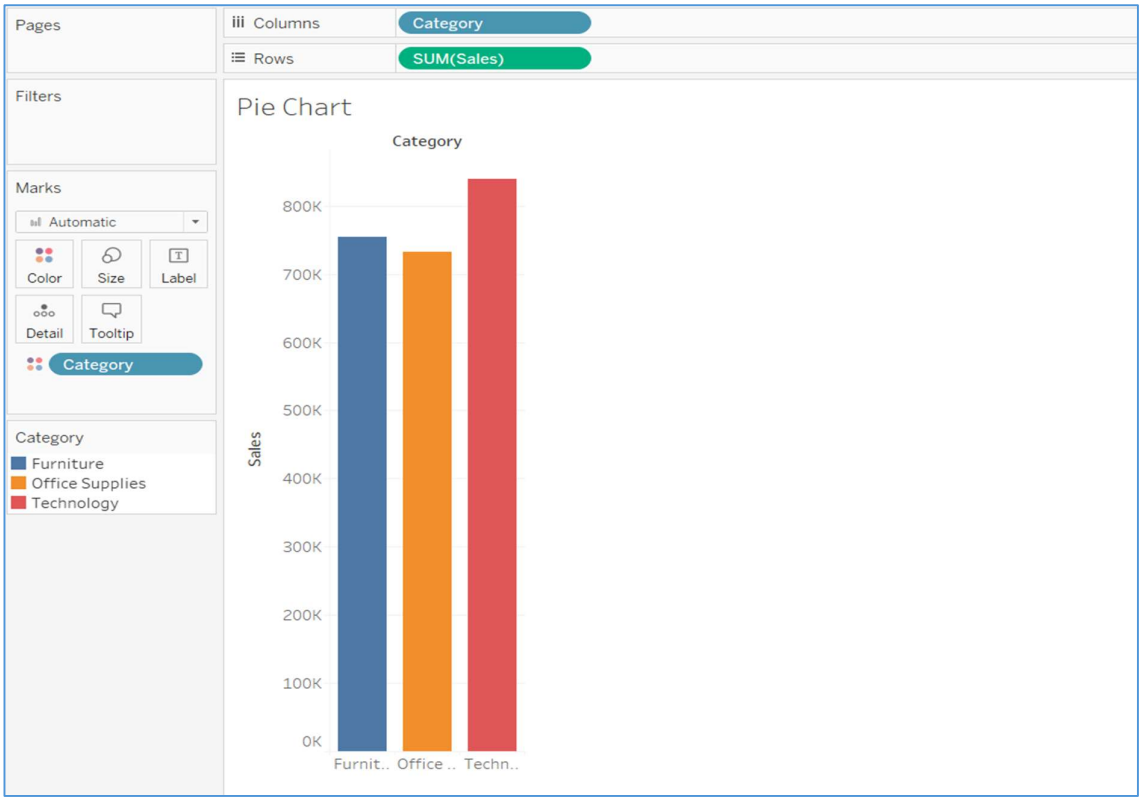
Tableau automatically separates the data into Dimensions and Measures. Dimensions are the categorical fields. These fields will create labels in the chart. Measures are the quantitative fields. These are the numbers we want to analyse. They create axis in the chart.



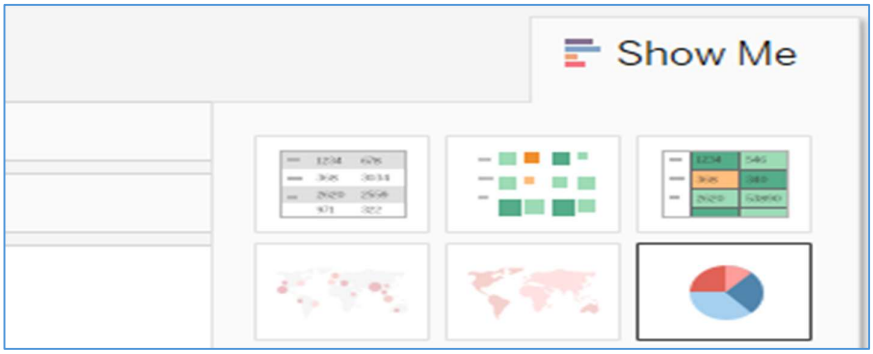
STEPS FOR CREATING STATISTICAL DATA GRAPHS VISUALIZATION

1. PIE CHART

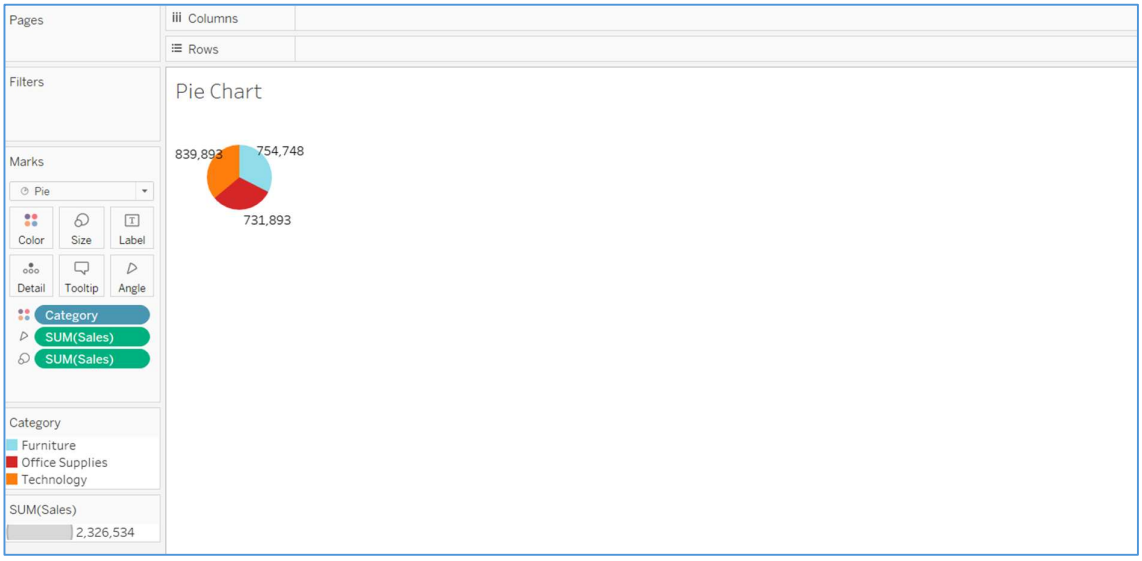
To Create a pie chart, we need to drag **Category** to the Columns shelf and Color shelf in the Mark card, drag **Sales** to the Rows shelf.

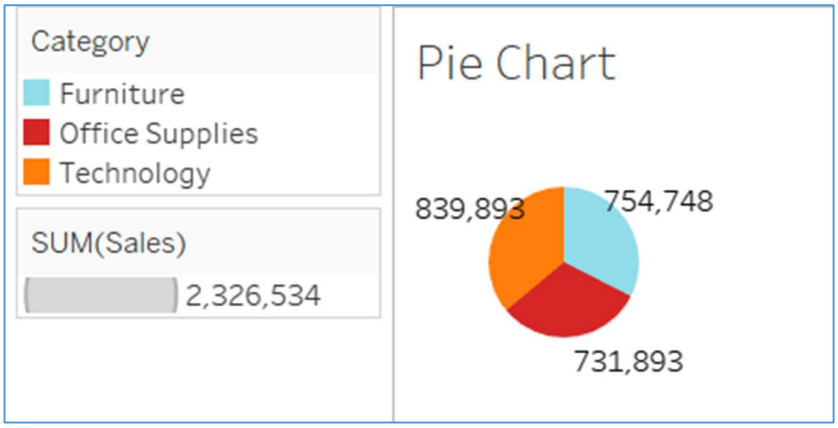


Above we can see **Net Sales Based on Category**, now we need to convert the graph into pie chart for which we need to go to the Show-me card and click on Pie Chart.



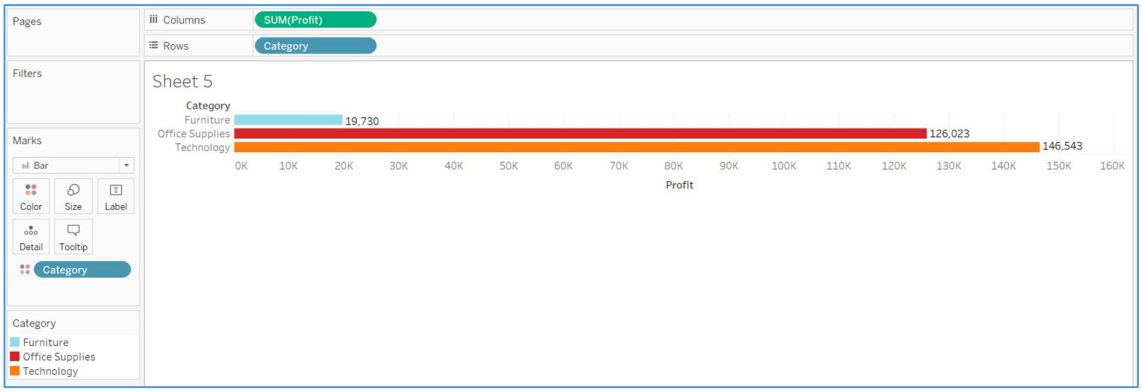
This will create a pie chart representing **Net Sales Based on Category**, as shown below:



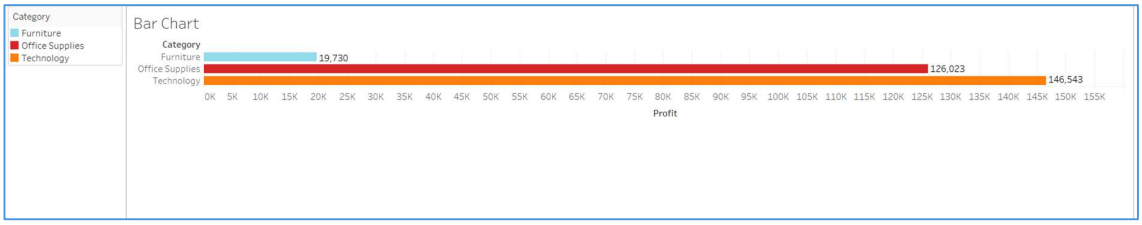


2. BAR CHART

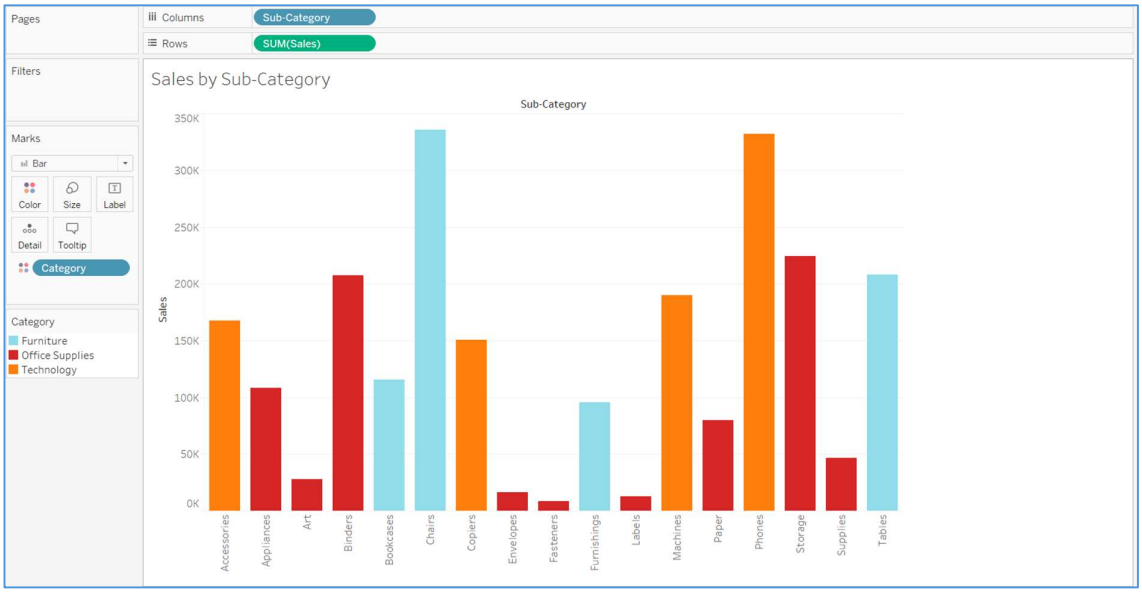
To create a Bar graph, we need to drag **Category** to the Rows shelf and Color shelf in the Mark card, drag **Profit** to the Column shelf. Change the chart type from the Automatic to Bar in the Marks card.



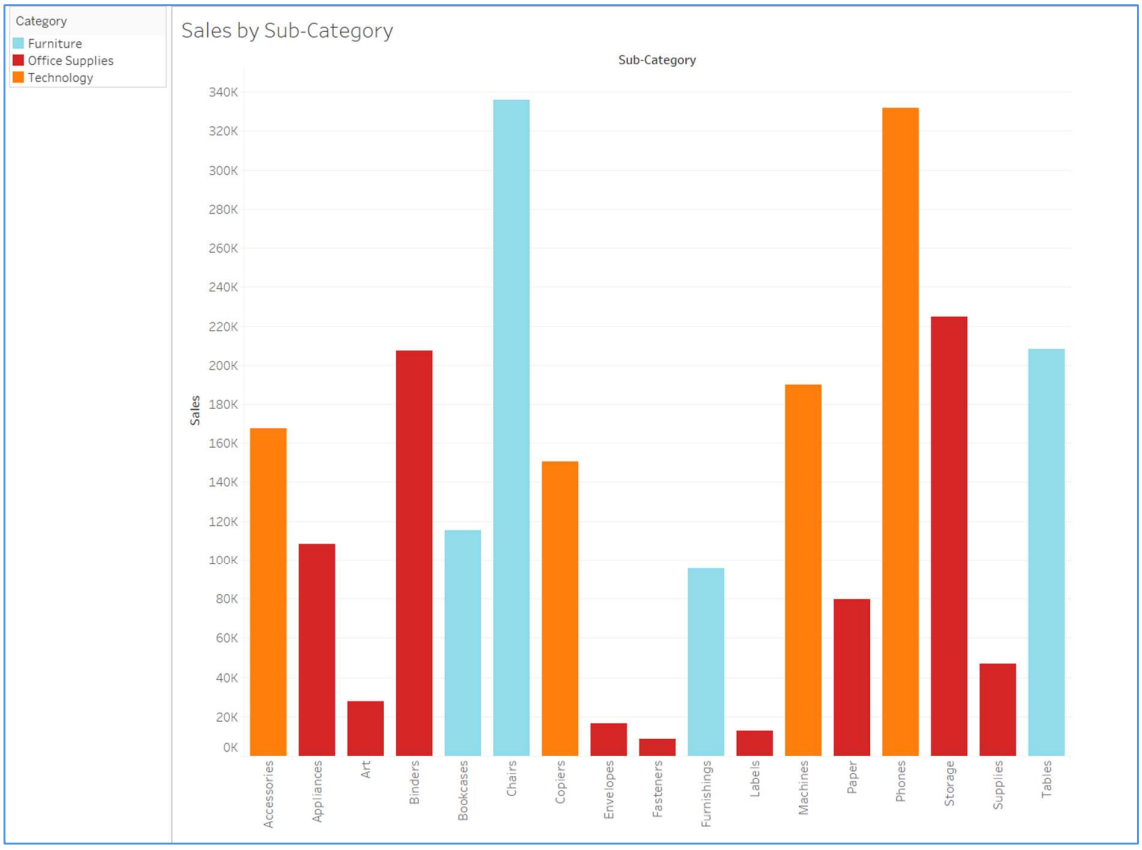
This will create a bar chart representing **Net Profit Based on Category**, as shown below:



Another form of Bar Graph where we have dragged **Sub-Category** to the Column shelf and **Category** to Color shelf in the Mark card, drag **Sales** to the Rows shelf. Change the chart type from the Automatic to Bar in the Marks card.

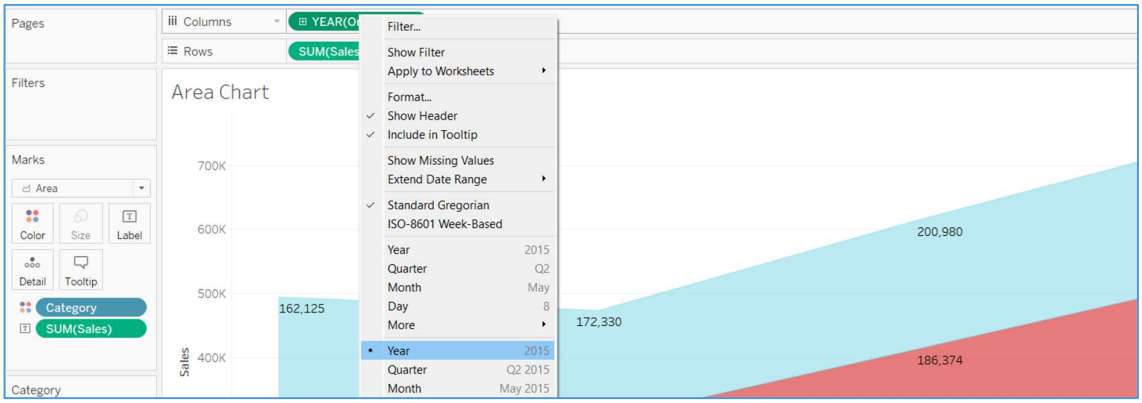


This will create a bar chart representing **Sales by Sub-Category**, as shown below:

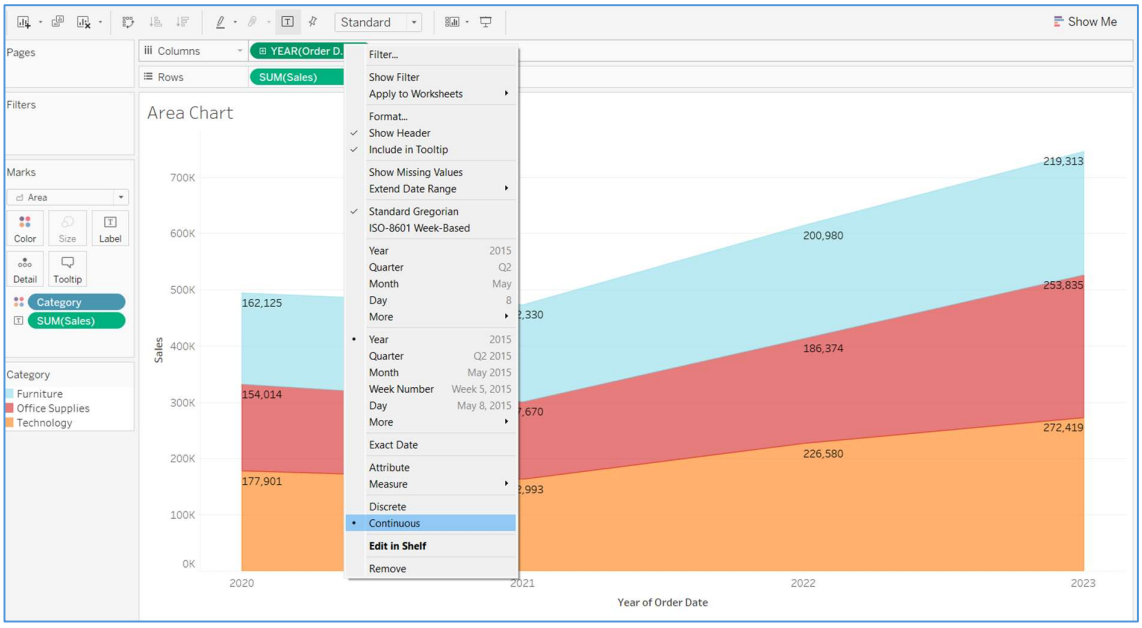


3. AREA CHART

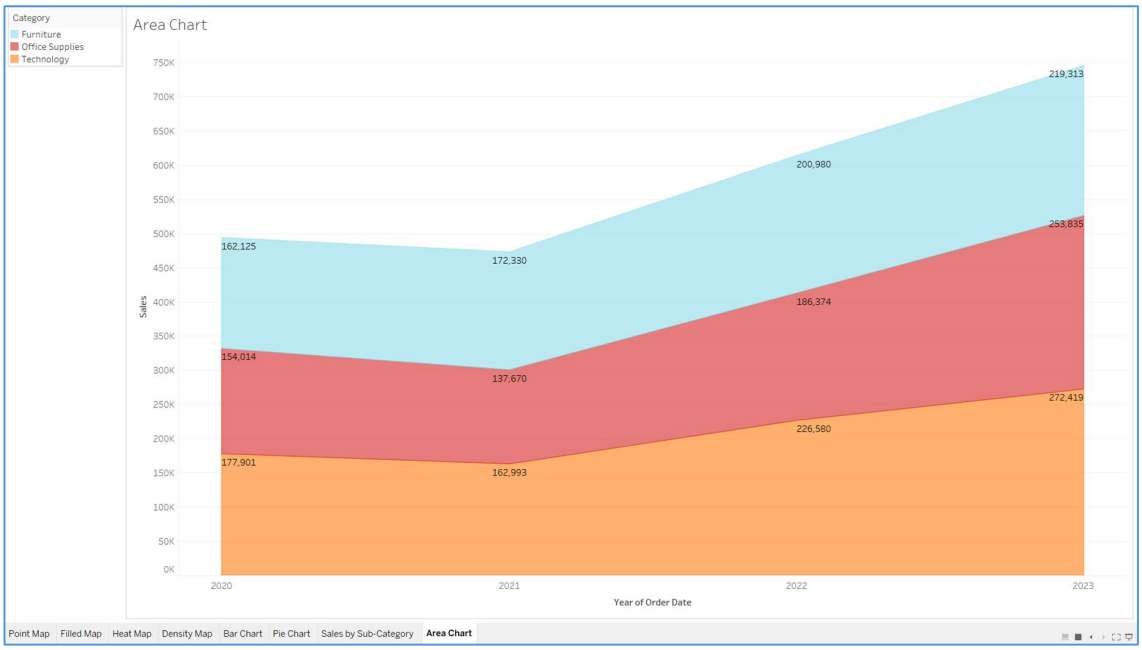
To create a Area graph, we need to drag **Sales** to the Rows shelf and **Category** to the Color shelf in the Mark card, drag **Order Date** to the Column shelf. Change the chart type from the Automatic to Area in the Marks card. Then right click on the **Year (Order Date)** in the columns shelf and tick on **Year and Continuous**.



Drag Sales to the labels in the Marks card to add labels in the Area graph.



This will create a area chart representing **Sales Growth over the years**, as shown below:



Category Analysis Dashboard



PRACTICAL 4

Aim: Generating map-based data using visualization tool

WHAT ARE MAPS?

Maps help you visualize and understand geographic data and reveal spatial patterns within your dataset. They are particularly useful when working on big data projects. Tableau offers mapping functionalities that cater to different types of geographic analyses.

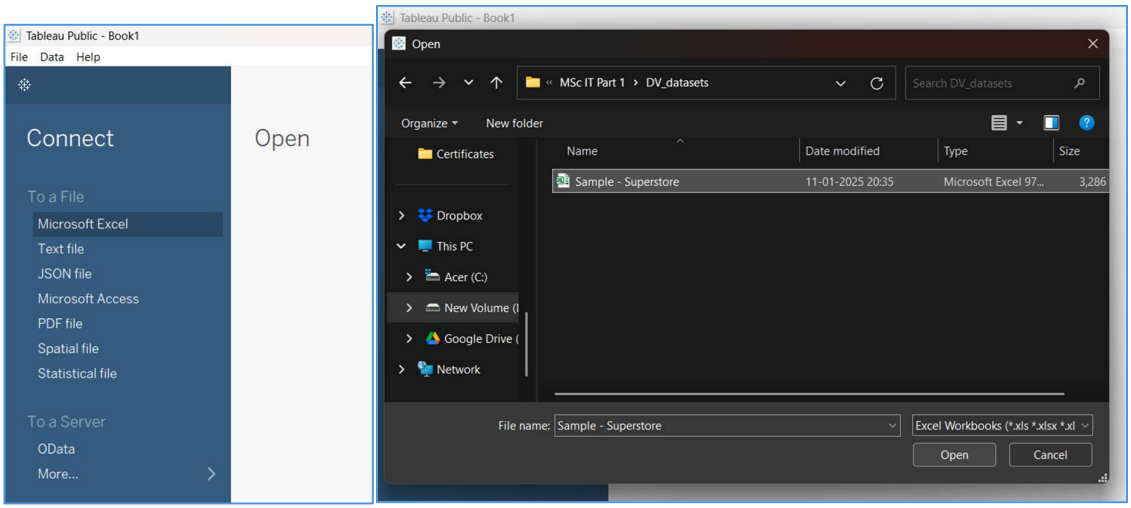
CONNECTING TO EXCEL FILE

Before connecting to excel file, we need to download the sample superstore file from tableau community or from Kaggle which is freely available

Dataset – Sample Superstore

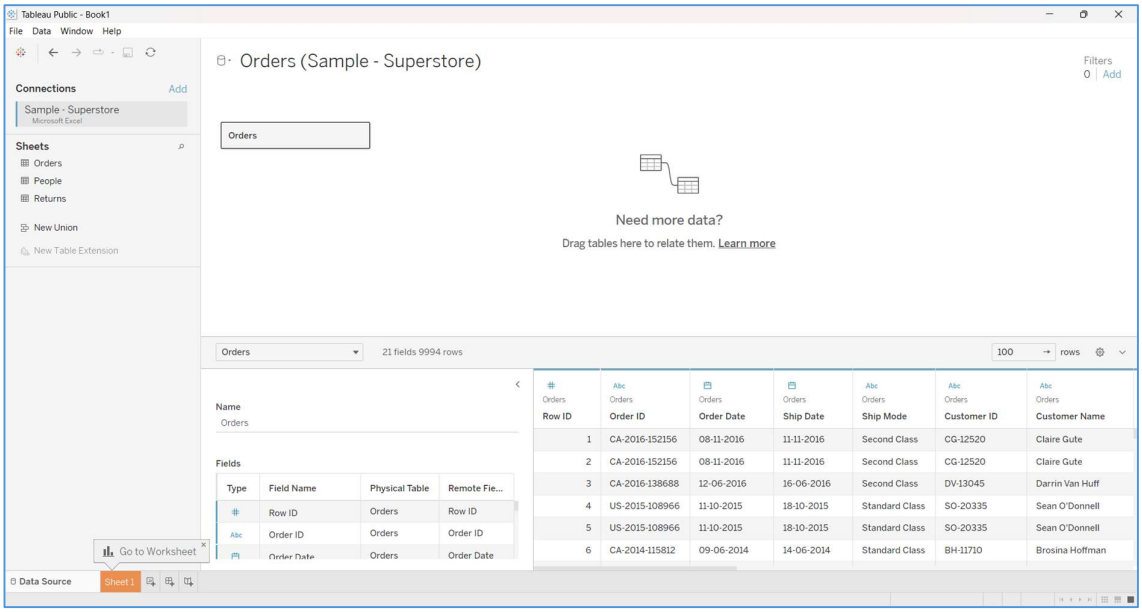
To connect to an excel file, click “Microsoft Excel” on the left-hand side. Navigate to the file on your computer and double click to open it. For this practical, we will use a sample file that comes with the installation called “superstore”. You should open the appropriate file that you will be working with.

Note: Always remember to change the file type of any downloaded dataset or excel file into excel workbook



Now you are in the data connection window. It looks like the following: -

Notice that I have three sheets in this file- Orders, People, and Returns. I can simply drag the table I want (in these practical we will drag or use the Orders table). If I drag more than one table, Tableau automatically creates the join between the tables.



OPEN WORKSHEET

Creating charts based on the data we connected is easy. At the bottom of the page, click on a sheet (sheet 1) and we will see the following screen:

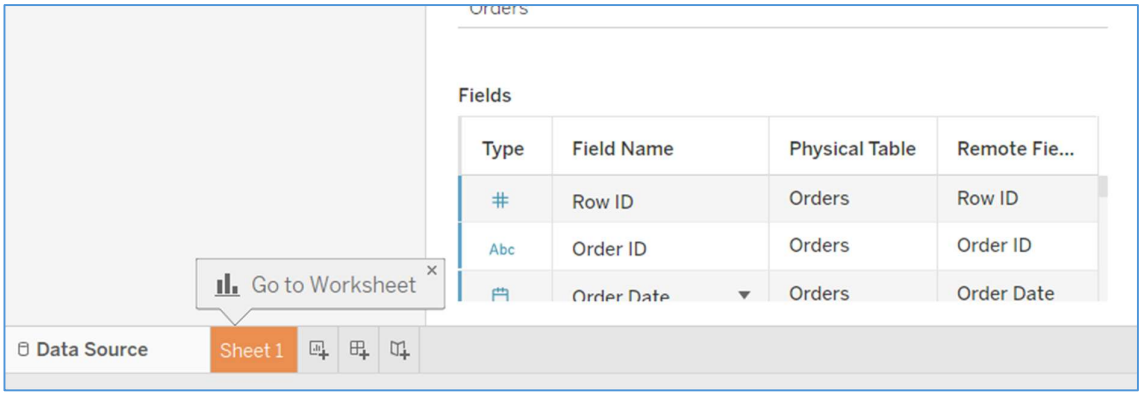
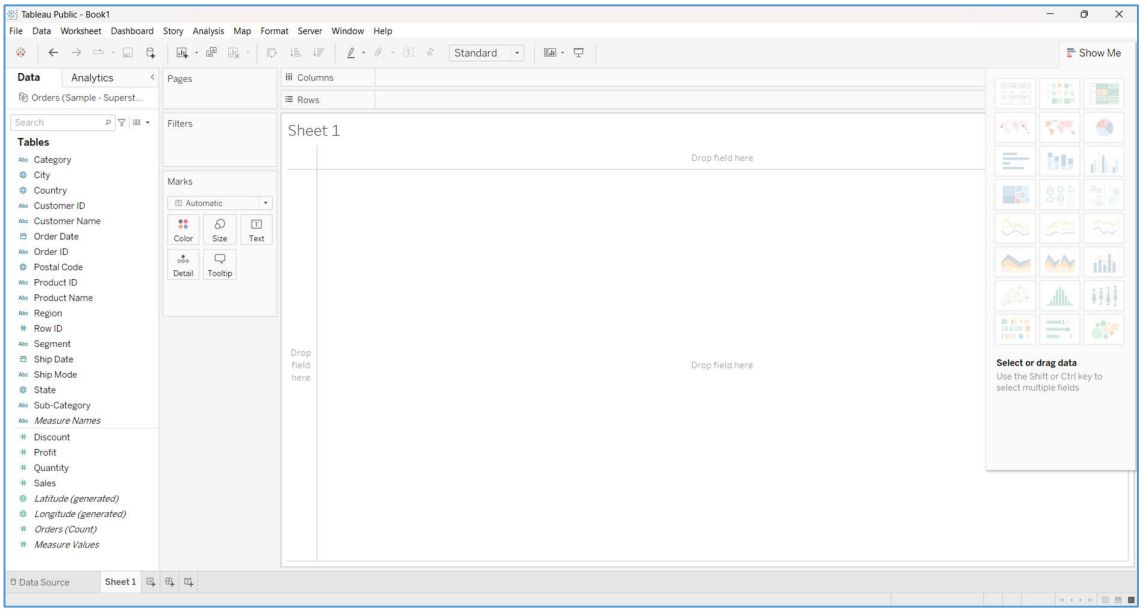
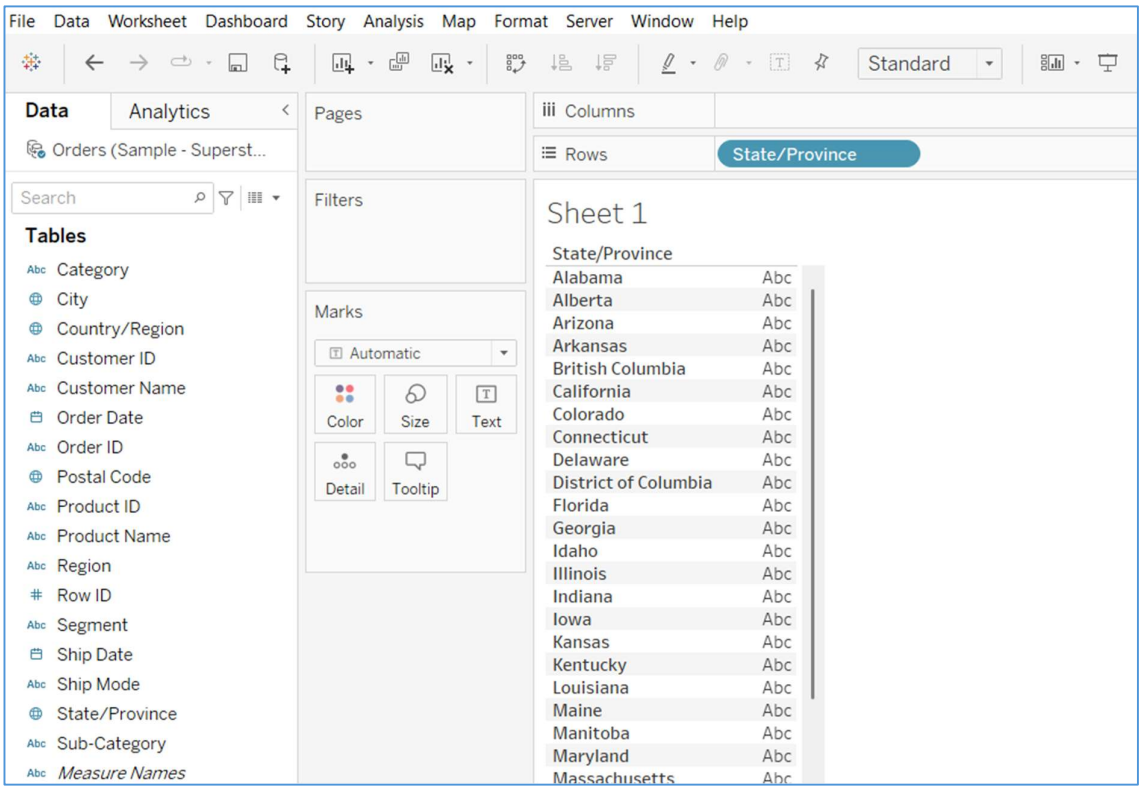


Tableau automatically separates the data into Dimensions and Measures. Dimensions are the categorical fields. These fields will create labels in the chart. Measures are the quantitative fields. These are the numbers we want to analyse. They create axis in the chart.

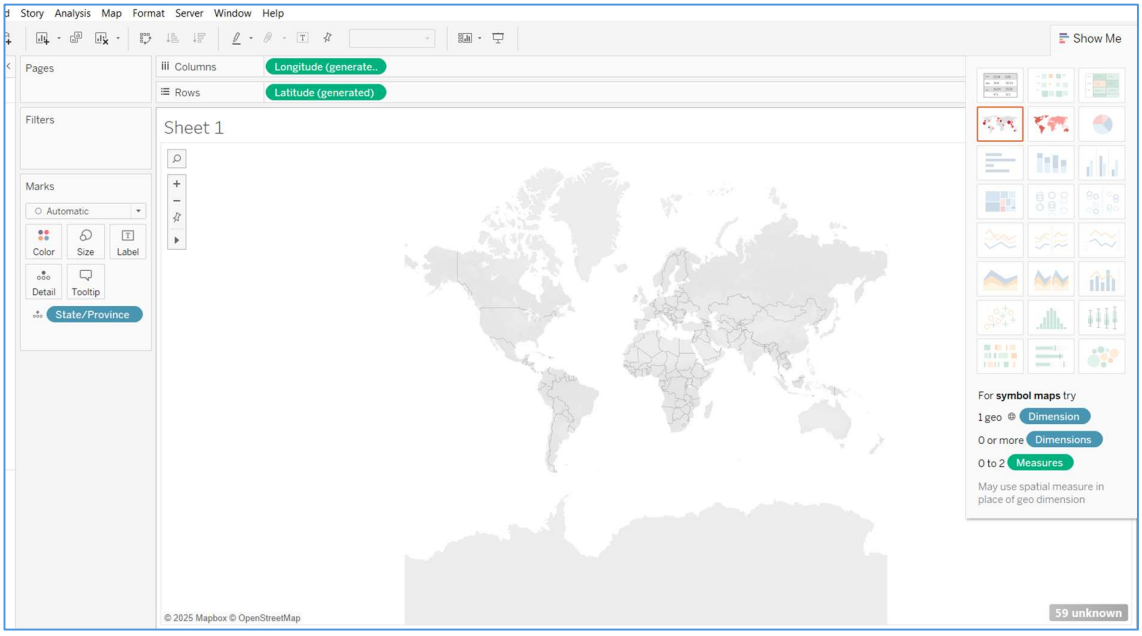


STEPS FOR CREATING MAP BASED DATA VISUALIZATION

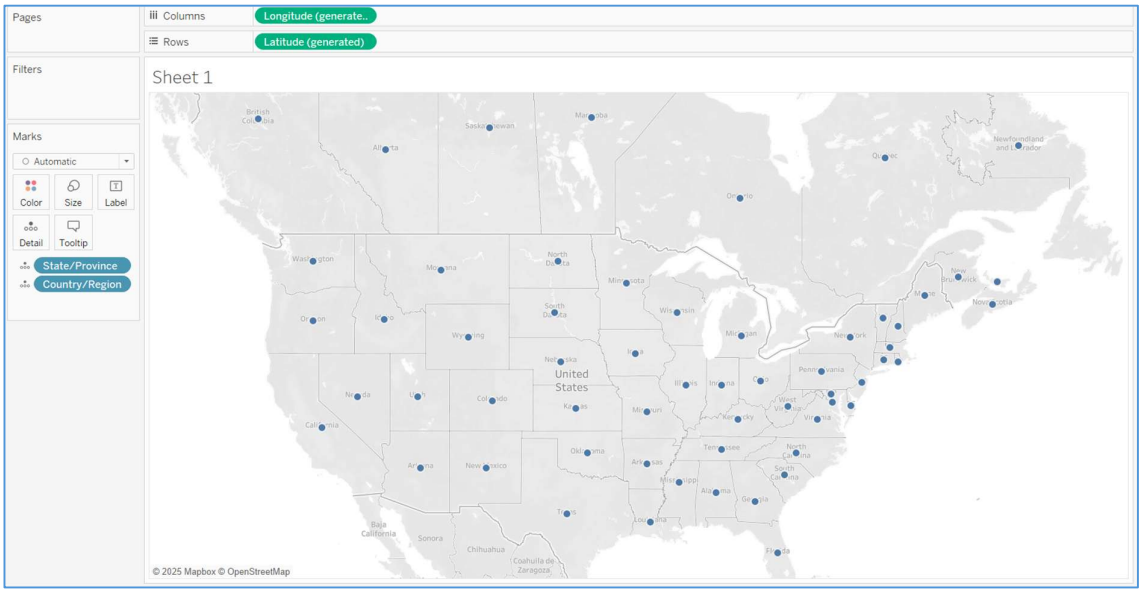
Step 1: Drag the **State/Province or City** field to the Rows shelf and the **Sales** variable to the Rows shelf.



Step 2: The next step is to convert your visual to a map. To do this, click on the **show-me** menu card on the top right and double click on **symbol map** it will show the world map by default and adds Longitude to Column Shelf, Latitude to Rows Shelf and State/Province to the details in the Maps card. After that drag and drop the Country field to the Maps card.



This will create a map in Tableau which will look like the following:



POPULAR TYPES OF MAPS IN TABLEAU

We can create various map types in Tableau to visualize and analyze our geographic data. This section will help us to understand the available map types and their use cases.

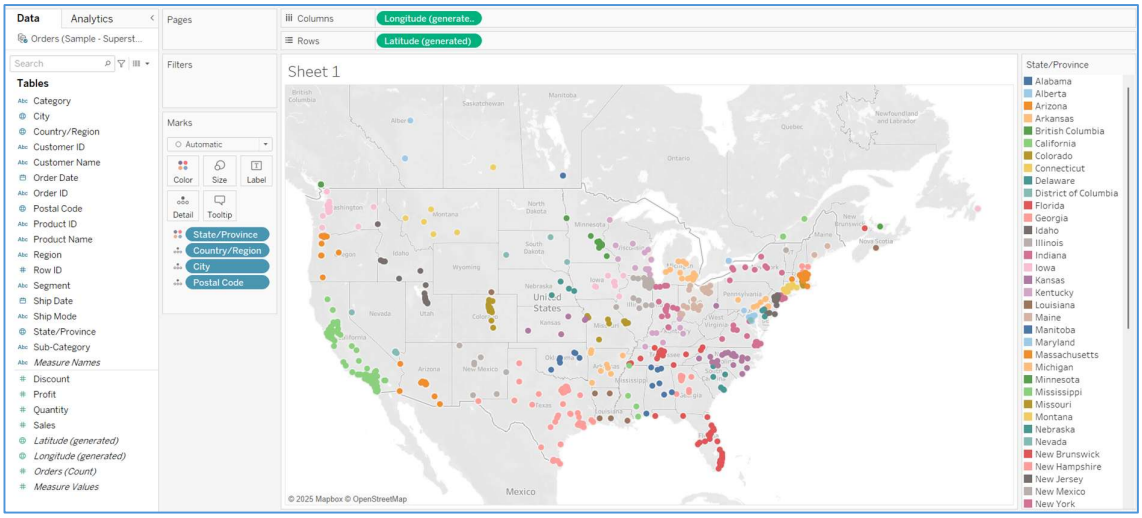
We'll go over the following map types:

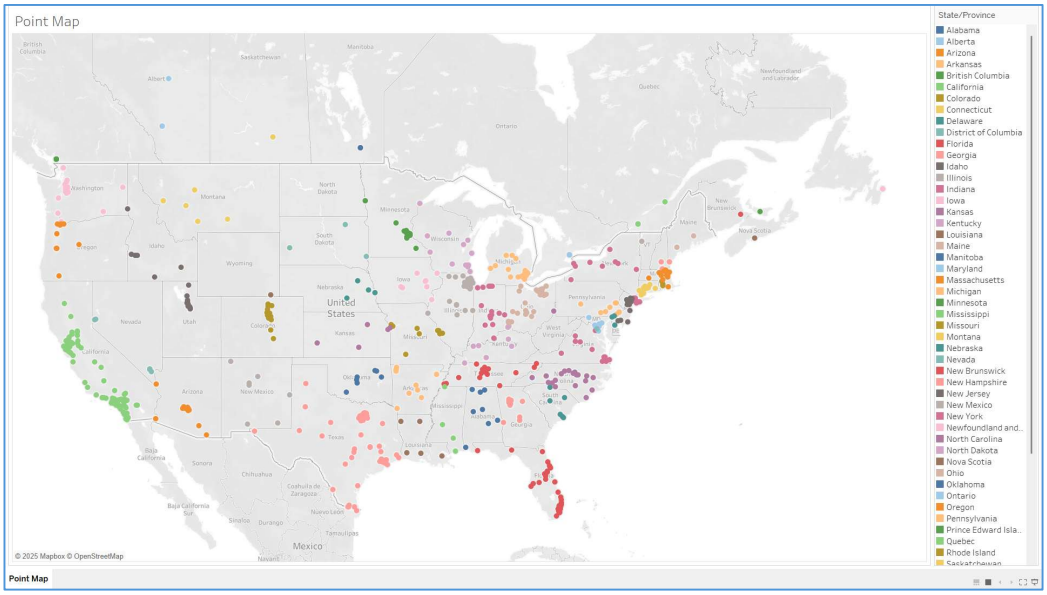
1. Point Map
2. Filled Map
3. Heat Map
4. Density Map

1. POINT MAP

A point map displays individual data points as symbols on the map. This type of map is useful when you want to show the location of specific items or events, such as store locations or crime incidents.

To create a point distribution maps in Tableau, we'll need one geographic dimension, such as **latitude and longitude** on the rows and column shelfs, a combination of other geographic fields like **city, postal code, state, and country** in the maps card. Below is a point distribution map based on different city points.



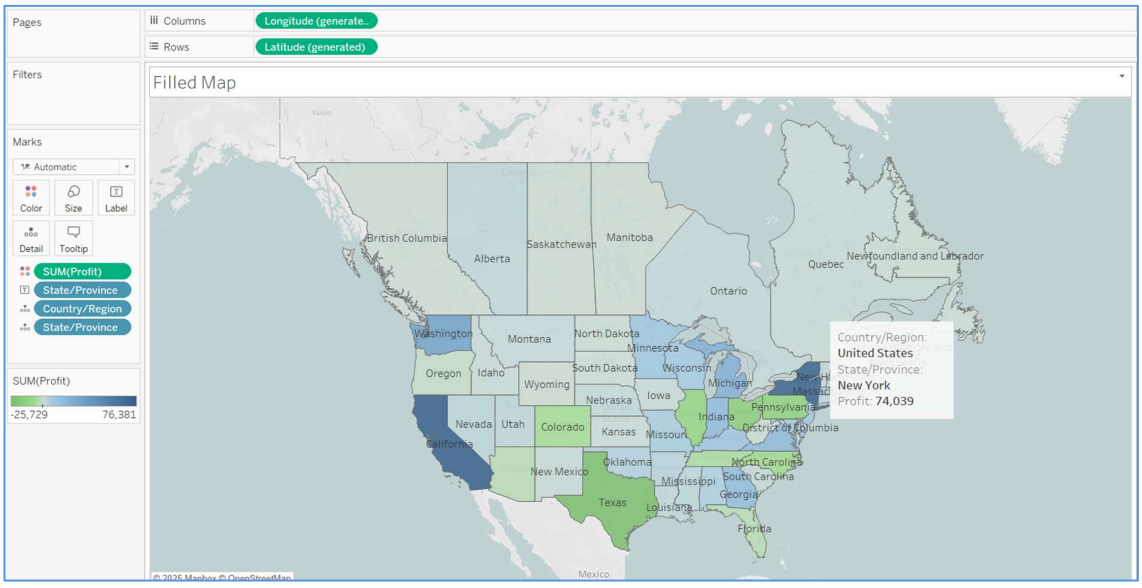


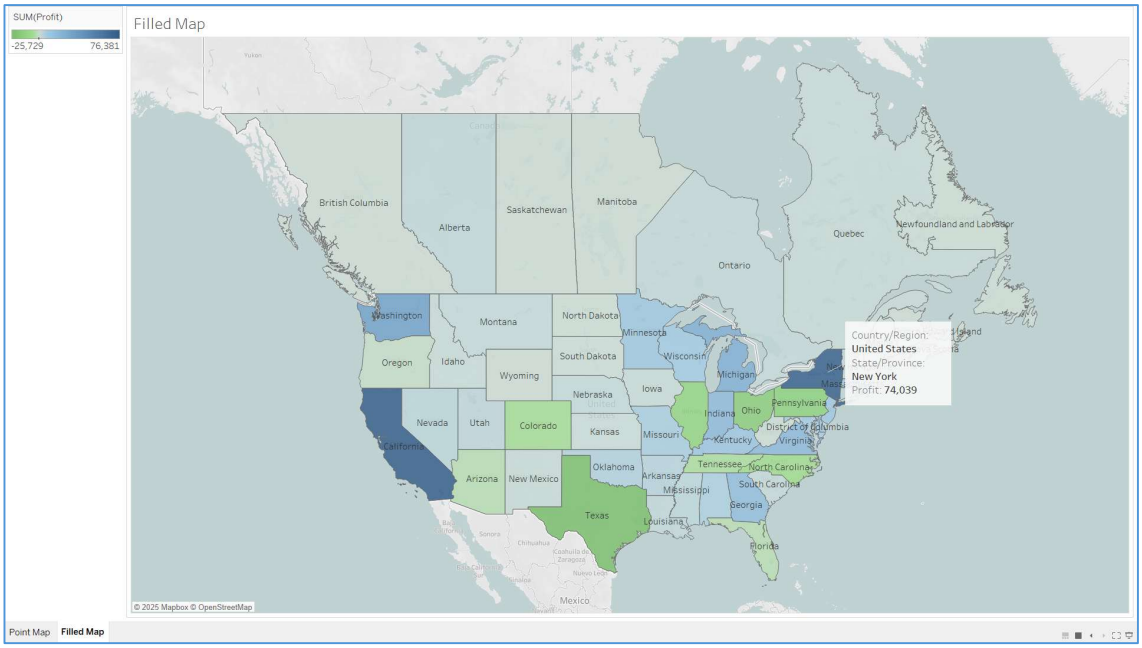
2. FILLED MAP

Filled maps in Tableau are a type of map visualization where geographical data is shaded or coloured according to a measure. This type of map is particularly useful for visualizing the geographic concentration of data, allowing for easy comparisons of a measure between different regions at a glance.

To create a filled map, drag a **geographical dimension (Longitude/ Latitude)** to the Columns / Rows shelf and a **measure (Profit)** to the Color shelf on the Marks card. A combination of other **geographic fields like city, postal code, state, and country** in the maps card.

The filled background maps are shown below:



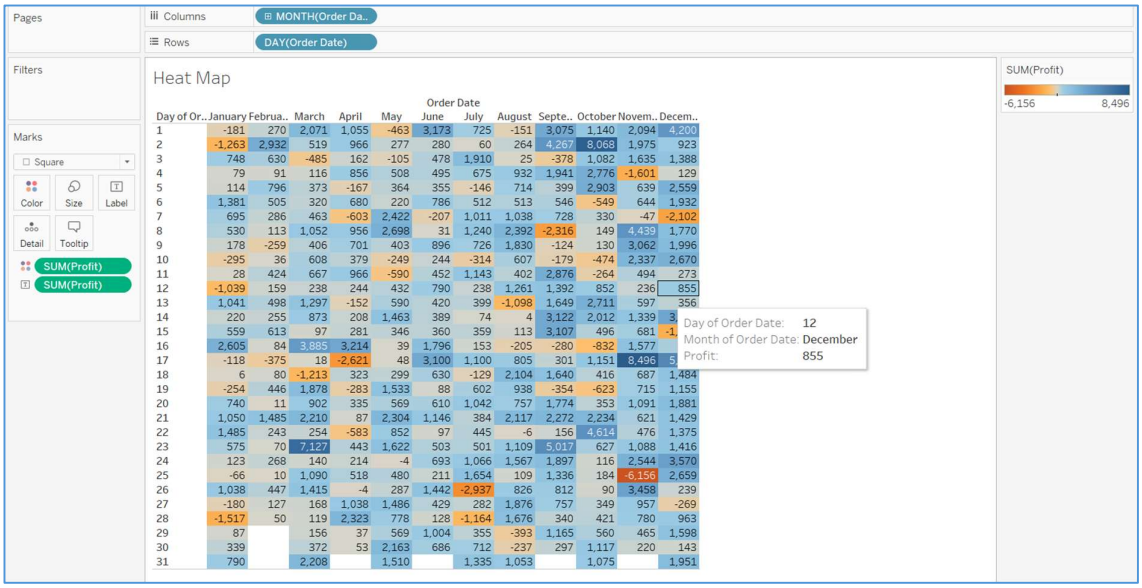


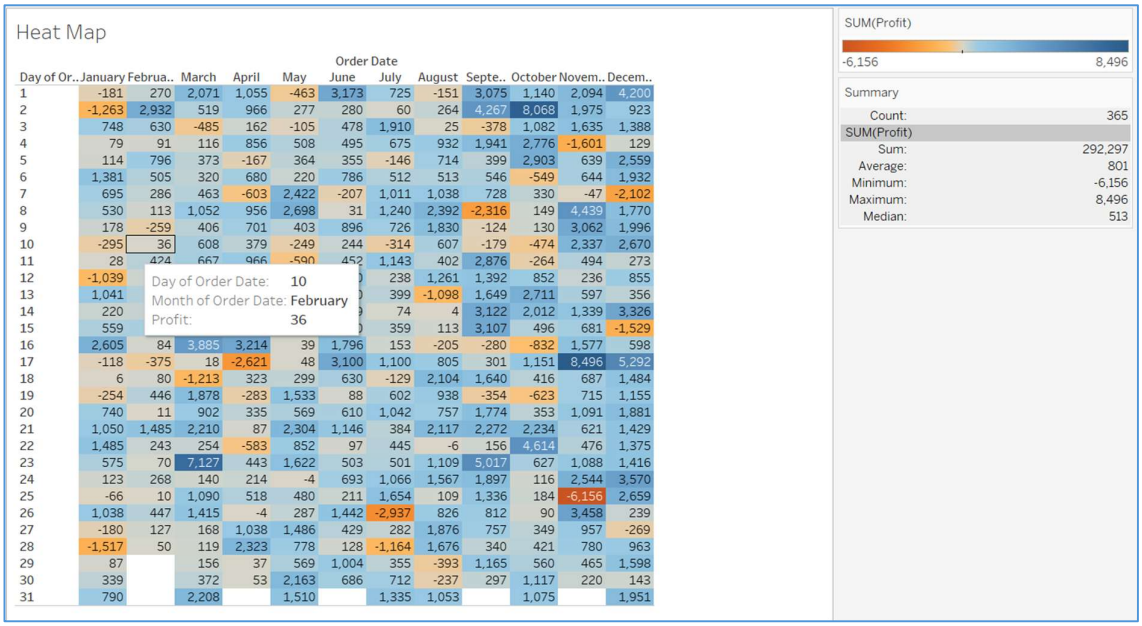
3. HEAT MAP

A heat map uses color gradients to represent the magnitude or frequency of variables in a two-dimensional table. It allows you to easily see patterns related to the intersection of two variables.

To create heat maps, drag one **dimension (Order Date)** to the Columns shelf (change it from year to month by right clicking on it) and **another dimension (Order Date)** to the Rows shelf (change it from year to days by right clicking on it). Then, drag a **measure (Profit)** to the Color shelf and same to the Label shelf on the Marks card. You can choose the **"Square" or "Text"** mark type from the Marks drop-down to better visualize your heat map.

The heat background map is shown below:



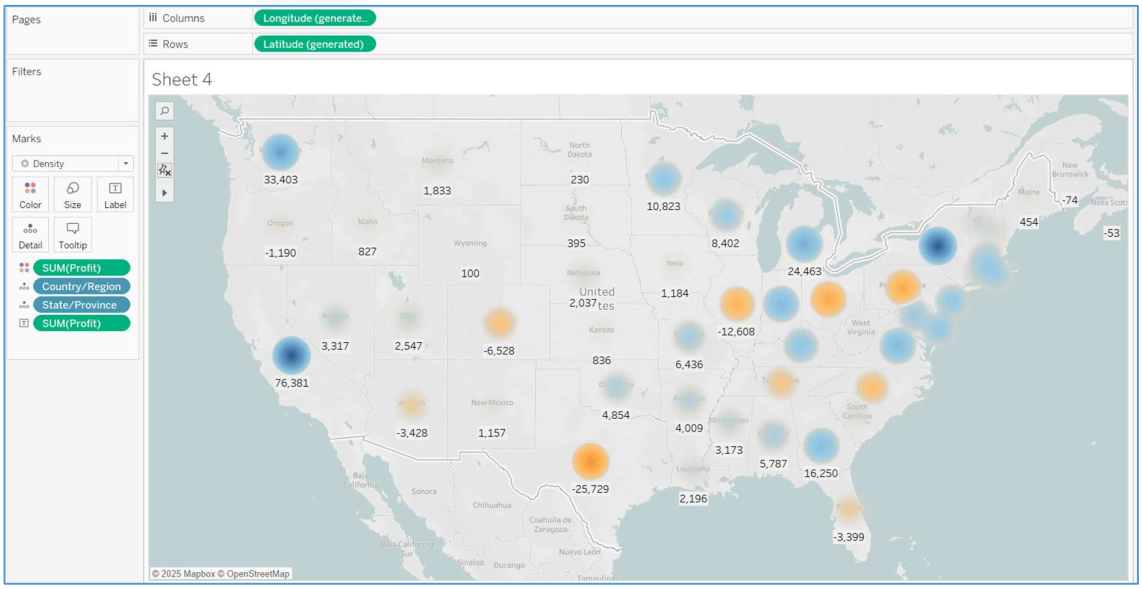


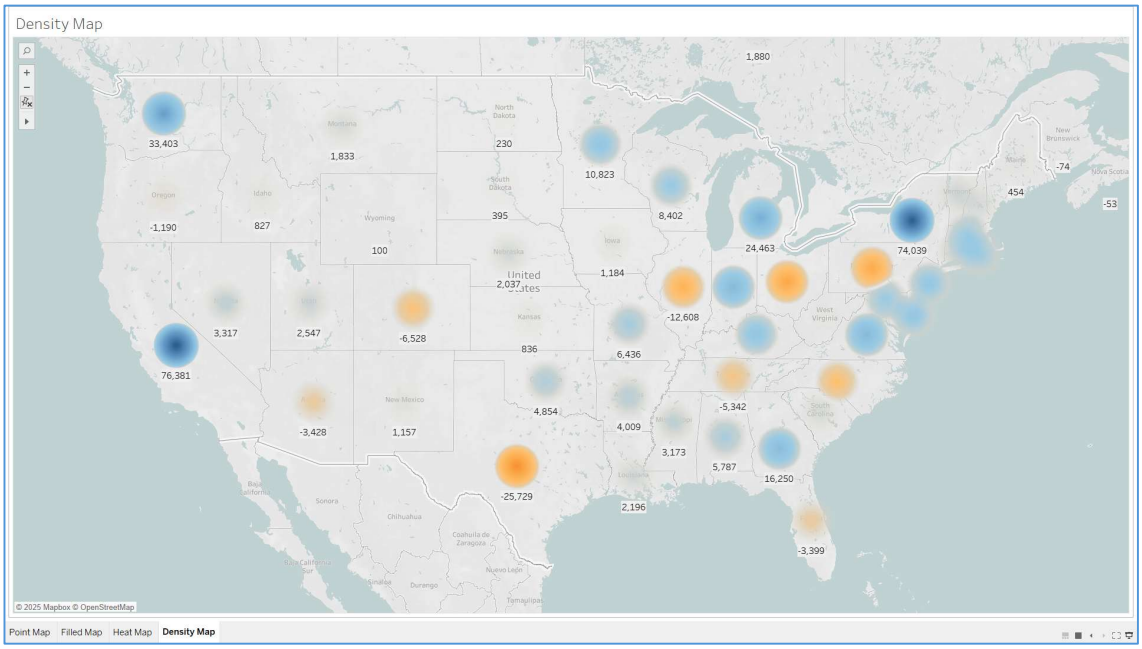
4. DENSITY MAP

A Density map in Tableau is used to display the concentration of data points in a geographic area or within a bounded context. Density maps use color gradients to show areas with higher or lower concentrations of data points.

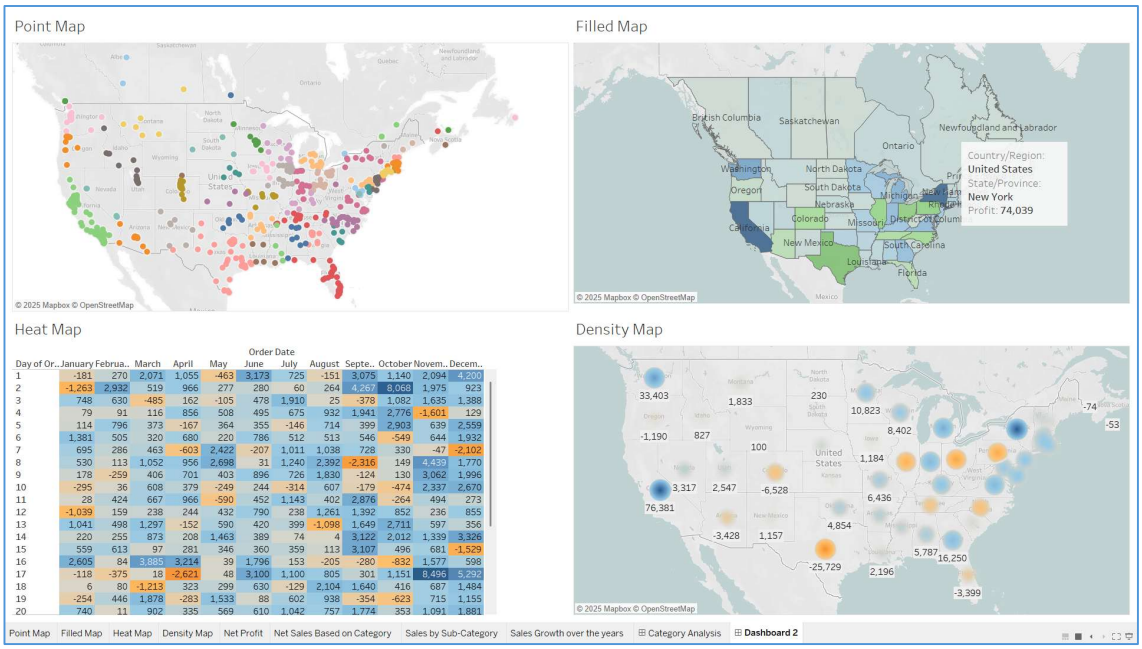
To create maps, drag your **latitude and longitude** coordinates or other relevant dimensions from the data pane to the Rows and/or Columns shelves, and then choose **"Density"** from the Marks card dropdown. Then, drag a **measure (Profit)** to the Color shelf and same to the label to represent the density. Tableau will then generate a color gradient that indicates the density of data points in the view.

This will create proportional symbol maps with density, as shown below:





Maps Dashboard



PRACTICAL 5

Aim: Demonstrate visualization of one, two, and multi-dimensional data

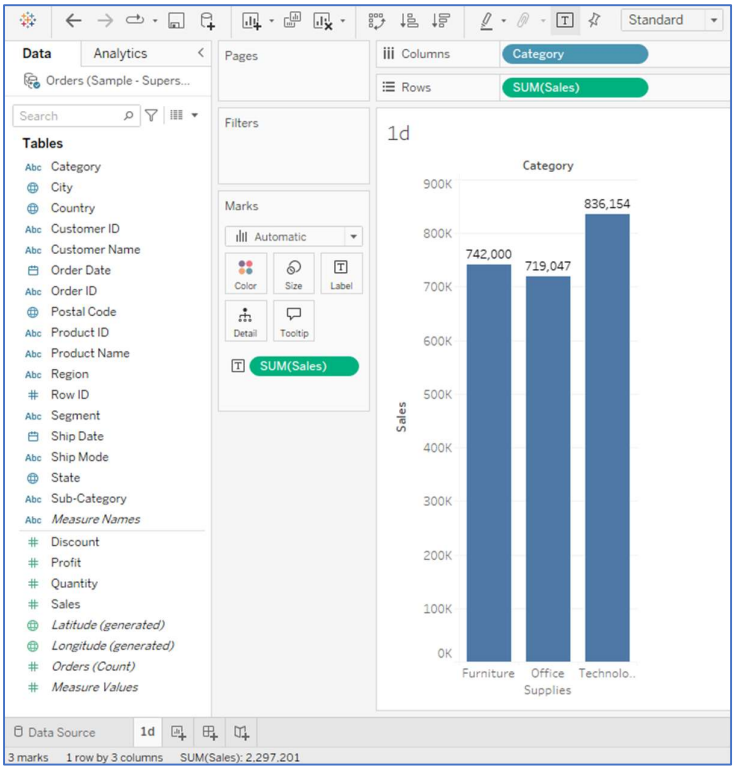
Since true 3D in the sense of volumetric rendering isn't directly supported by Tableau, we'll focus on creating visualizations that *represent* three dimensions using techniques like color, size, and position.

Step 1. Connecting to the Data

- Open Tableau Desktop.
- In the "Connect" pane, choose "Microsoft Excel" (or the appropriate connection type if your data is in a different format).
- Navigate to the **Sample - Superstore** Excel file (usually found in your Tableau Repository's "Data" folder).
- Drag the "Orders" sheet (or the relevant sheet you want to use) to the canvas.

Step 2. 1D Visualization (Example: Distribution of Sales by Category)

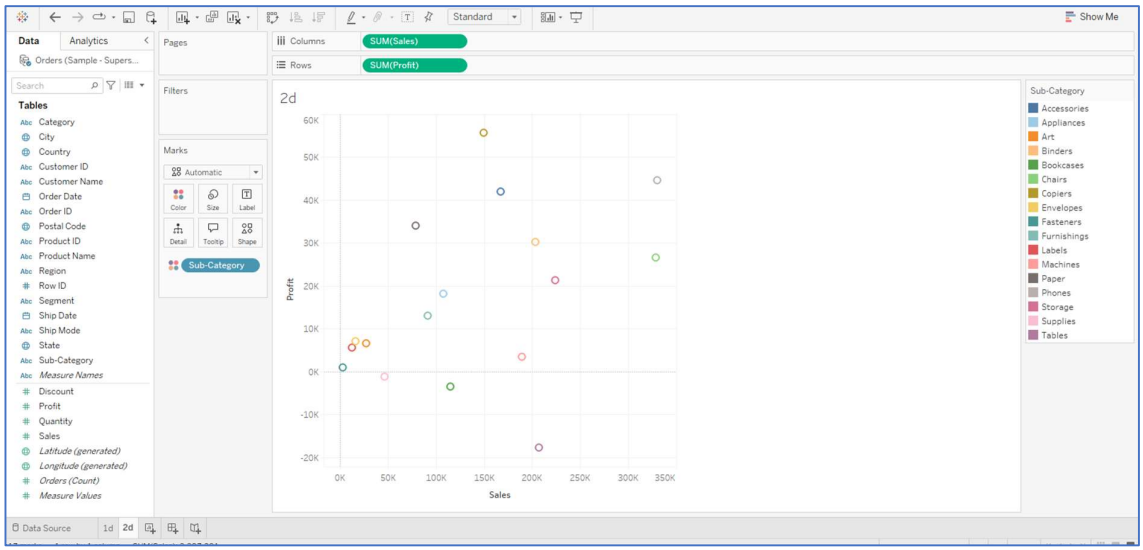
- Drag **"Category"** to the Columns shelf. This creates a basic bar chart with each category on the x-axis.
- Drag **"Sales"** to the Rows shelf. This determines the height of the bars, representing the total sales for each category.
- (Optional) Add a Label: Drag **"Sales"** to the Label card to display the sales value on each bar.
- (Optional) Sort: Click on the sorting icons in the axis headers to arrange the bars by sales.



Step 3. 2D Visualization (Example: Sales vs. Profit by Sub-category)

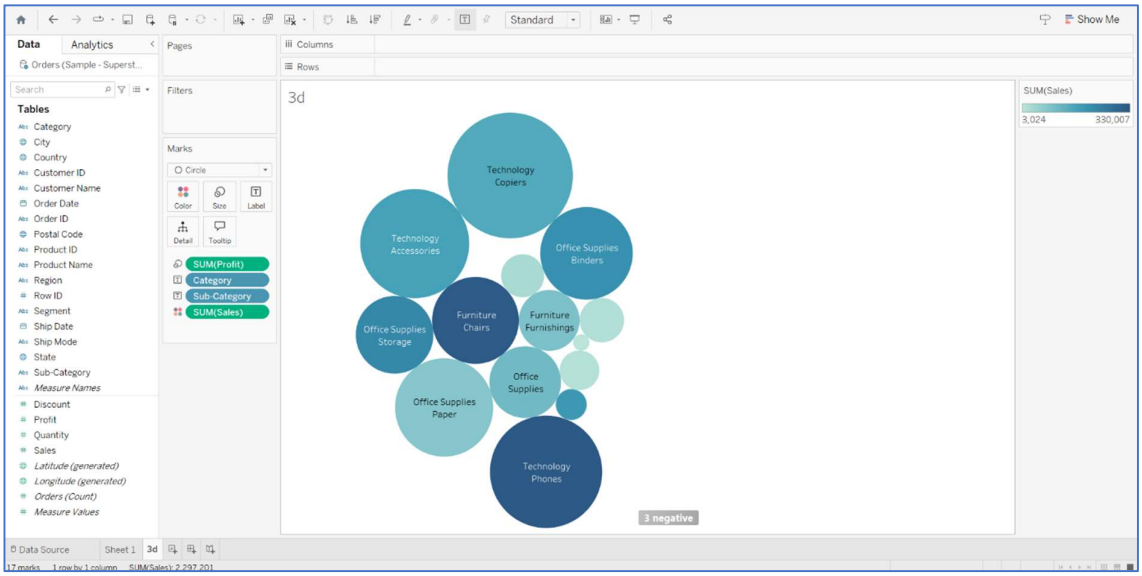
Drag and Drop:

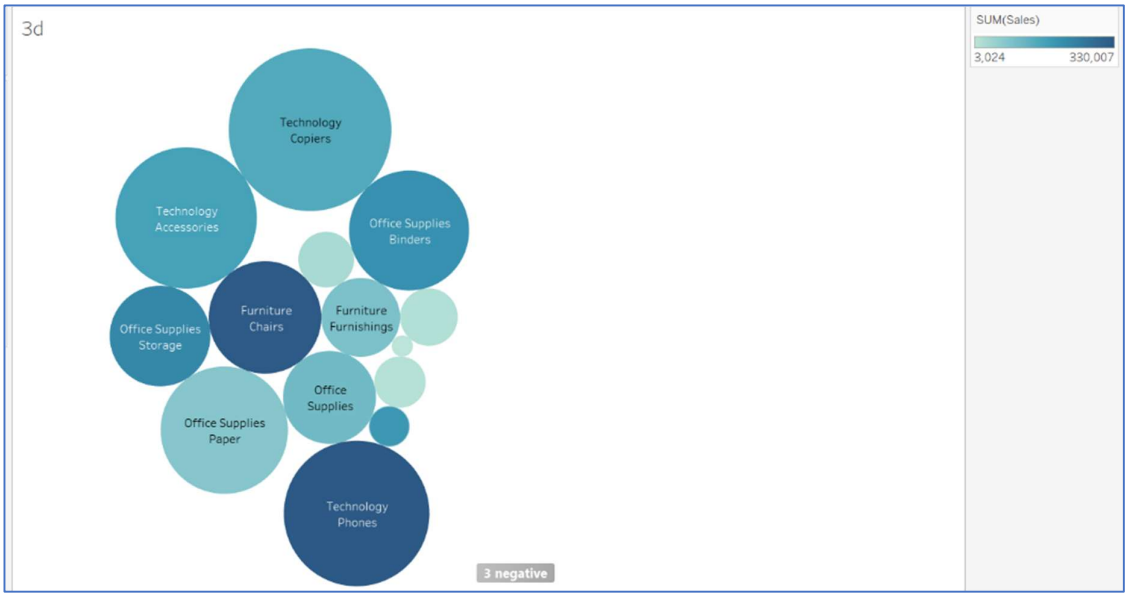
- Drag **"Sales"** to the Columns shelf.
- Drag **"Profit"** to the Rows shelf.
- Drag **"Sub-Category"** to the Color card.



Step 4. Representing 3D (Example: Sales, Profit by Category and sub-category)

- Drag **"Category"** to the Columns shelf. This will create separate columns for each category (Furniture, Office Supplies, Technology).
- Drag **"Sub-Category"** to the Rows shelf. This will further divide each category column into rows for the different sub-categories (Chairs, Tables, Bookcases, etc.).
- Drag **"Sales"** to the Size card in the Marks pane. This will make the bubbles vary in size according to the sales amount for each sub-category.
- Drag **"Profit"** to the Color card in the Marks pane.
- Select Bubble Chart from the Show me Section
- Click on the color legend that appears. Choose a diverging color palette (e.g., Red-Green or Orange-Blue Diverging). This will allow you to clearly see which sub-categories are profitable (e.g., green for positive profit) and which are not (e.g., red for negative profit). You can adjust the color steps to fine-tune the color breaks.
- If you want to display the sub-category names on the bubbles, drag "Sub-Category" to the Label card. You might need to adjust label size or allow overlapping labels if they are too crowded.


Data
Analytics
Orders (Sample - Superst...
Search
Tables
Category
City
Country
Customer ID
Customer Name
Order Date
Order ID
Postal Code
Product ID
Product Name
Region
Row ID
Segment
Ship Date
Ship Mode
State
Sub-Category
Measure Names
Discount
Profit
Quantity
Sales
Latitude (generated)
Longitude (generated)
Orders (Count)
Measure Values
Pages
Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
SUM(Profit)
Category
Sub-Category
SUM(Sales)
Columns
Rows
Standard
Show Me
3d
Technology Copiers
Technology Accessories
Office Supplies Binders
Office Supplies Storage
Furniture Chairs
Furniture Furnishings
Office Supplies
Office Supplies Paper
Technology Phones
3 negative
SUM(Sales)
3,024
330,007
Data Source
Sheet 1
3d


3d
Technology Copiers
Technology Accessories
Office Supplies Binders
Office Supplies Storage
Furniture Chairs
Furniture Furnishings
Office Supplies
Office Supplies Paper
Technology Phones
3 negative
SUM(Sales)
3,024
330,007

Practical 6

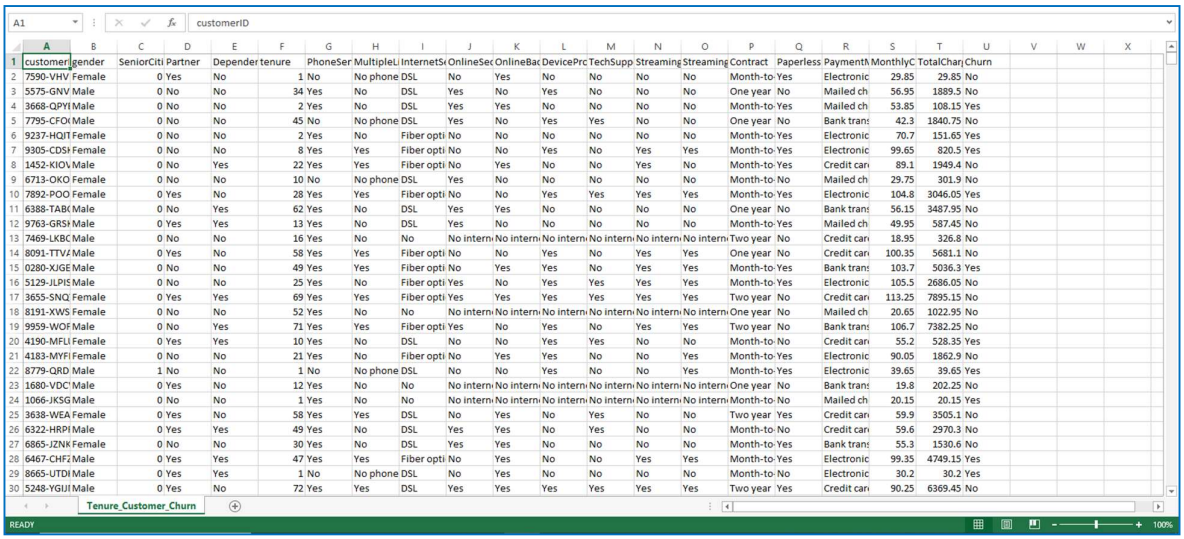
Aim: Visualizing tenure, monthly charges, total charges, and other individual columns using a scatter plot in tableau

WHAT IS SCATTER PLOT?

A scatter plot is a type of plot using Cartesian coordinates to display values for typically two variables for a set of data. If the points are coded (color/shape/size), one additional variable can be displayed. Scatter plots are commonly used in statistical analysis. They are an extremely effective way to compare multiple measures for a dimension with many distinct values. The basic case is to compare two measures with x and y axes. More measures can be added by Tableau's visual elements, such as size and color.

CONNECTING TO EXCEL FILE

Launch Tableau and connect to your data source (e.g., Excel, CSV, database). 'Tenure_Customer_Churn.csv'

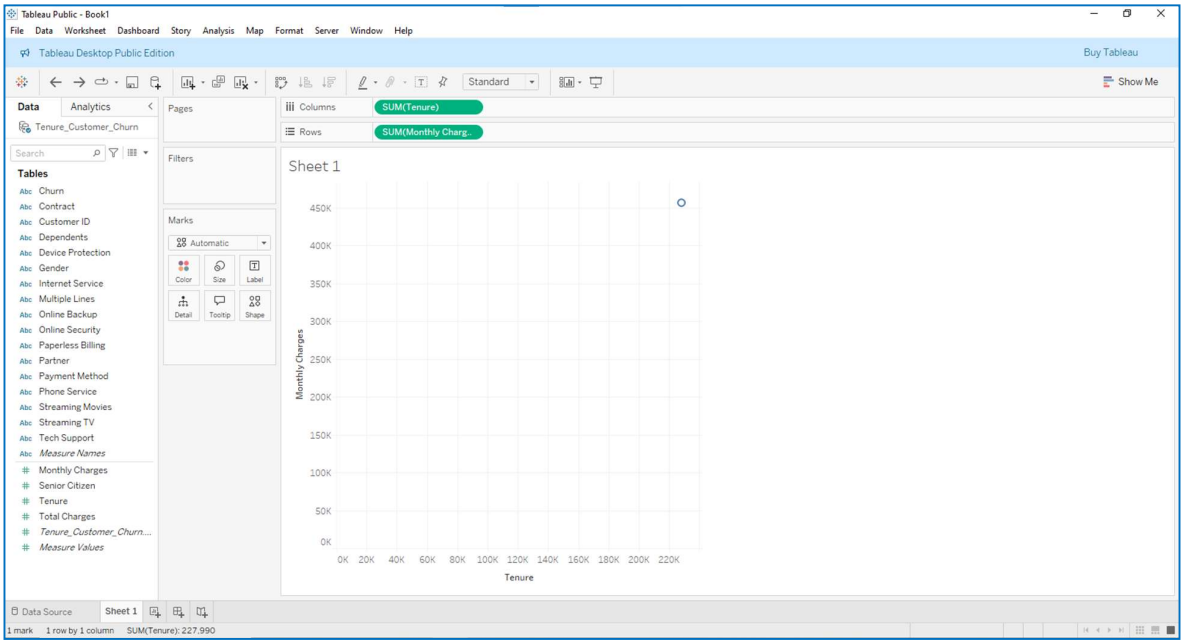


STEPS TO BE FOLLOWED

Click on the "New Worksheet" icon at the bottom of the Tableau interface.

Step 1. Drag Measures to Columns and Rows:

- Drag the Tenure measure to the Columns shelf.
- Drag the Monthly Charges measure to the Rows shelf.

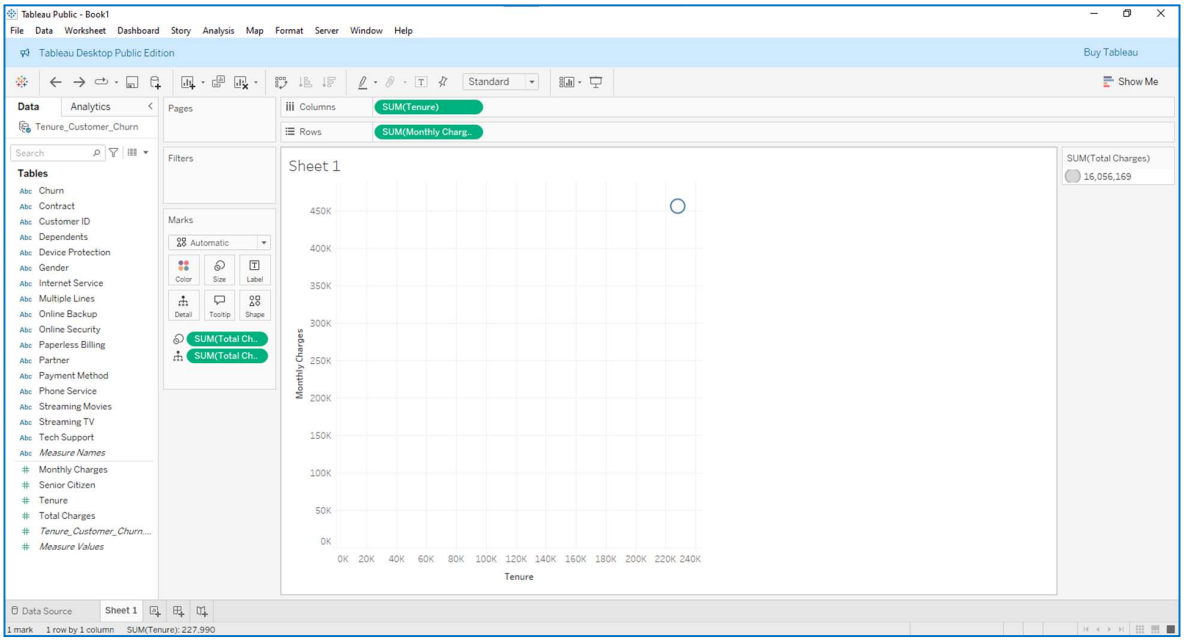


Step 2. Add Additional Measures:

- To add more measures like Total Charges, drag them to the Detail shelf on the Marks card. This will add more information to each data point.

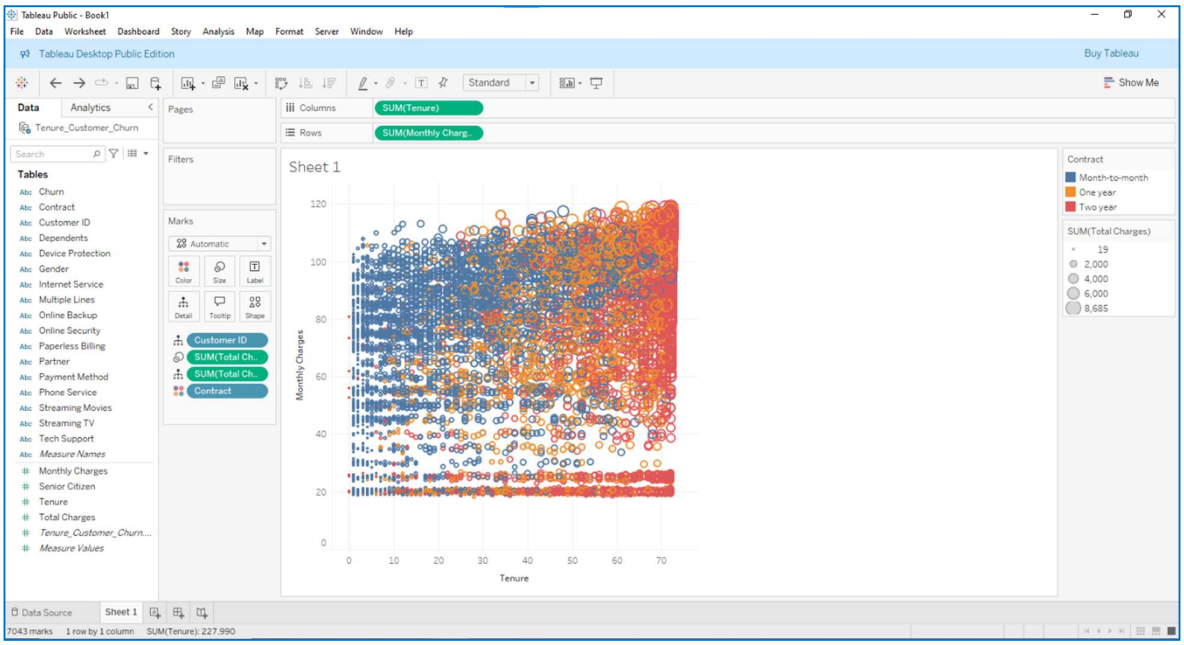
Step 3. Customize the Scatter Plot:

- Click on the "Show Me" panel on the right side of the interface and select the Scatter Plot option.
- You can customize the appearance of the scatter plot by adjusting the size, color, and shape of the marks. For example, you can drag the Total Charges measure to the Size shelf to vary the size of the marks based on total charges.



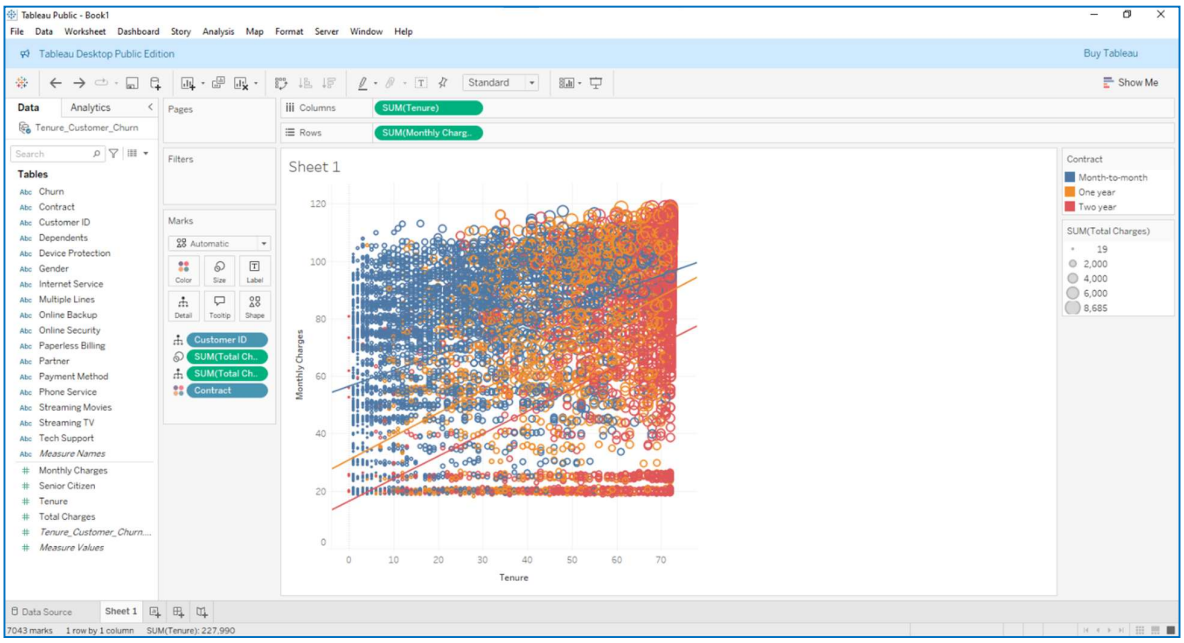
Step 4. Add Dimensions for More Detail:

- Drag dimensions like Customer ID or Contract Type to the Detail shelf to add more granularity to the scatter plot.



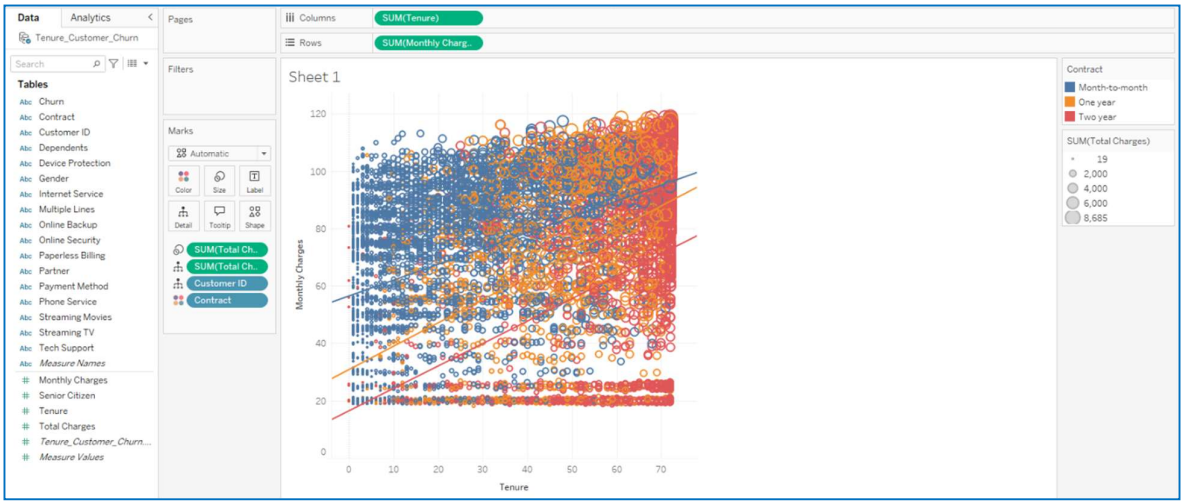
Step 5. Add Trend Lines (Optional):

- To add trend lines, go to the Analytics pane and drag the Trend Line option to the scatter plot. This will help you see the overall trend and relationship between the variables.



Step 6. Format the Scatter Plot:

- Right-click on the scatter plot and select "Format" to customize the fonts, colors, and other visual elements.



Practical 7

Aim: Demonstrate visualization of text and text document

CONCEPT AND CHARACTERISTICS

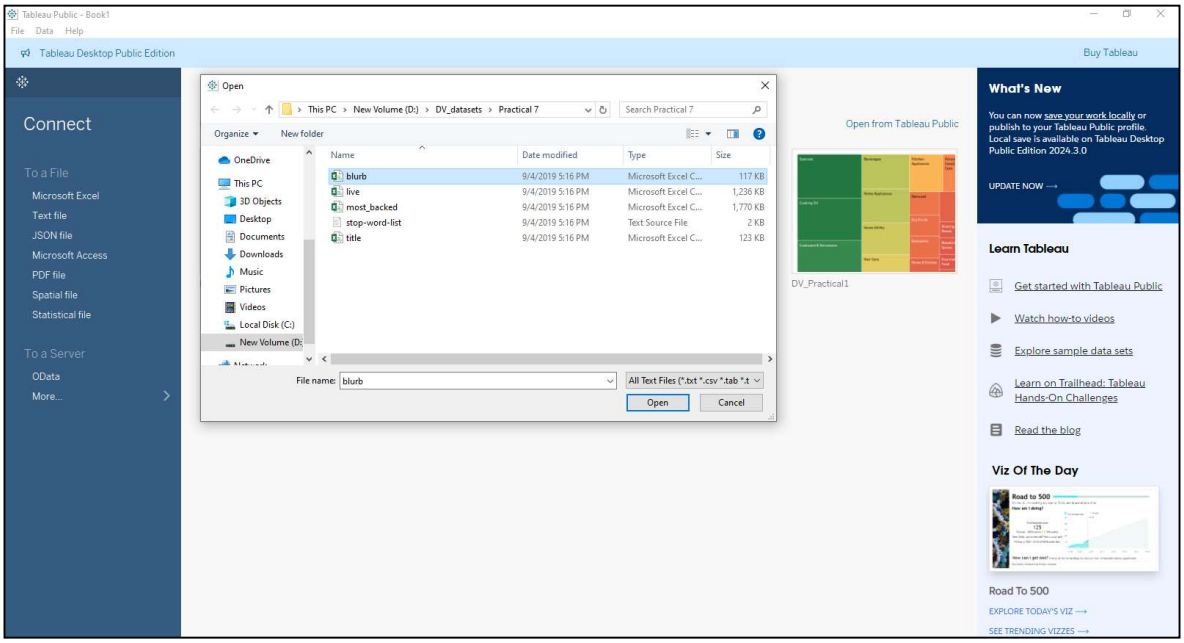
We'll use real-life datasets to build a word cloud which is a technique for text data visualization step by step. Meanwhile, we will draw some conclusions from Tableau visualization. The word cloud, called a *tag cloud* more broadly, is a novelty visual representation of text data. The basic unit is usually called a *tag* and in most cases displays as word. The tags form the shape of a cloud. That's the origin of the name.

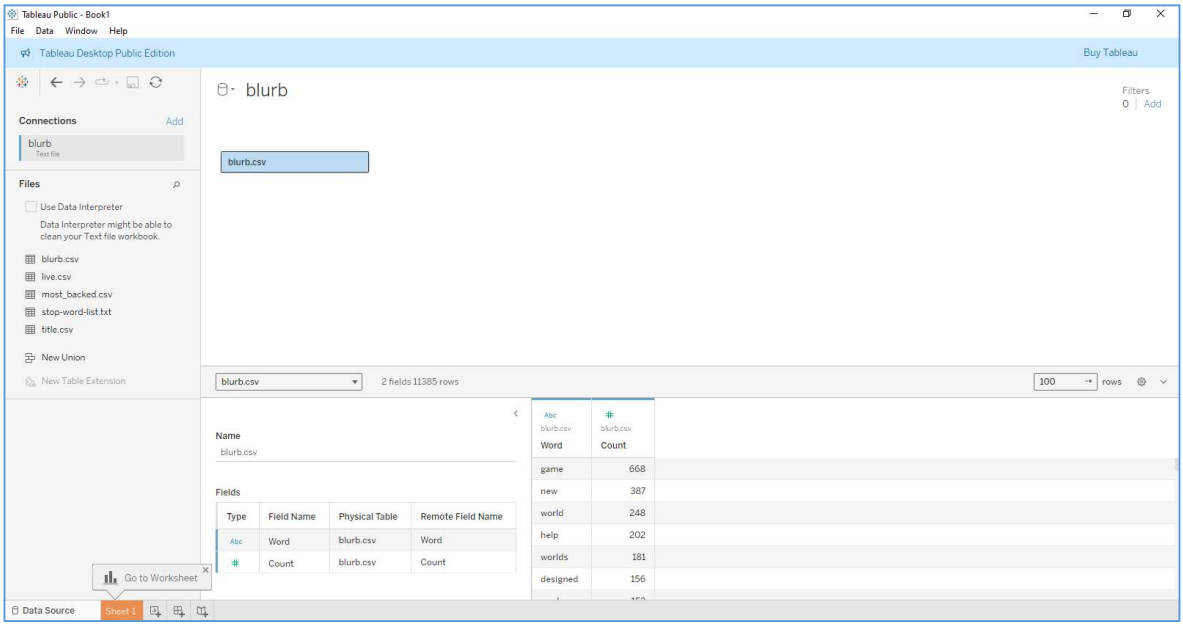
DATASET

In this guide, we'll use the dataset Kickstarter Project Statistics from Kaggle Dataset. This dataset contains the top 4000 most backed projects ever on Kickstarter. You can download "most_backed.csv" which is preprocessed to “blurb.csv” on the home page. In this guide, we will analyze the popular keywords in project blurbs.

LIST OF WORDS AND COUNT

In this example, we connect to the data source text file named **"blurb.csv".**

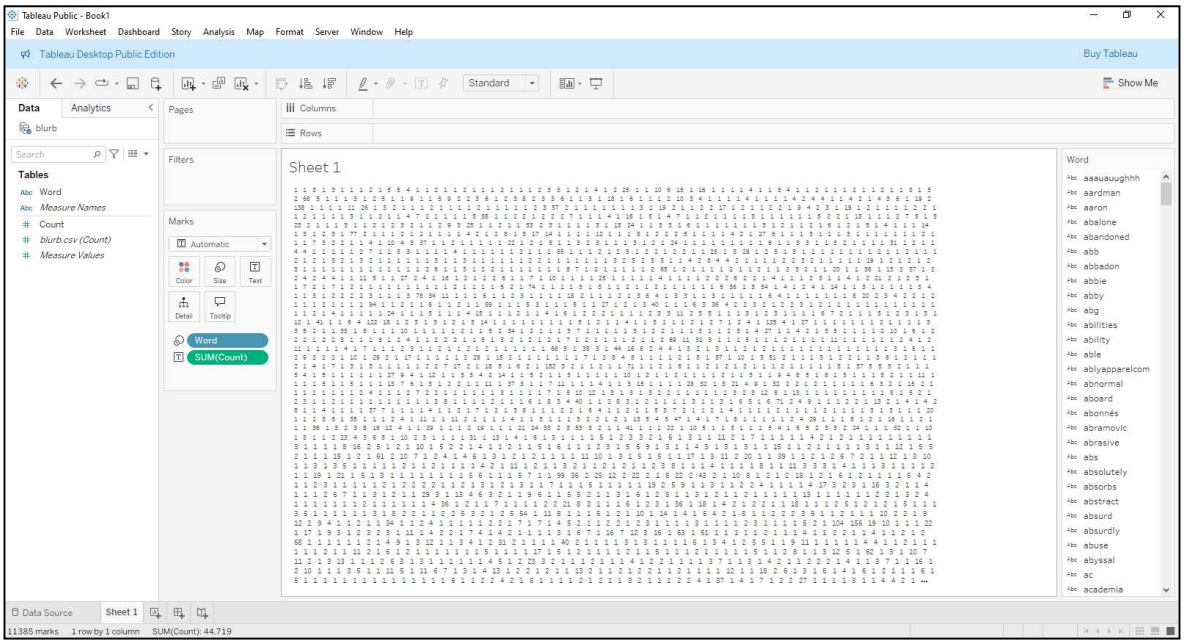




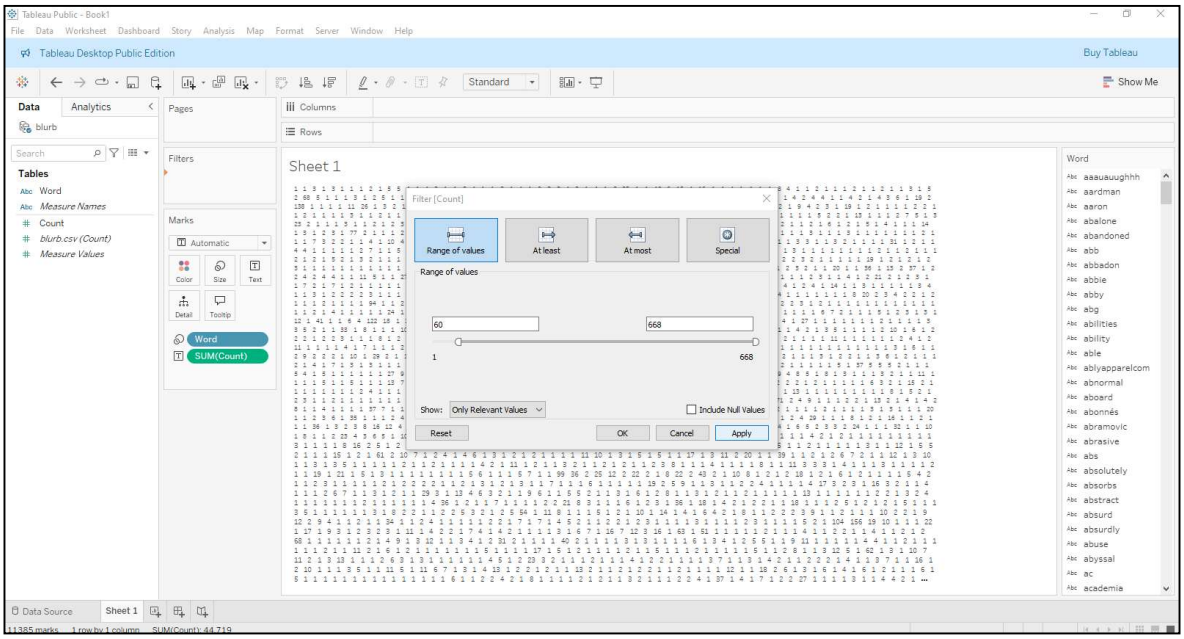
STEPS TO BE FOLLOWED

Step 1. When we drag our data into the view, Tableau will recommend a treemap by default.

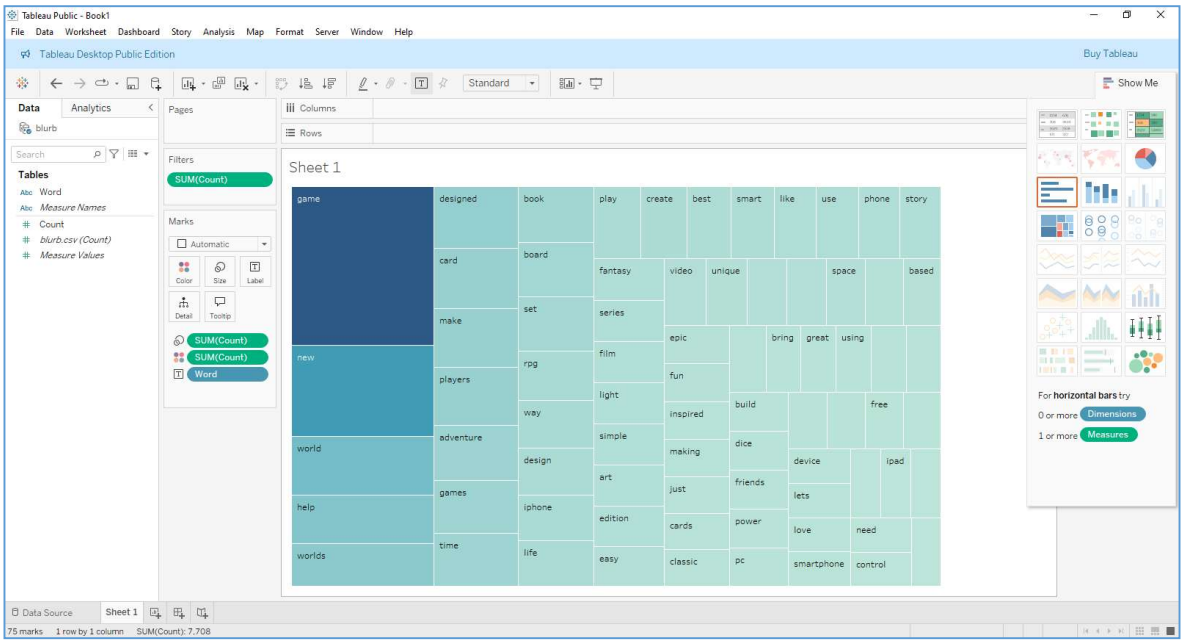
a. Drag Count into Marks – Label / Text.
b. Drag Word into Marks - Size.



Drag Count to the Filters > Select sum and next > Select range to Values 60 to 668 > Apply & Ok

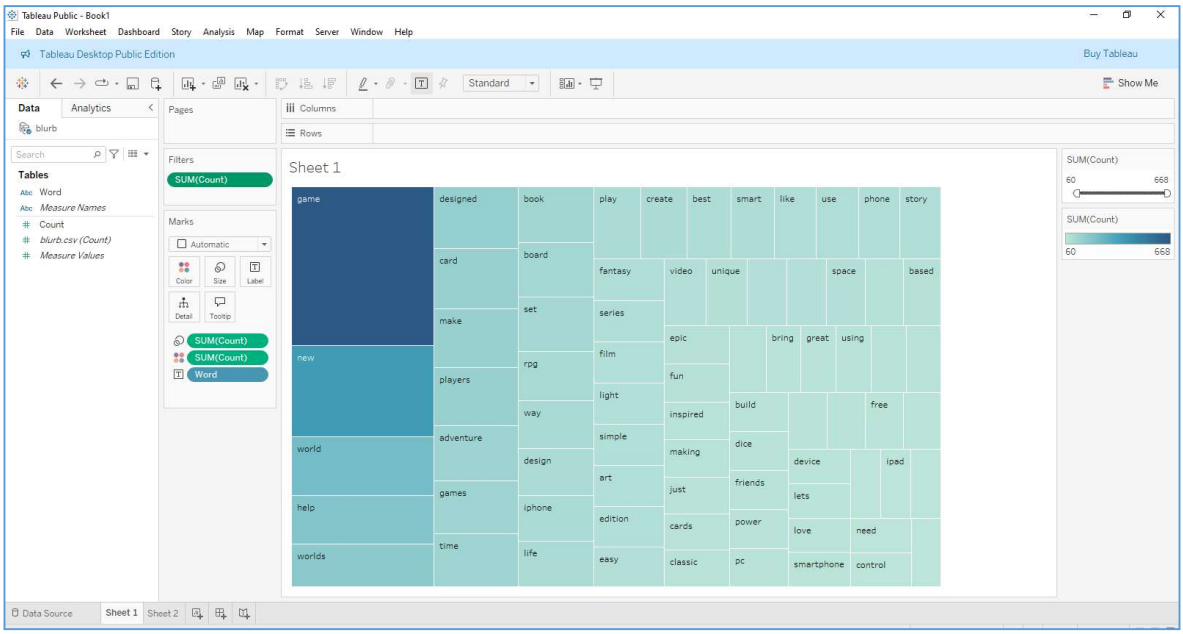


Select Treemap on Show me

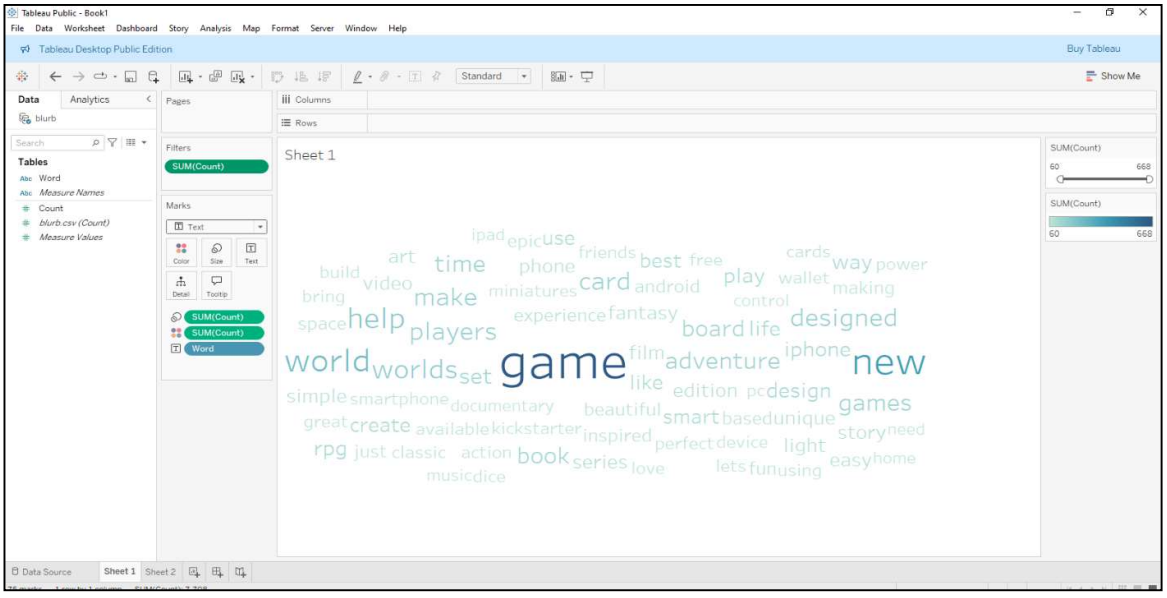


Step 2. Before we change it into a word cloud, we need filter by count. Because there are so many words, if we don't filter, it will take a lot of time to generate a word cloud chart.

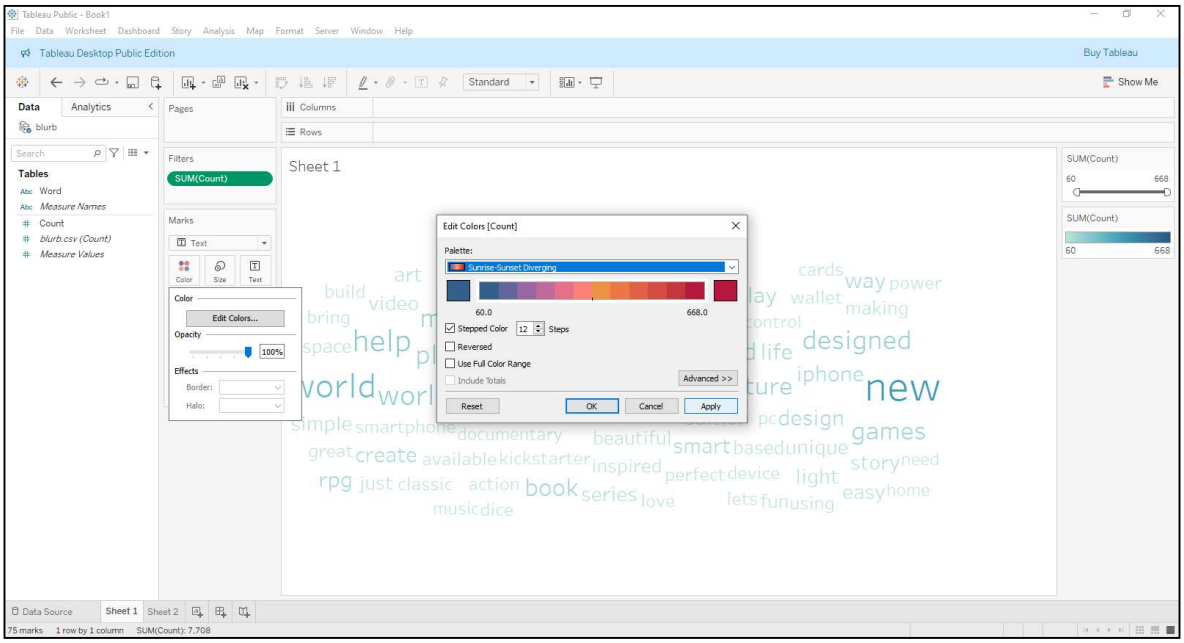
Right-click on the filter and check Show Filter.



Step 3. Now we can convert the marks type to Text. Tableau has built-in support for word clouds, which are automatically arranged into cloud shapes. But we cannot customize the shape.

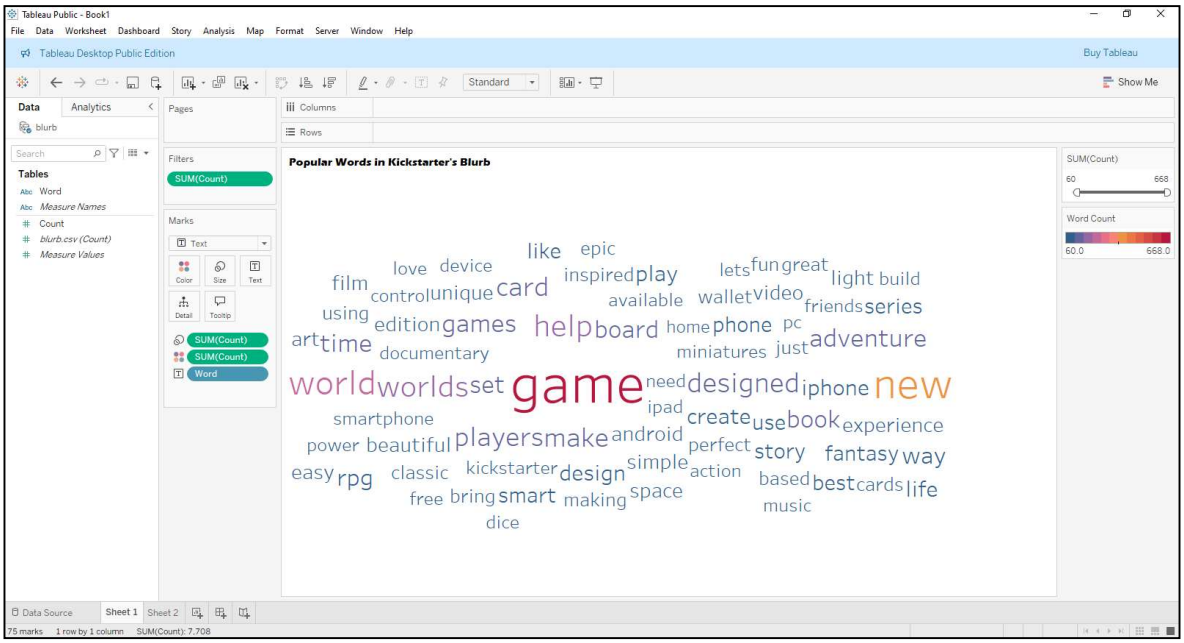


Step 4. Click on Color > Edit Color > Select desired palette > tick on stepped color (12) steps > Apply & Ok

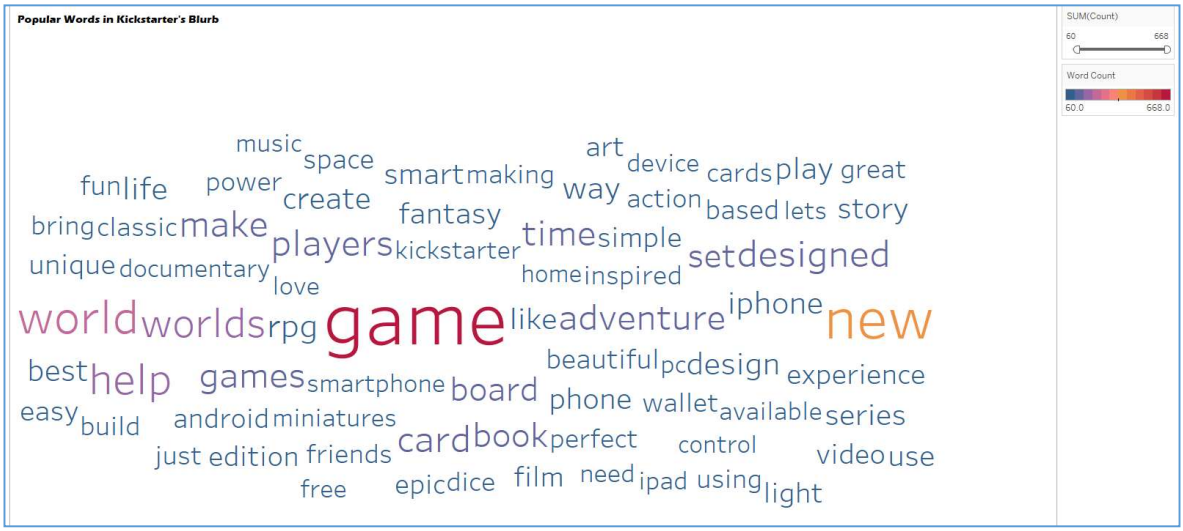


Step 5. In the last step, let's polish this chart: Edit title to "Popular Words in Kickstarter's Blurb" & rename the color legend to "Word Count".

Here is the final chart:



POPULAR WORDS IN KICKSTARTER'S BLURB BASED ON COUNT



PRACTICAL 8

Aim: Create a Map view with appropriate dataset using tableau

WHAT ARE MAPS?

Maps help you visualize and understand geographic data and reveal spatial patterns within your dataset. They are particularly useful when working on big data projects. Tableau offers mapping functionalities that cater to different types of geographic analyses.

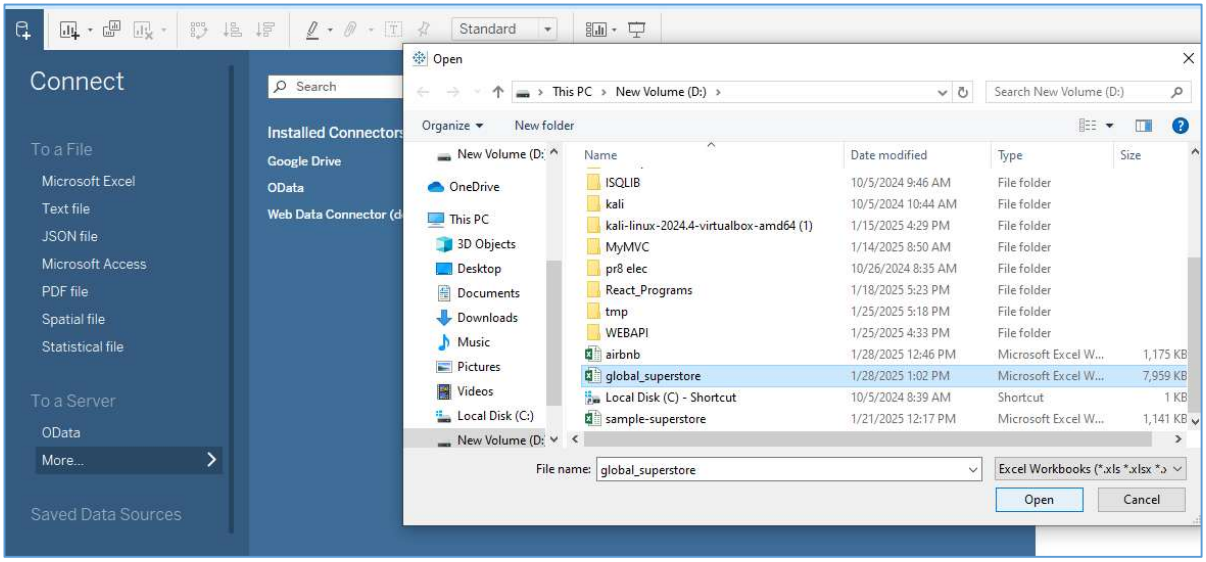
CONNECTING TO EXCEL FILE

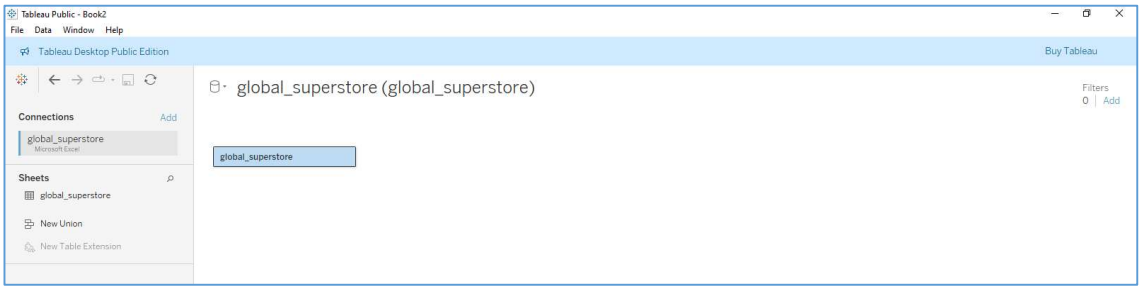
Before connecting to excel file, we need to download the global superstore file from tableau community or from Kaggle which is freely available.

Dataset – Global Superstore

To connect to an excel file, click “Microsoft Excel” on the left-hand side. Navigate to the file on your computer and double click to open it.

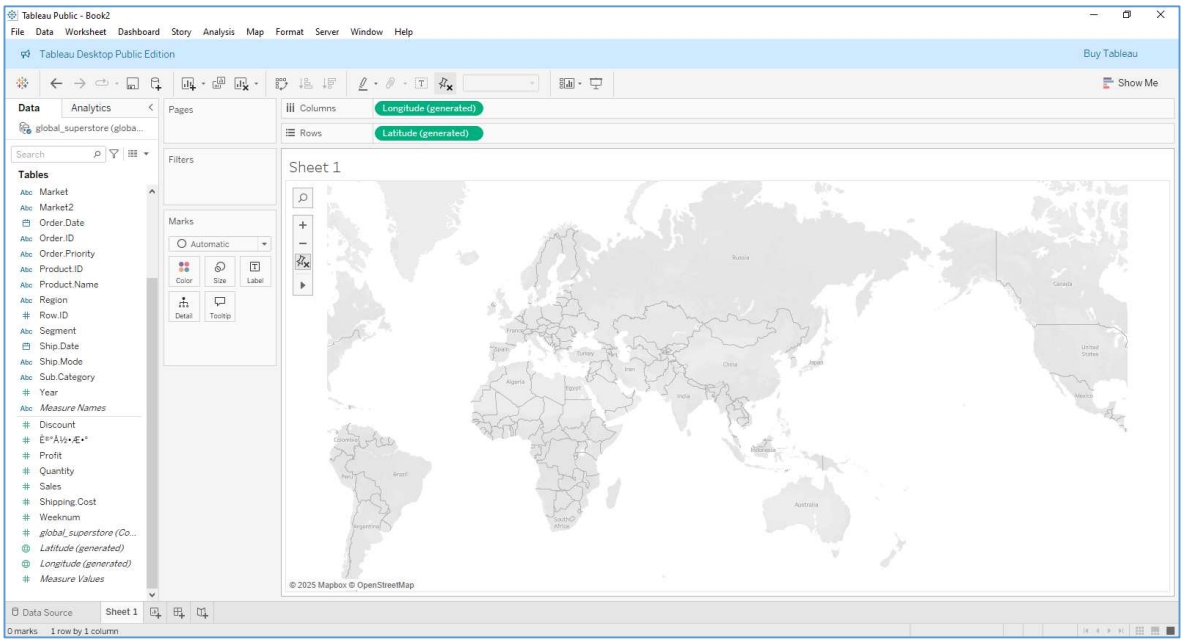
Note: Always remember to change the file type of any downloaded dataset or excel file into excel workbook



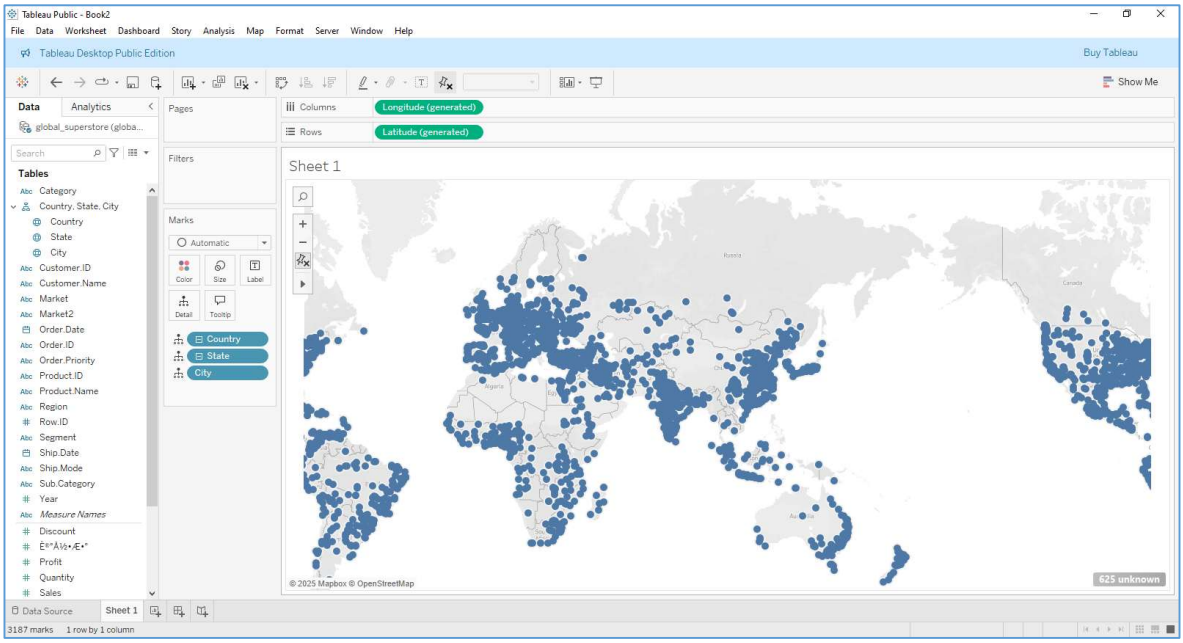


STEPS FOR CREATING MAP BASED DATA VISUALIZATION

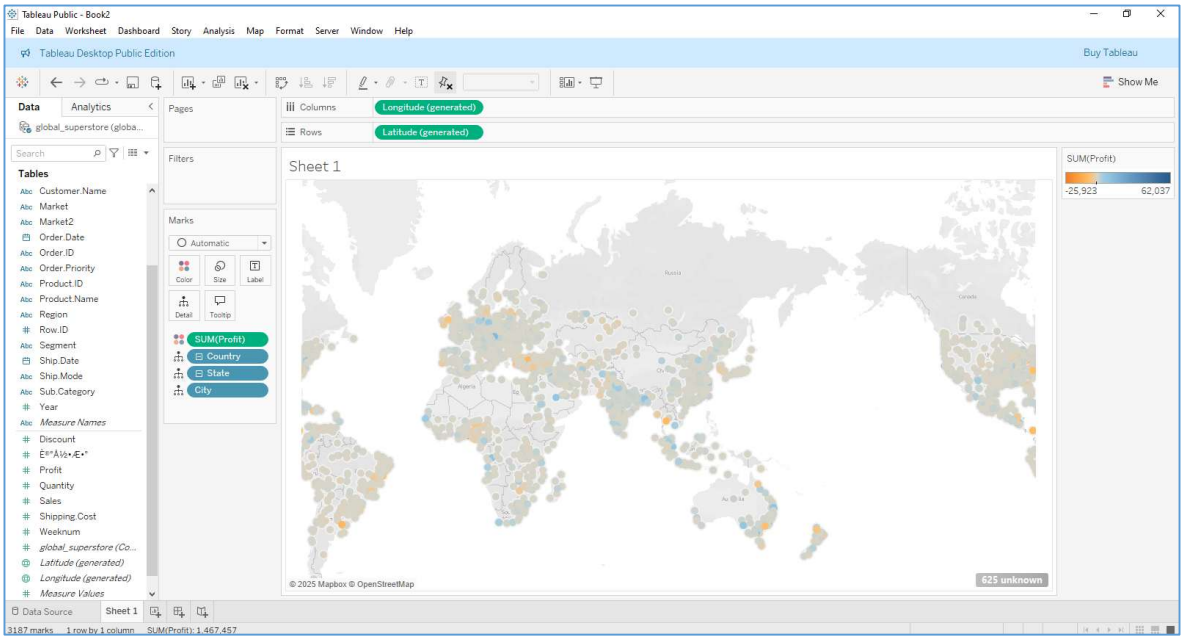
Step 1: Drag Longitude on the Columns shelf and Latitude in the Rows shelf.



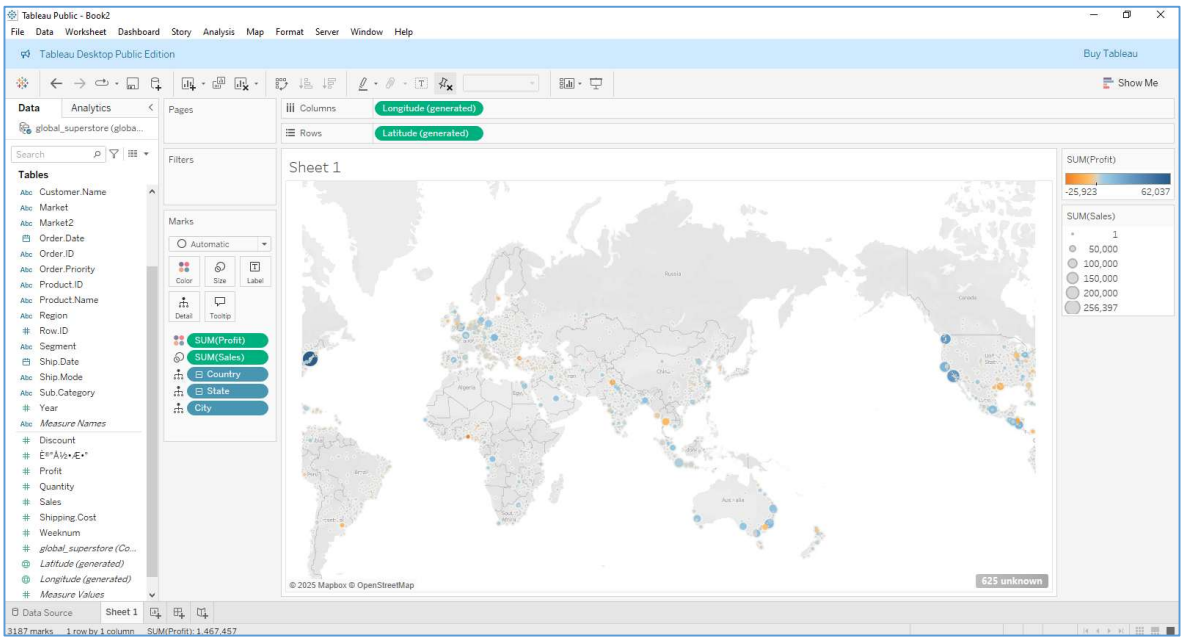
Step 2: Drag Country, State, and City in the Details shelf present inside Marks shelf.



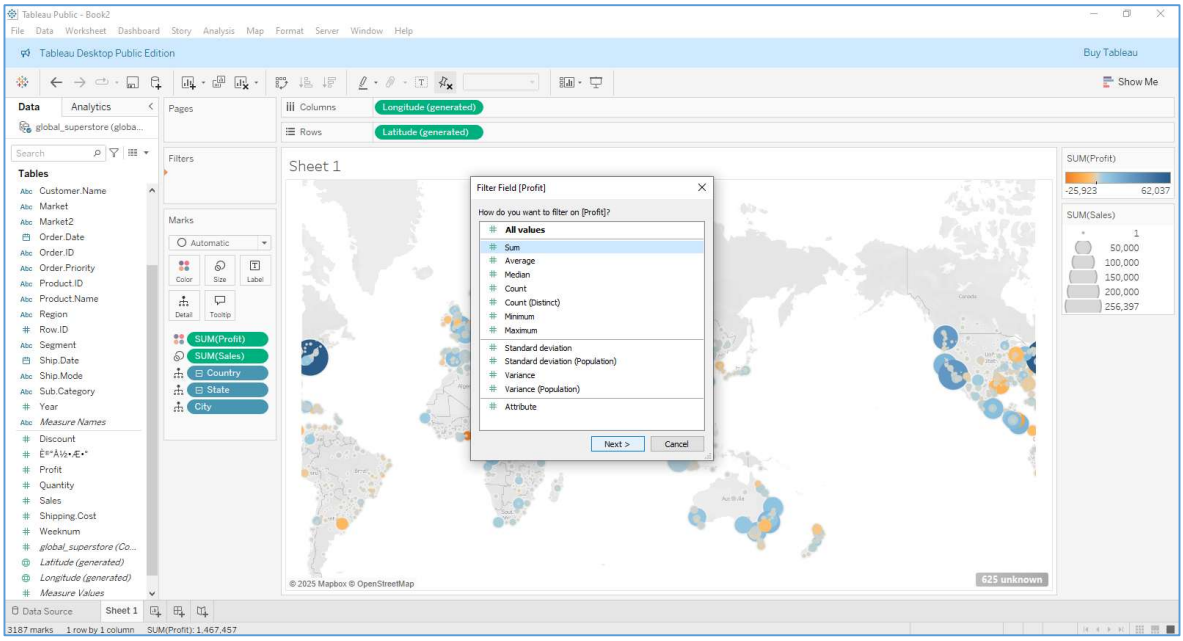
Step 3: Drag the Profit measure into the Color present inside Marks shelf



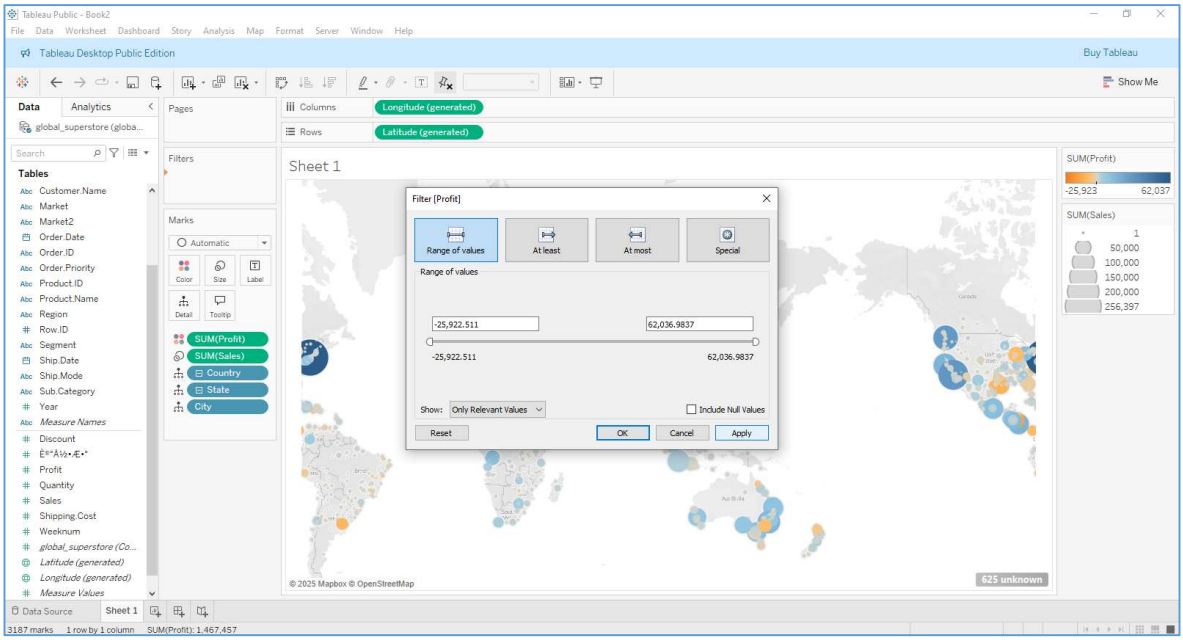
Step 4: Drag the Sales measure into the Size present inside Marks shelf



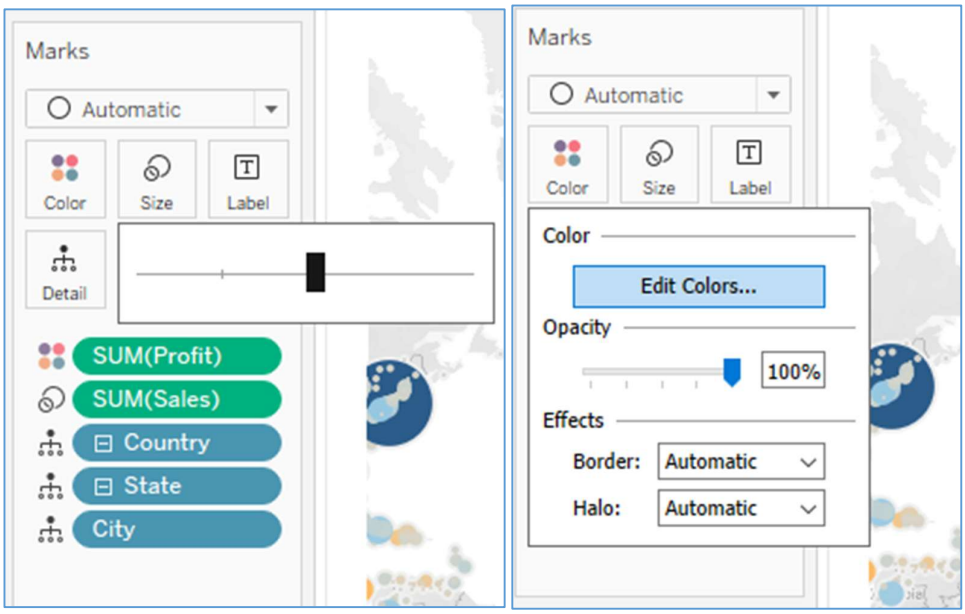
Step 5: Drag the Profit measure into the Filters a pop up will appear select Sum and click next.



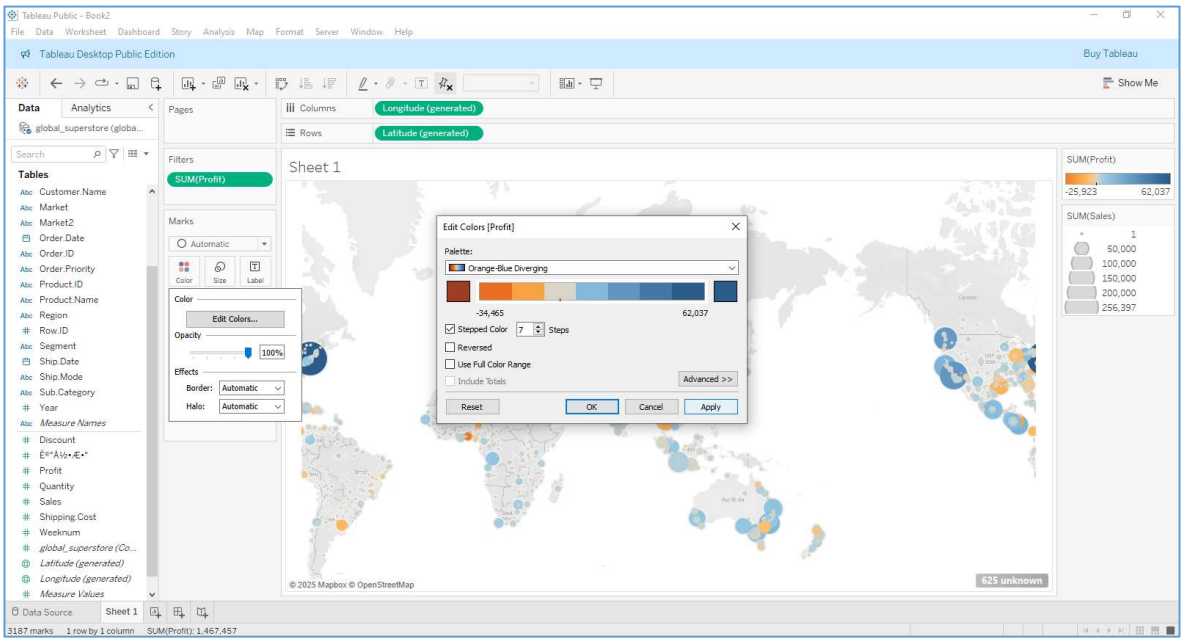
Click on Apply and OK



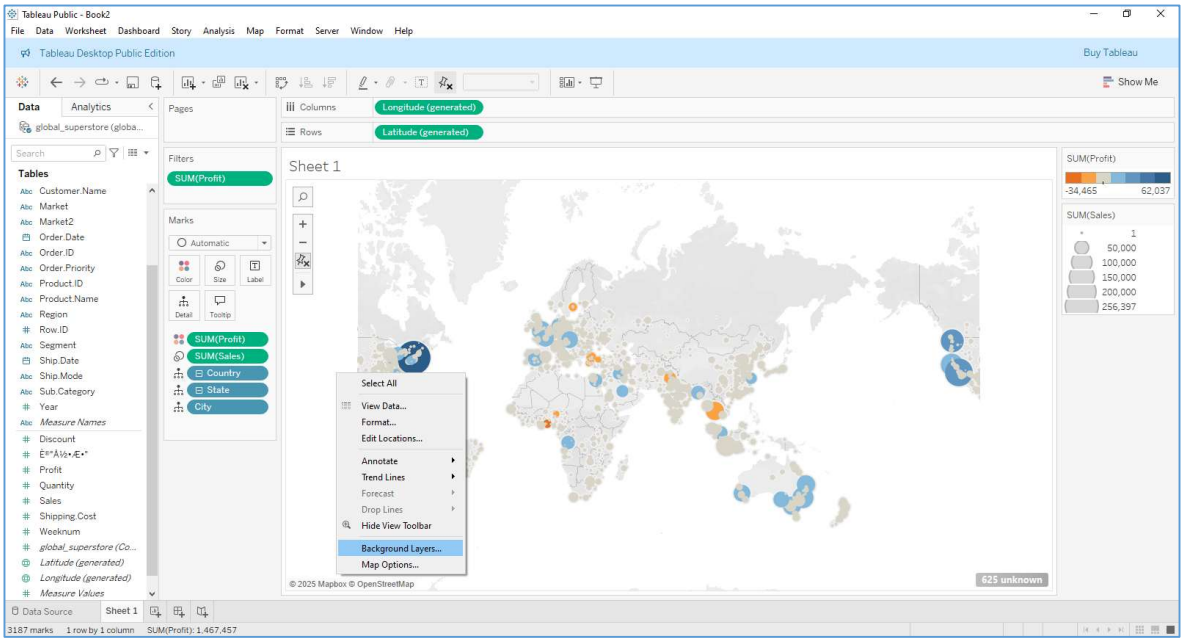
Step 6: You can change the Size and colors from the marks shelf



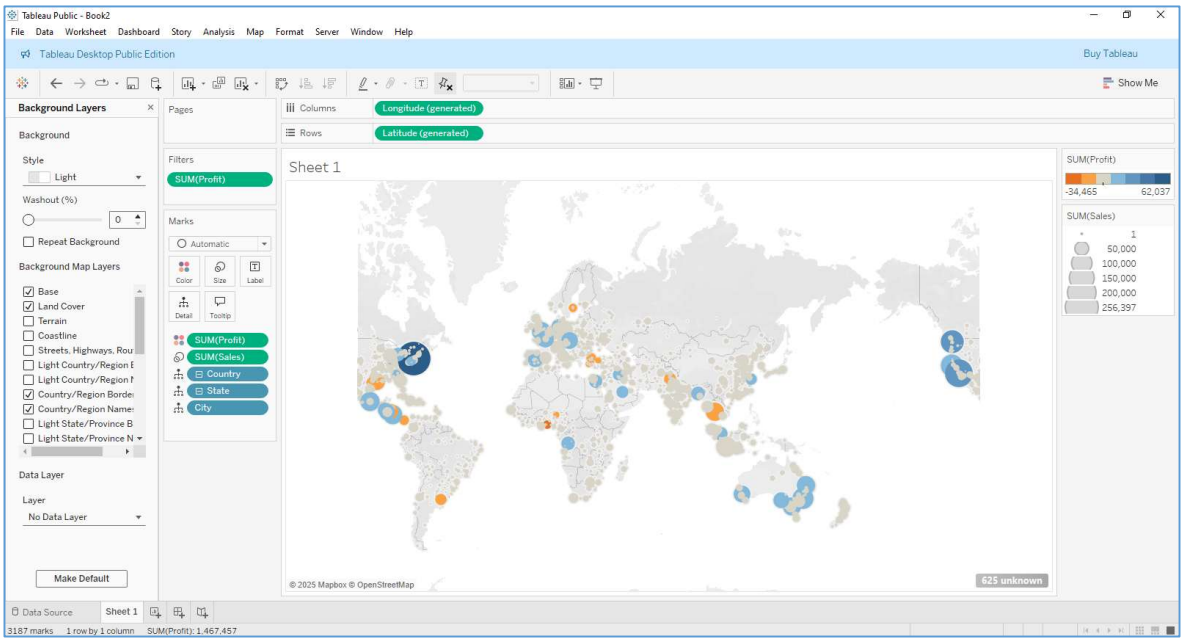
You can click on edit color to change the colors using palette dropdown and change the stepped color to 7 then Apply & OK



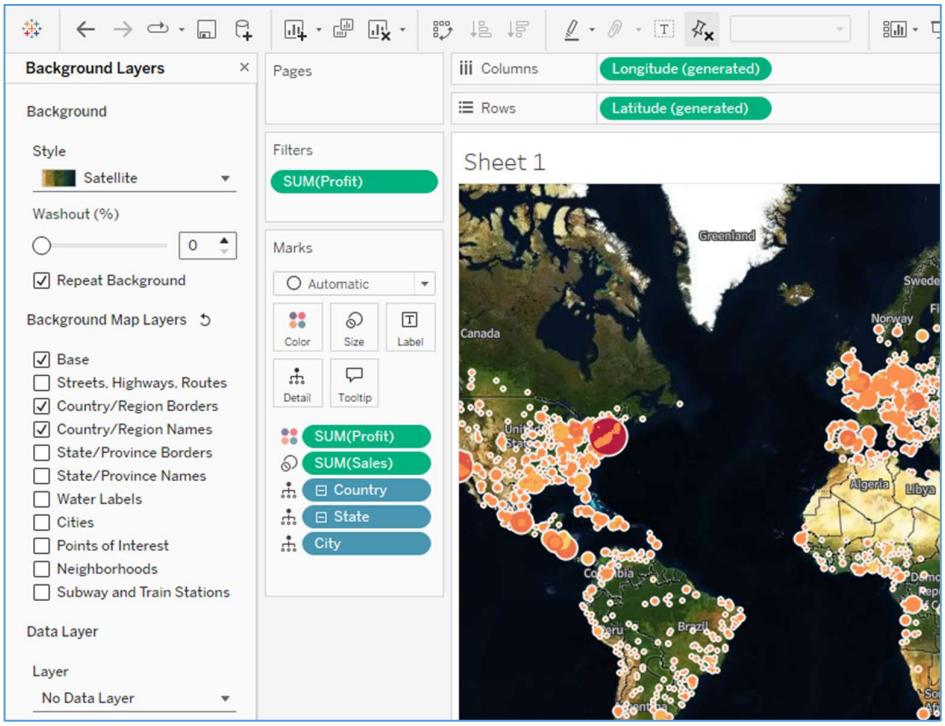
Step 7: Now right click anywhere on the map and select background layers



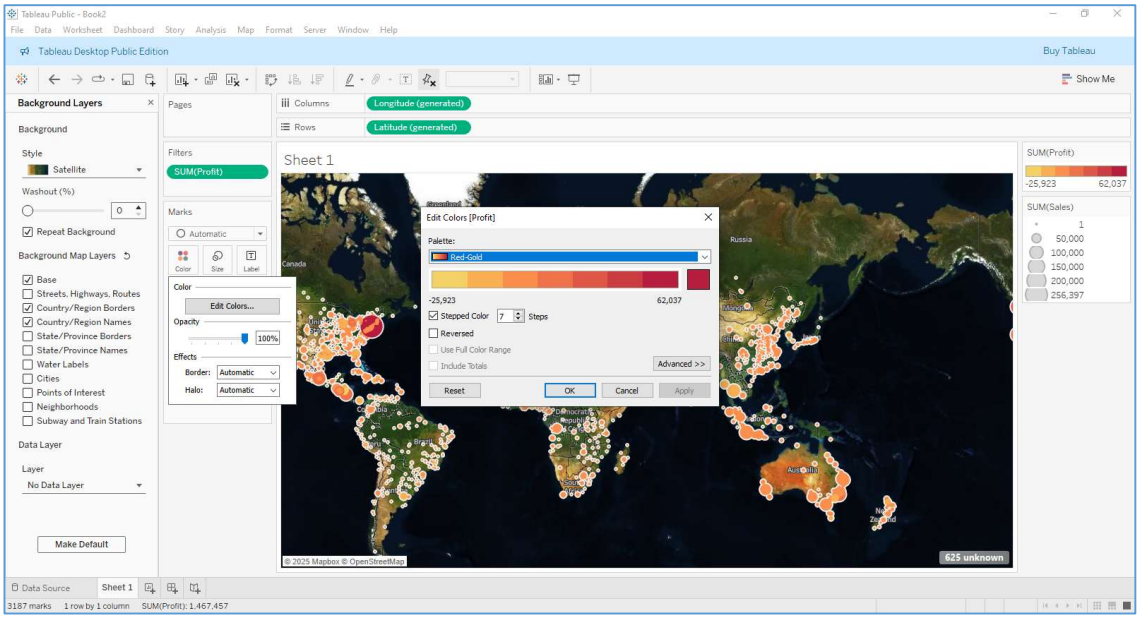
On the left side a new Background Layers pan will appear.



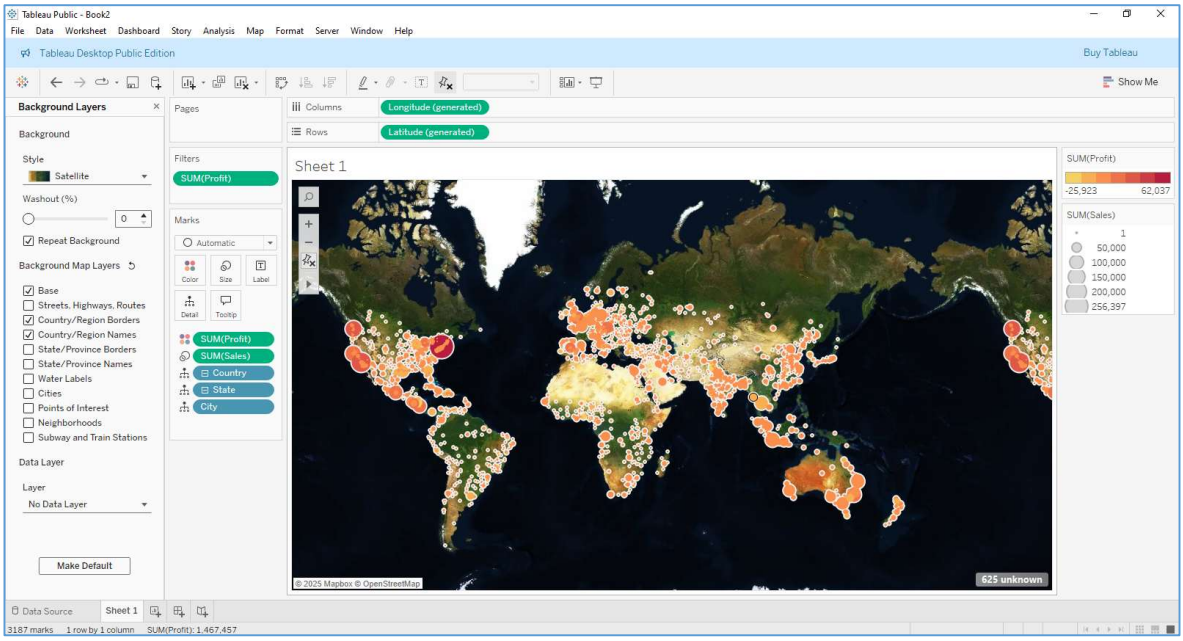
Change the style to Satellite, tick repeat background, tick on Country/Region Borders, Names



Adjust the colors accordingly...



Final map looks something like this



Map View showing global superstore profit based on sales bubble size.



PRACTICAL 9

Aim: Demonstrate visualization of graphs, trees, clusters

WHAT ARE GRAPHS?

Graphs (or charts) are used to represent data visually, making it easier to understand patterns, trends, and relationships. Different types of graphs are used depending on the data and purpose.

Types of Graphs?

1. Bar Chart
2. Histogram
3. Line Graph
4. Pie Chart
5. Scatter Plot
6. Area Chart
7. Box Plot (Box-and-Whisker Plot)
8. Network Graph
9. Heatmap

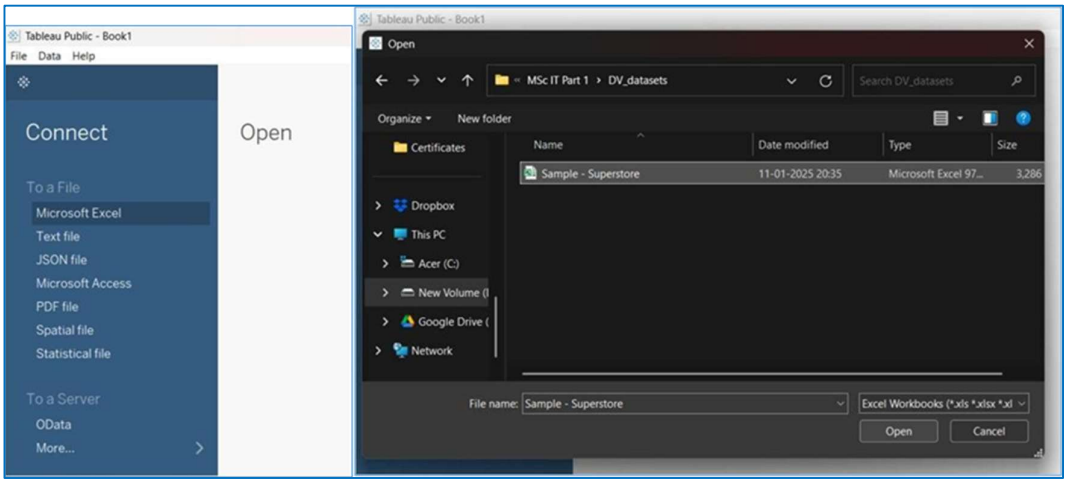
CONNECTING TO EXCEL FILE

Before connecting to excel file, we need to download the sample superstore file from tableau community or from Kaggle which is freely available

Dataset – Sample Superstore

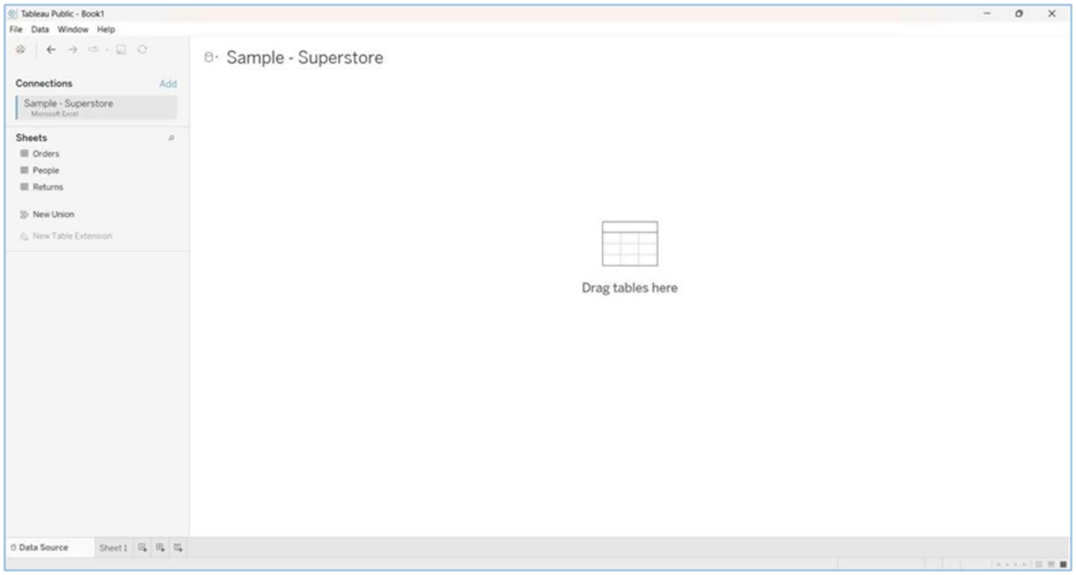
To connect to an excel file, click “Microsoft Excel” on the left-hand side. Navigate to the file on your computer and double click to open it. For this practical, we will use a sample file that comes with the installation called “superstore”. You should open the appropriate file that you will be working with.

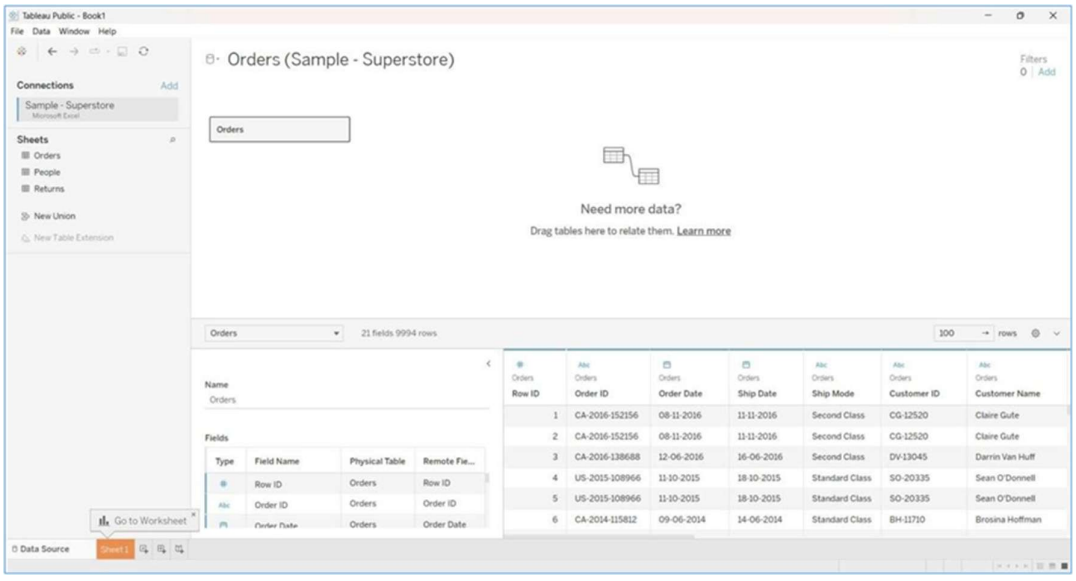
Note: Always remember to change the file type of any downloaded dataset or excel file into excel workbook



Now you are in the data connection window. It looks like the following: -

Notice that I have three sheets in this file- Orders, People, and Returns. I can simply drag the table I want (in these practical we will drag or use the Orders table). If I drag more than one table, Tableau automatically creates the join between the tables.





OPEN WORKSHEET

Creating charts based on the data we connected is easy. At the bottom of the page, click on a sheet (sheet 1) and we will see the following screen:

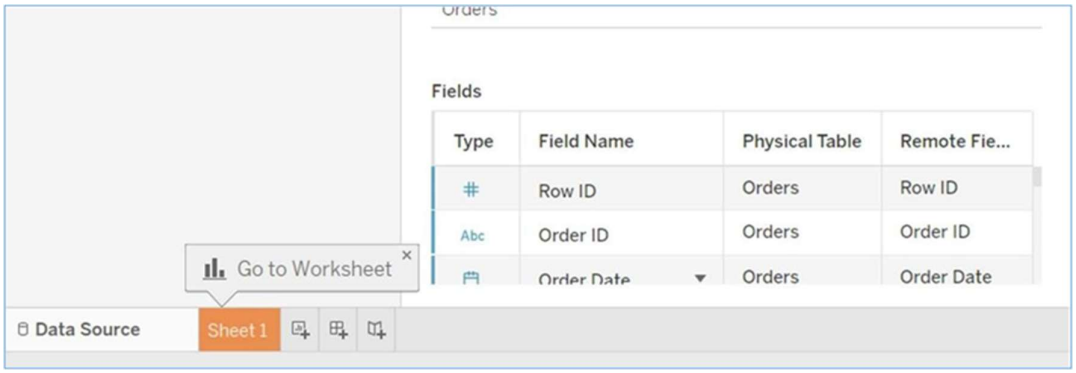
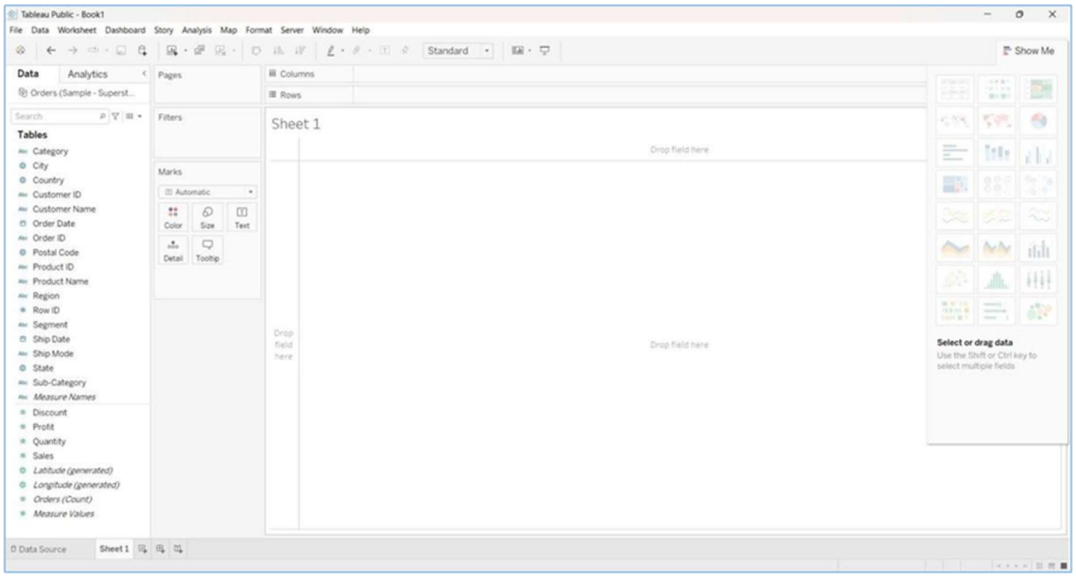
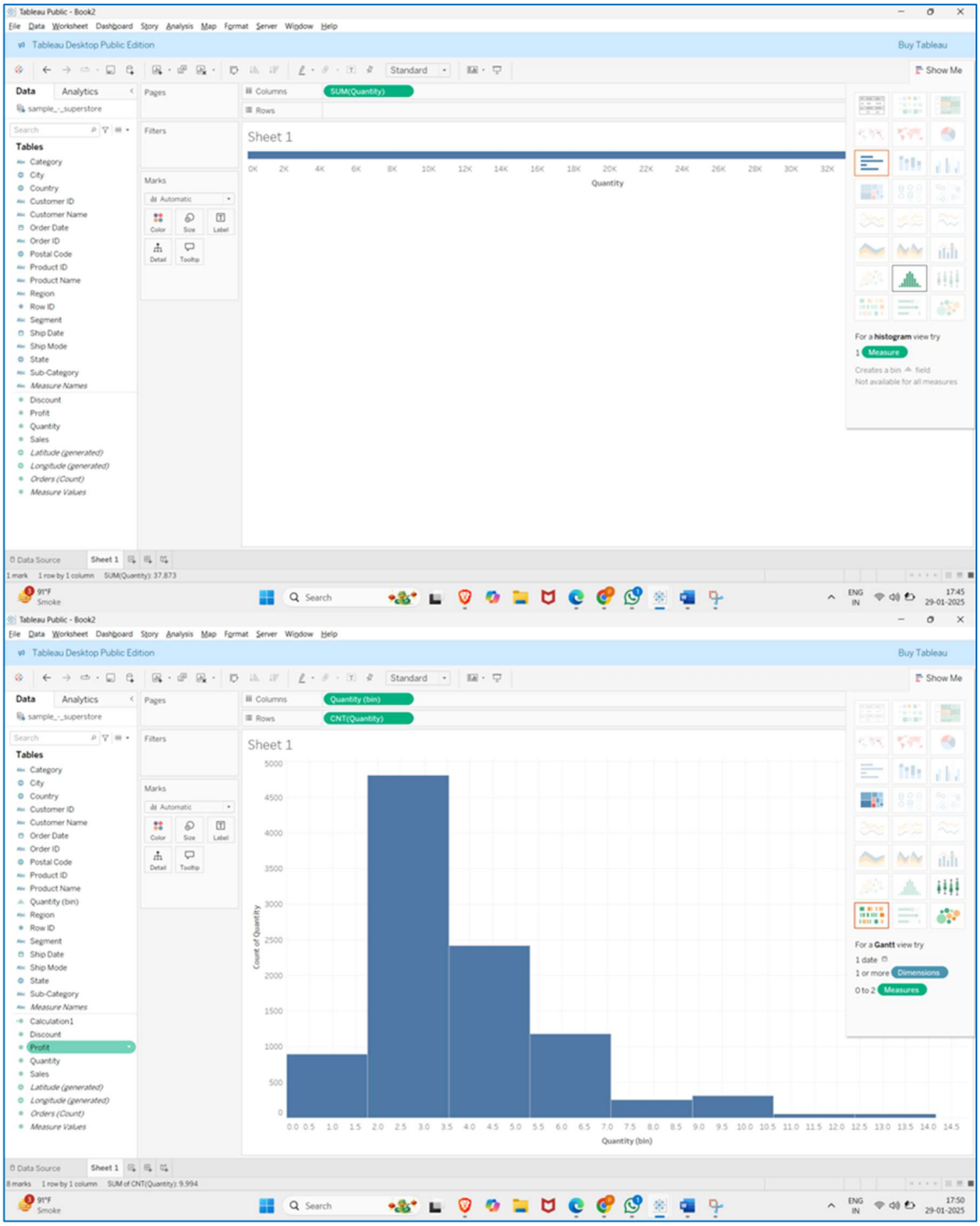


Tableau automatically separates the data into Dimensions and Measures. Dimensions are the categorical fields. These fields will create labels in the chart. Measures are the quantitative fields. These are the numbers we want to analyse. They create axis in the chart



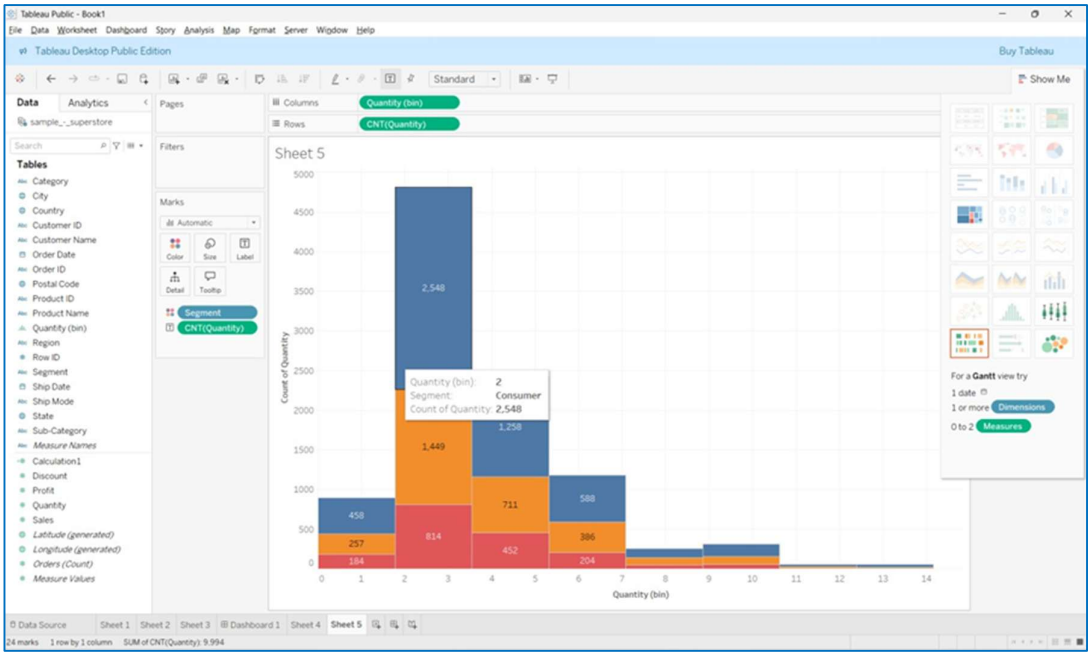
STEPS FOR CREATING HISTOGRAM CHART

Step 1: Drag Quantity to Columns. Click Show Me on the toolbar, then select the histogram chart type. The histogram chart type is available in Show Me when the view contains a single measure and no dimensions.

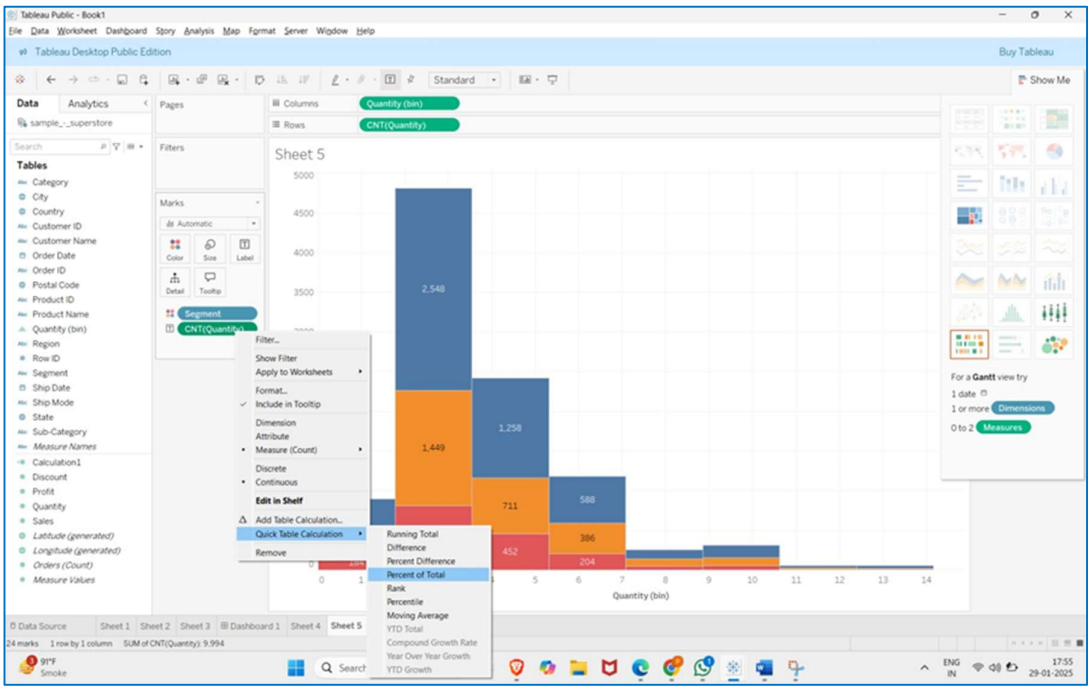


Step 2: Drag Segment to Color. Hold down the Ctrl key and drag the CNT(Quantity) field from the Rows shelf to Label.

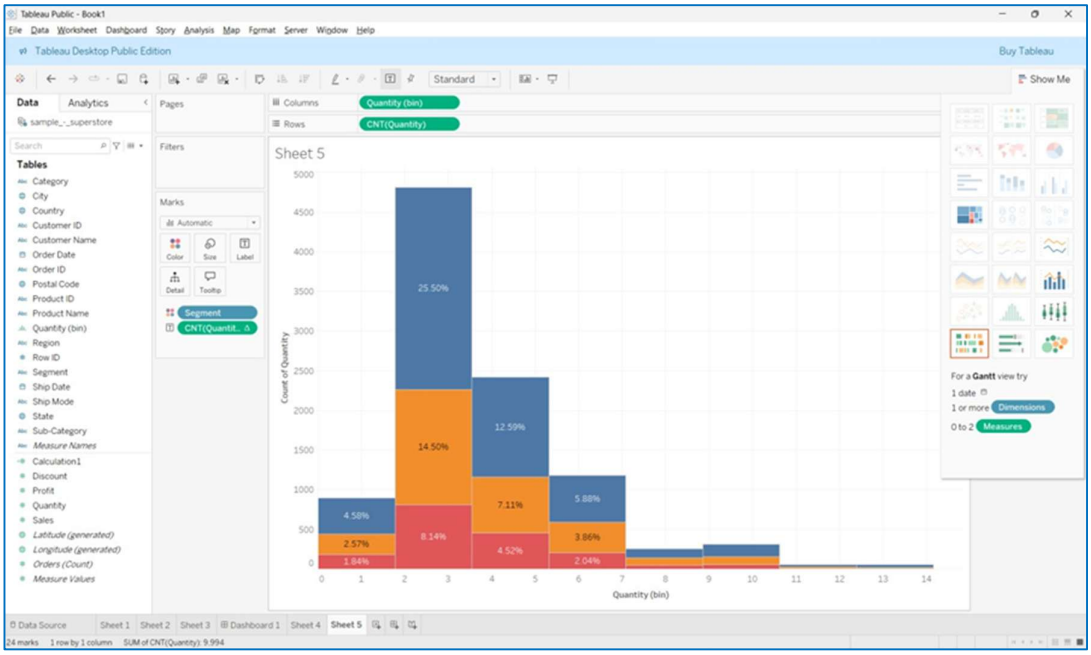
Holding down the Ctrl key copies the field to the new location without removing it from the original location.



Step 3: Right-click the CNT(Quantity) field on the Marks card and select Quick Table Calculation > Percent of Total.



Histogram

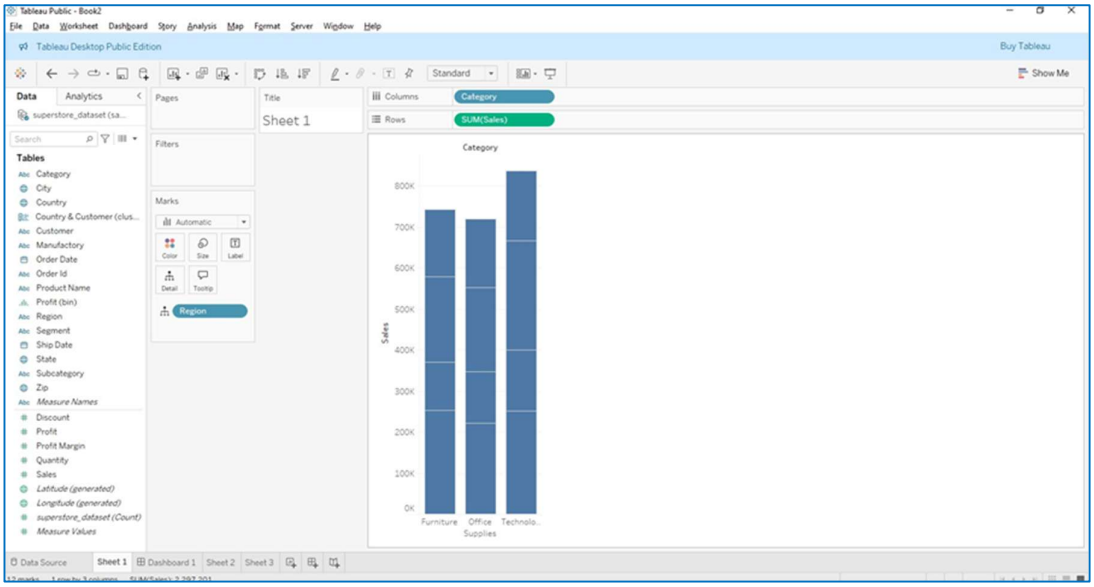


create a box-and-whisker plot. The goal is to visualize the continuous measure, Sales, against the dimension Category.

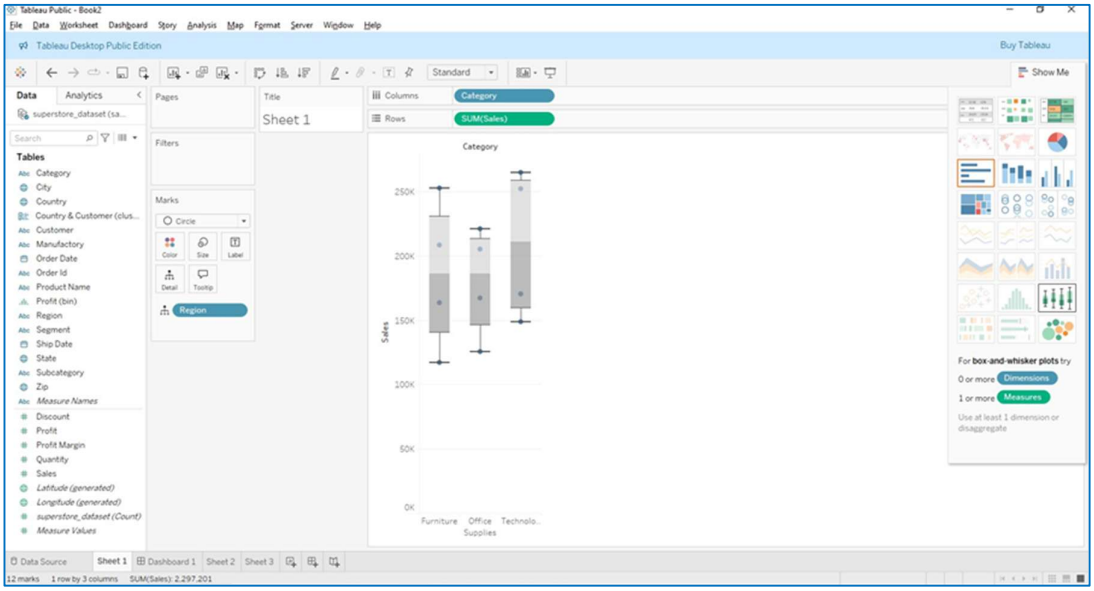
STEPS FOR CREATING BOX-AND-WHISKER PLOT

Step 1: create a bar chart with the dimension (Category) and measure of interest (Sales).

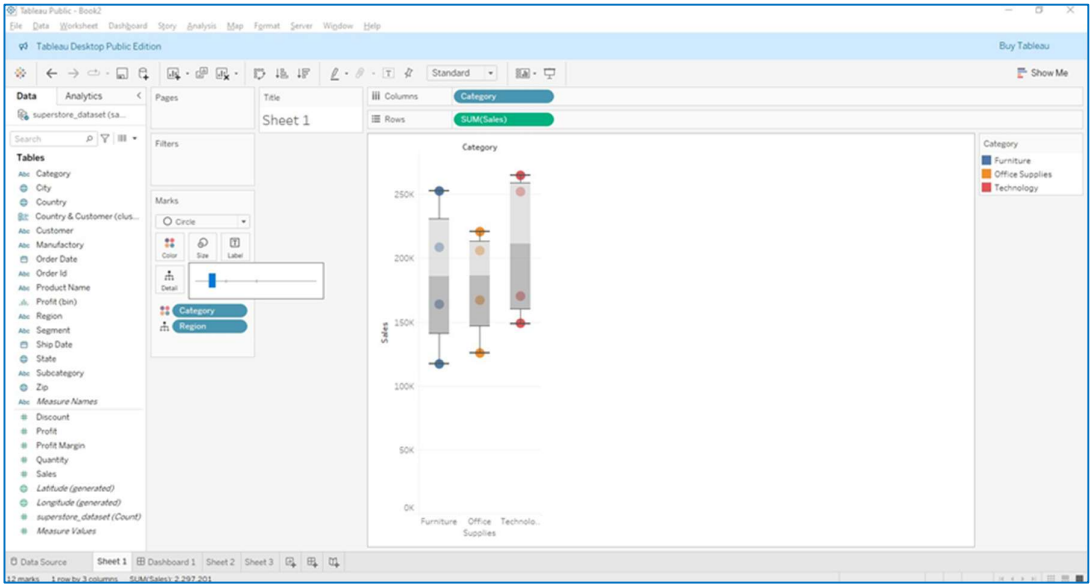
To do that, we'll drag the category dimension in the Columns shelf, and sales in the Rows section. Also drag the dimension Region to the Detail Marks Card.



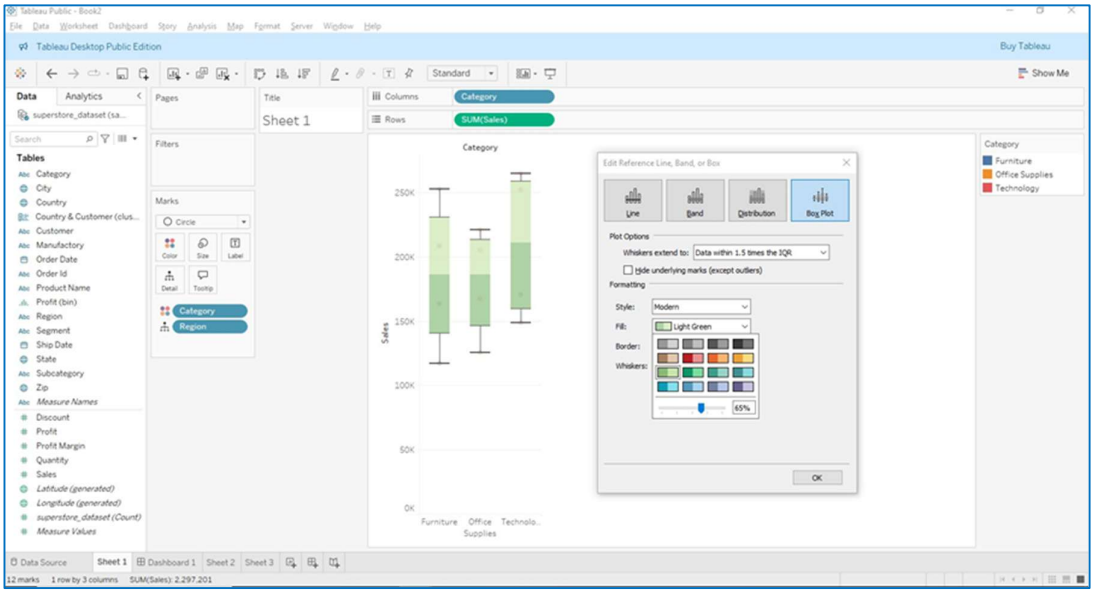
Step 2: The next step is to go to the Show Me option on the right corner of Tableau and select the box-and-whisker plot from the options.



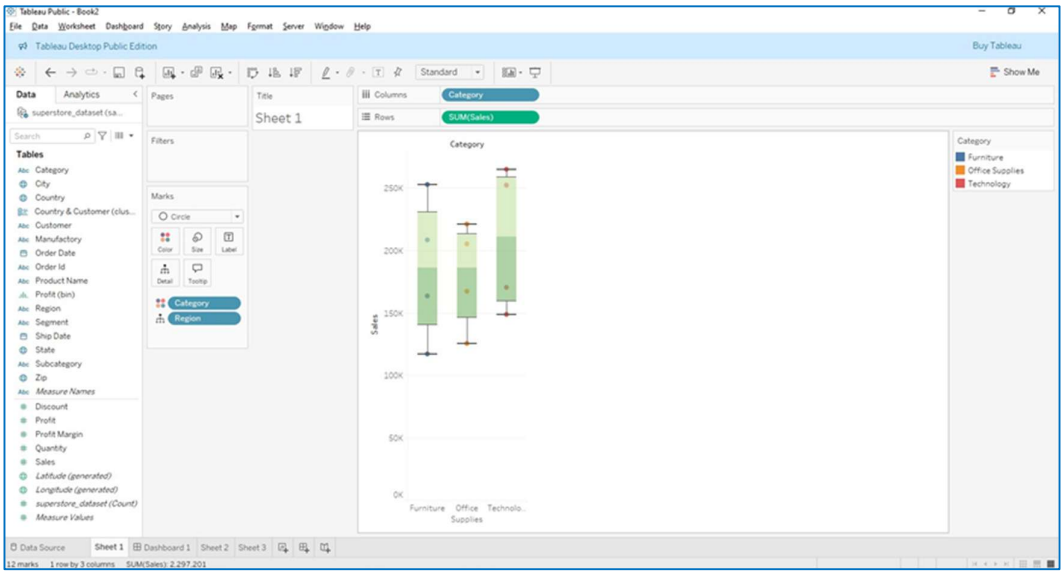
Step 3: If the data points are not properly visible, we can increase the circle size and reduce the opacity of the Color fill in the Marks tab.



Step 4: The aesthetics of the box and whisker plot can be altered by using the various formatting options. One way to do that is to right-click on the chart, which takes us to the option to edit. Clicking on that will open a new window.



Boxplot

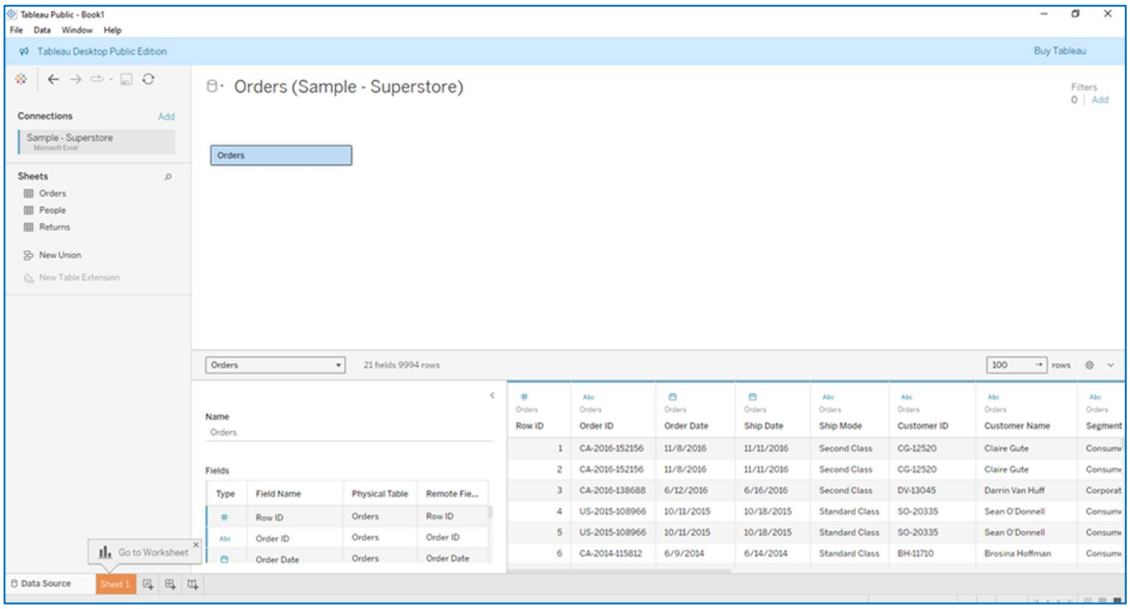


WHAT IS TREE?

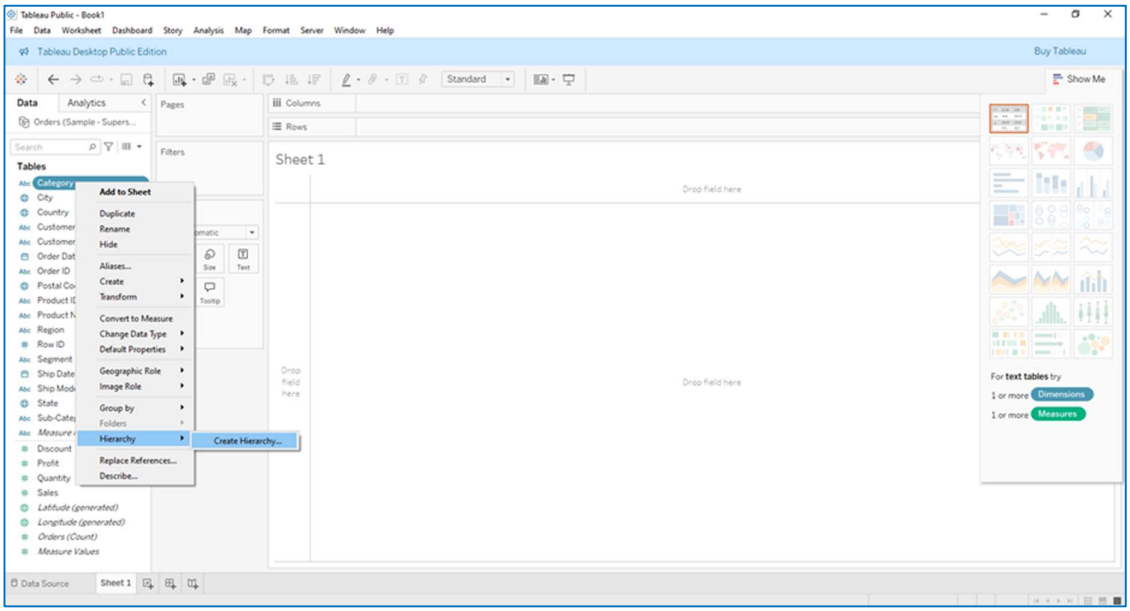
A tree in data visualization is a graphical representation of hierarchical data or relationships, where elements (called nodes) are connected by lines (called edges) to depict parent-child or ancestor-descendant relationships. Trees are used to organize and display data in a structured, branching format, making it easier to understand complex relationships or nested information.

STEPS FOR CREATING TREES

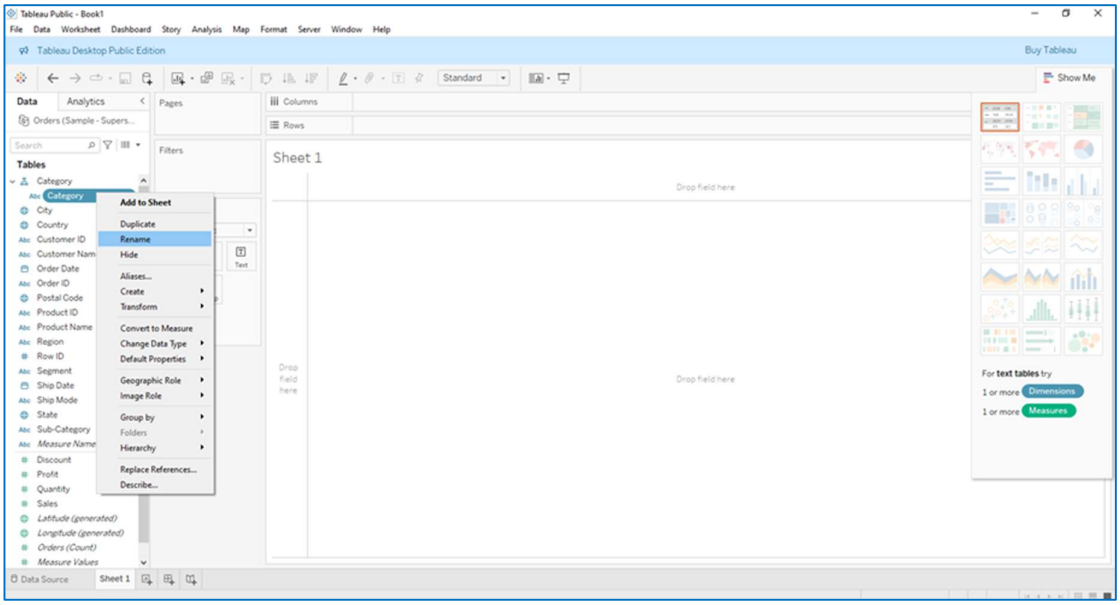
Load the sample_Superstore.xls. Click on the new sheet below.



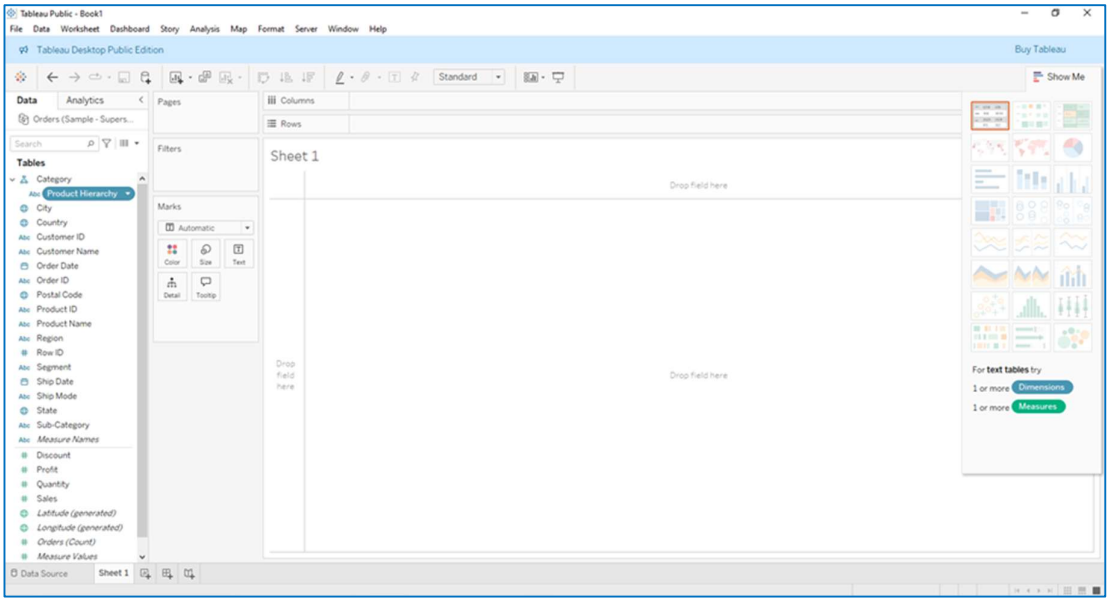
Now, in the new sheet Right-click on category to create its hierarchy as shown below.



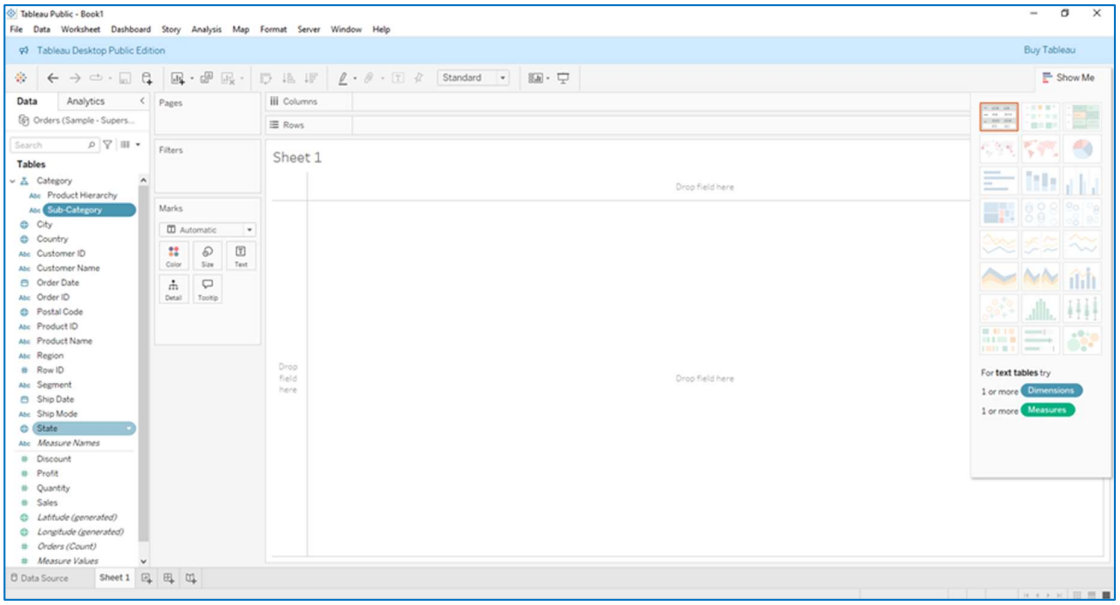
Rename the hierarchy as ‘product_hierarchy’.



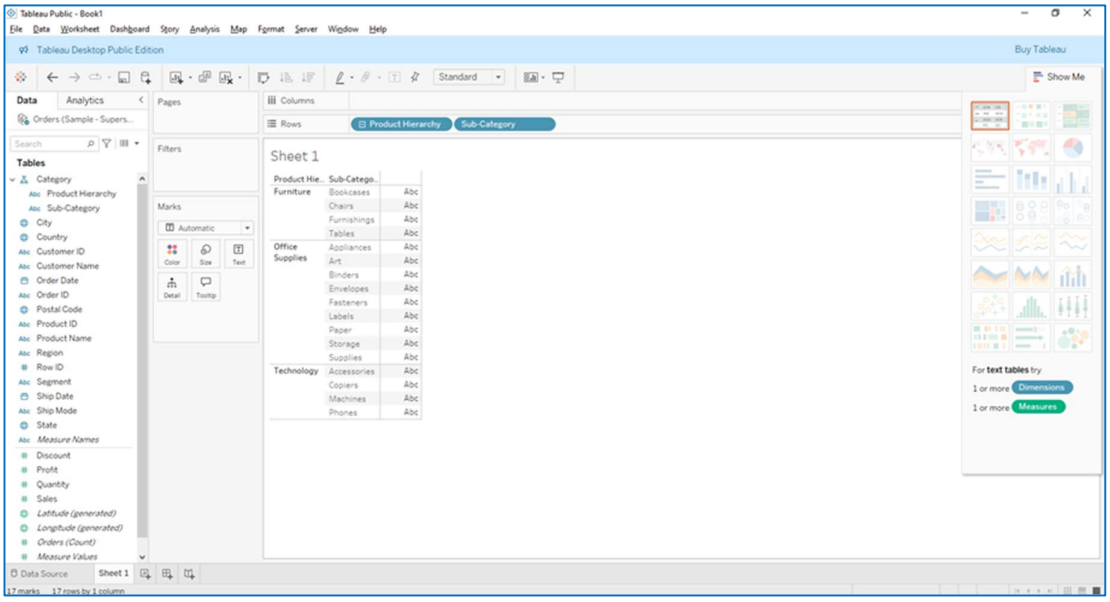
Product _Hierarchy is created successfully



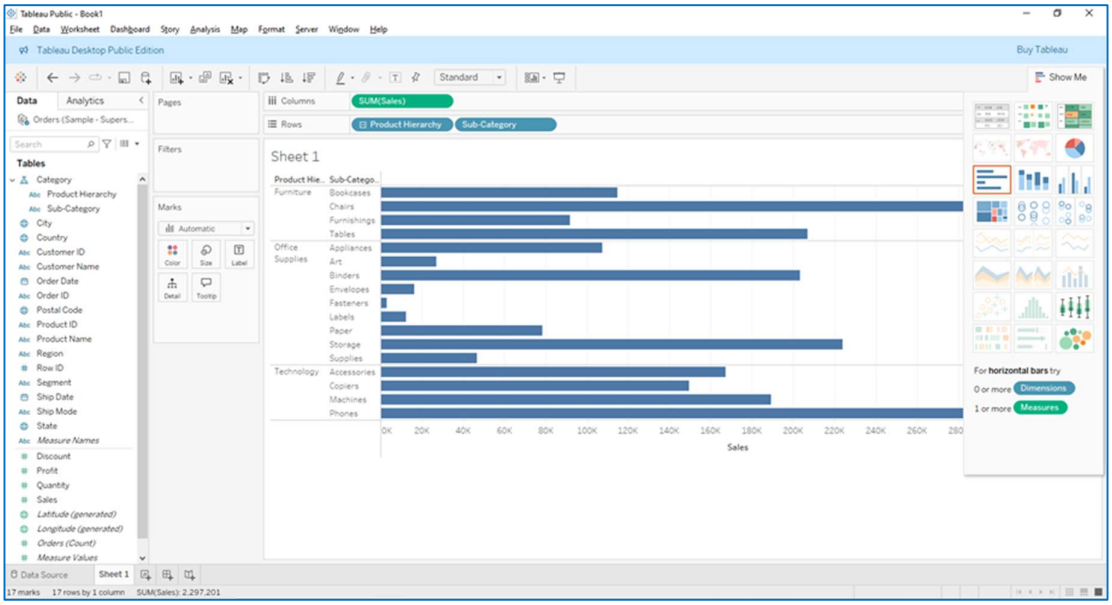
Drag the sub-category into product hierarchy which we created new.



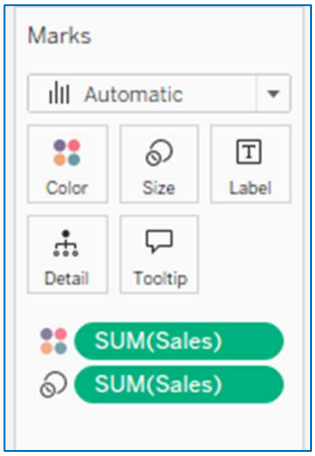
Add the product_hierarchy into the rows after that click of the + sign of product_hierarchy for sub-category

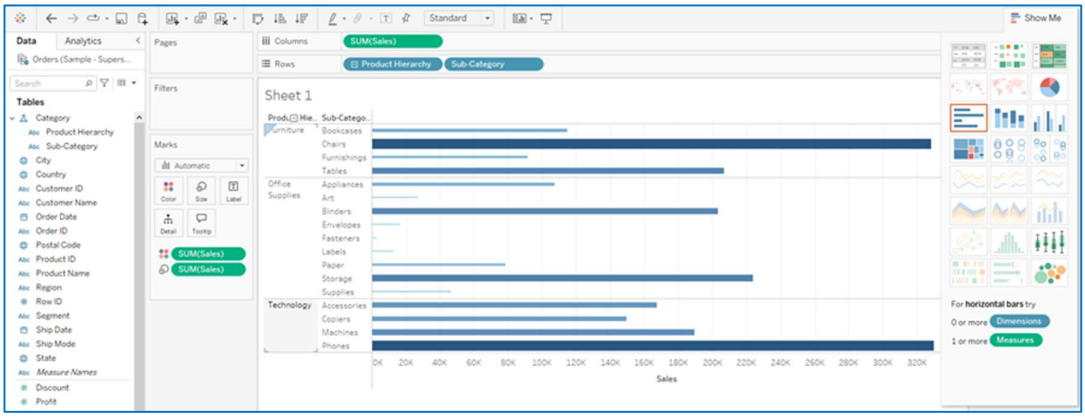


Add the Sales into the columns



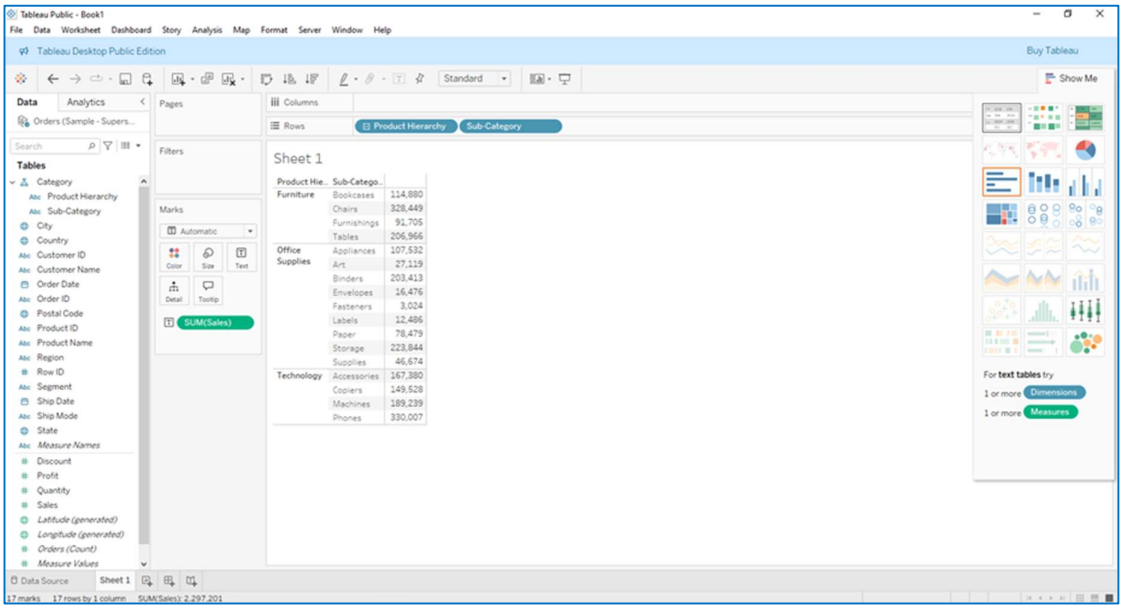
Also add Sales in color and size as shown below



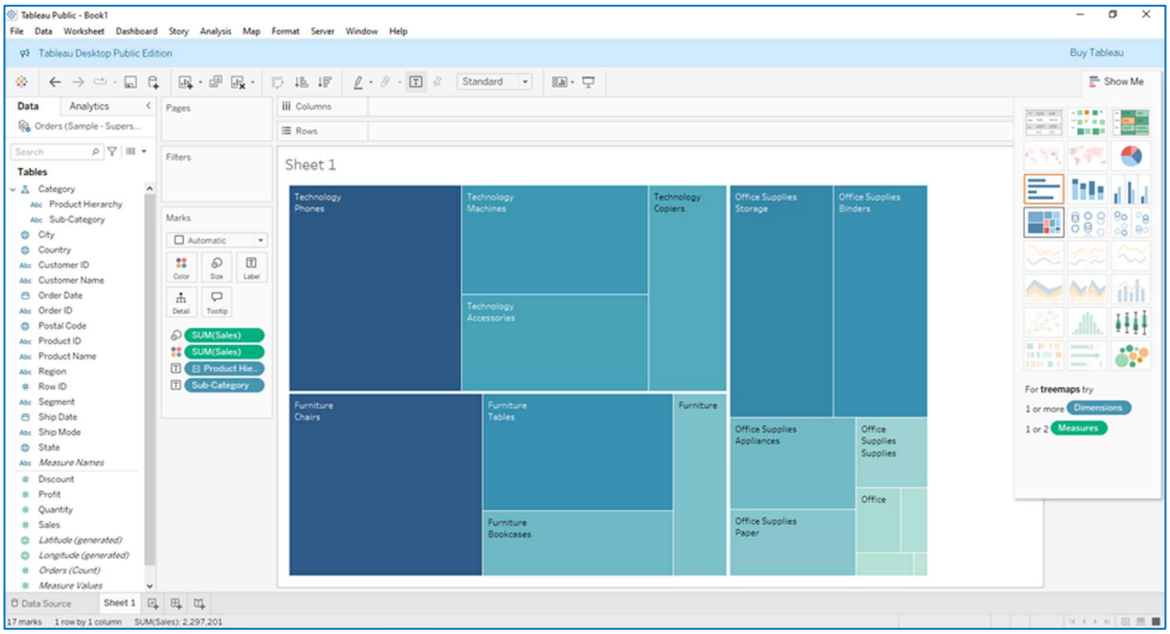


Click on Show Me tab select the first visual i.e. text tables

Here, we can see hierarchical structure.



To create tree maps, select the tree maps in Show-Me tab

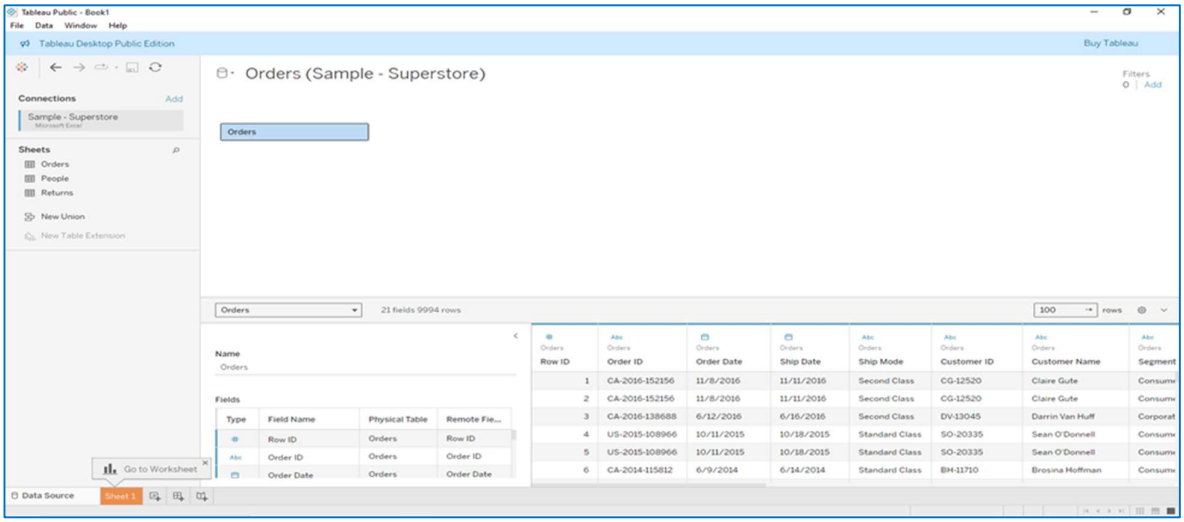


WHAT ARE CLUSTERS?

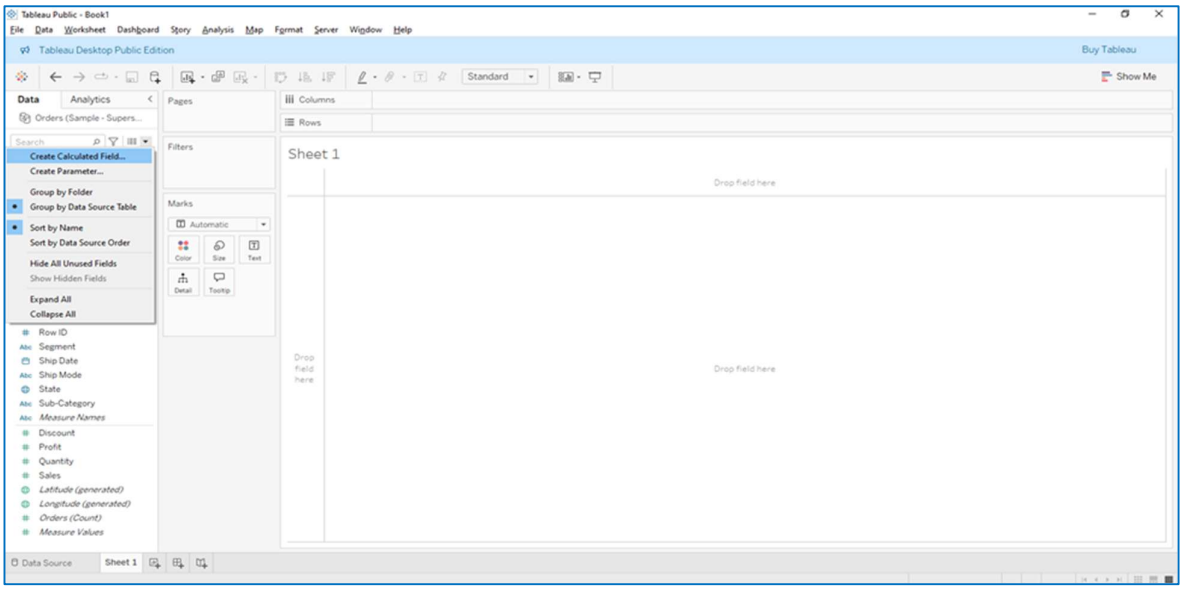
A cluster in data visualization refers to a group of data points that are similar to each other based on specific characteristics or metrics. Clusters are identified using clustering algorithms, and they help reveal patterns, structures, or groupings in data by visually separating distinct subsets. Clustering is commonly used to simplify complex datasets, enabling easier interpretation and analysis.

STEPS FOR CREATING CLUSTERS (K-MEAN CLUSTERING)

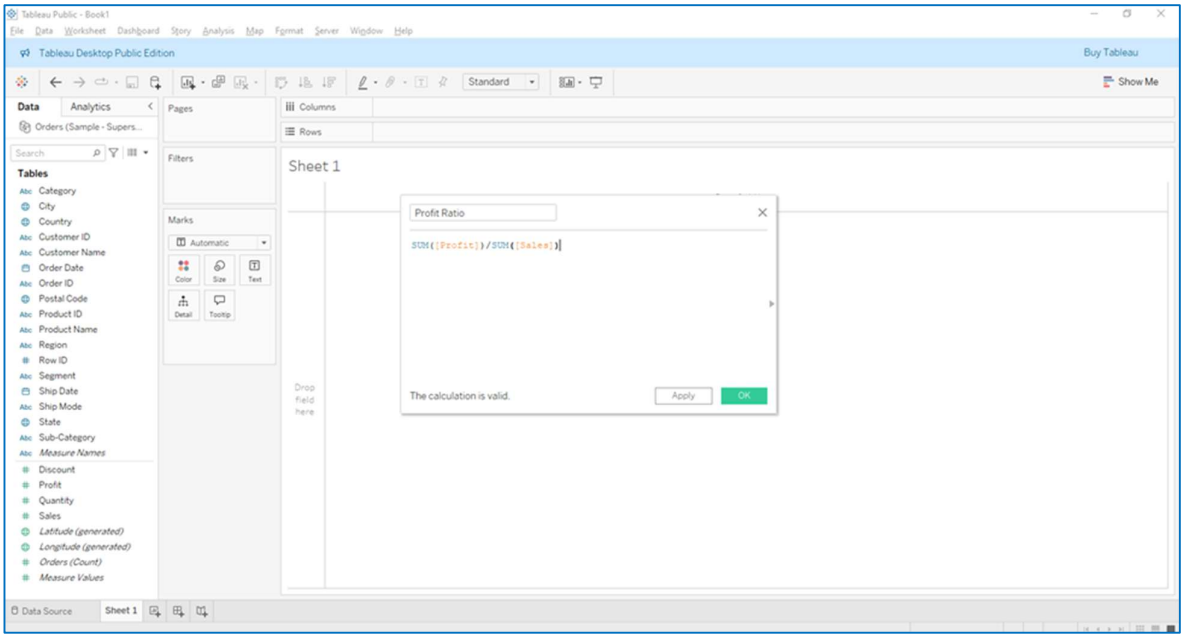
Load the sample_Superstore.xls. Click on the new sheet below.



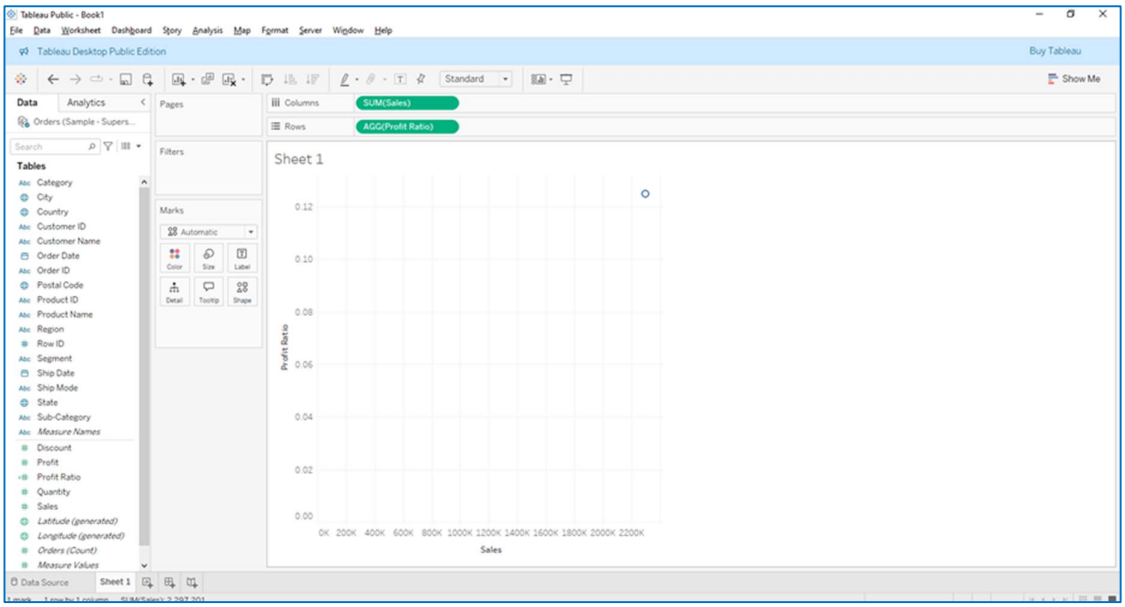
The first step to create clusters is to make a scatter plot. Use the variable Sales and the calculated field Profit Ratio. To create a calculated field, right click on the down arrow in the Data pane and select Create Calculated Field from the options. This is shown below.



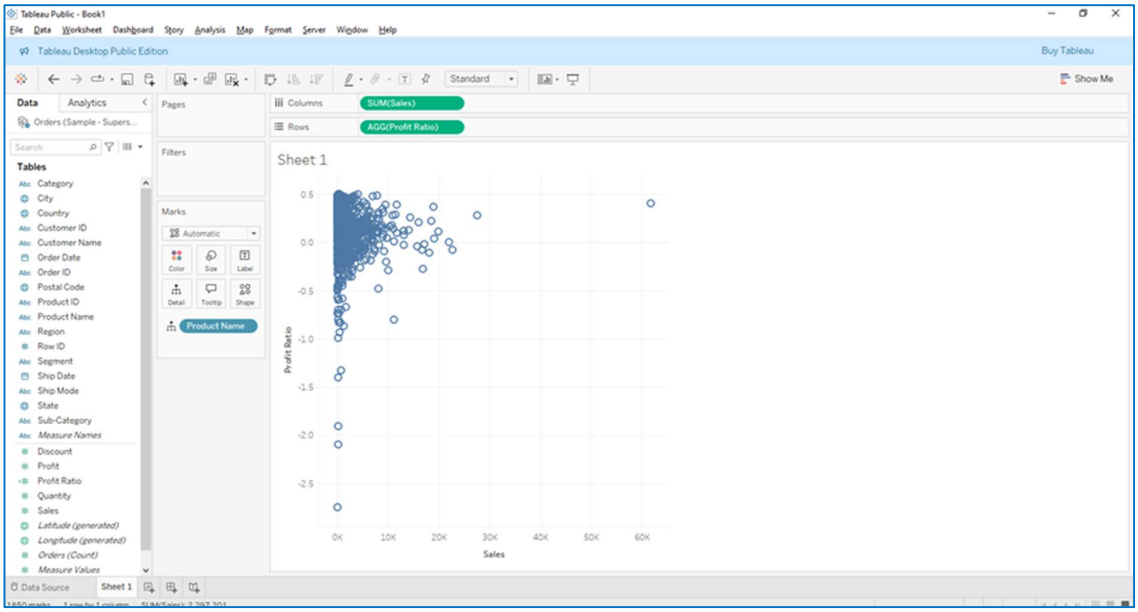
Write the logic in the field as shown below, and name it Profit Ratio.



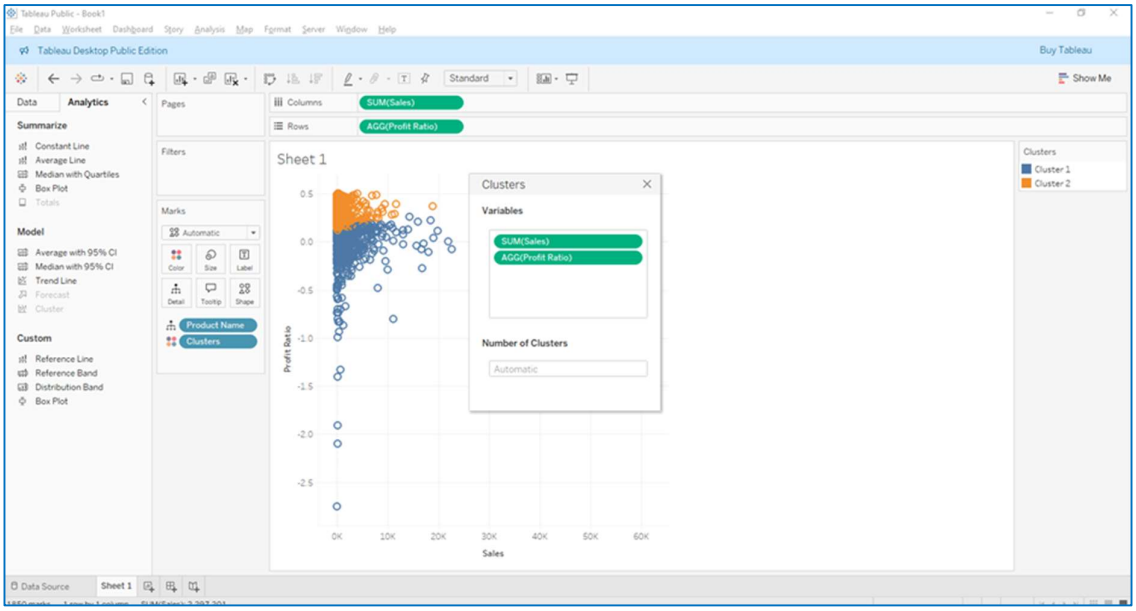
Drag Sales into the Columns shelf and Profit Ratio into the Rows shelf.



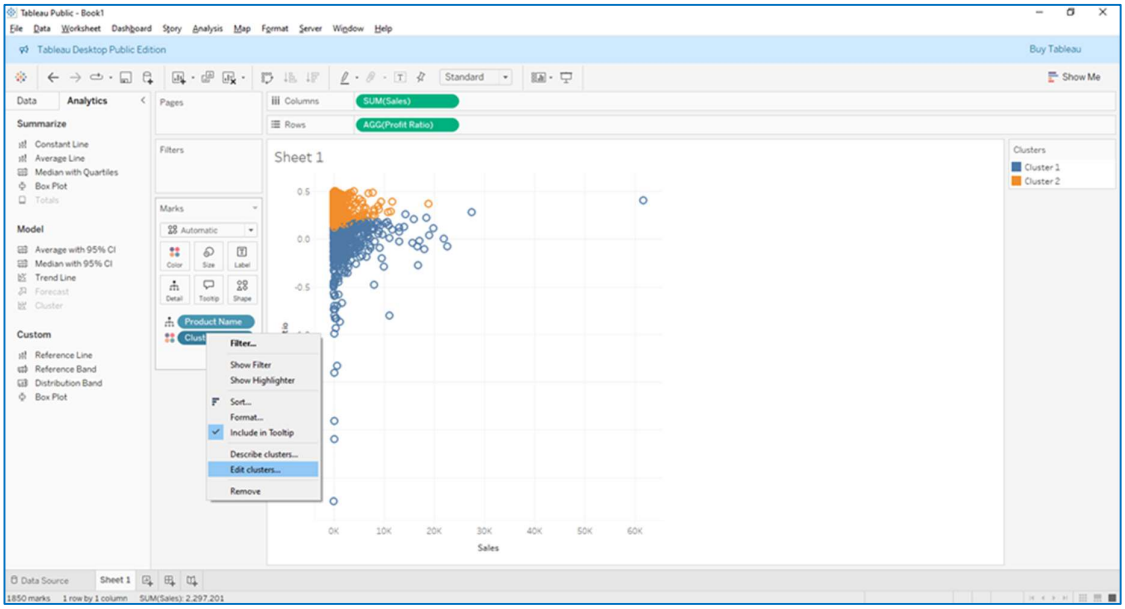
Next, drag Product Name into the Detail card of the Marks pane.



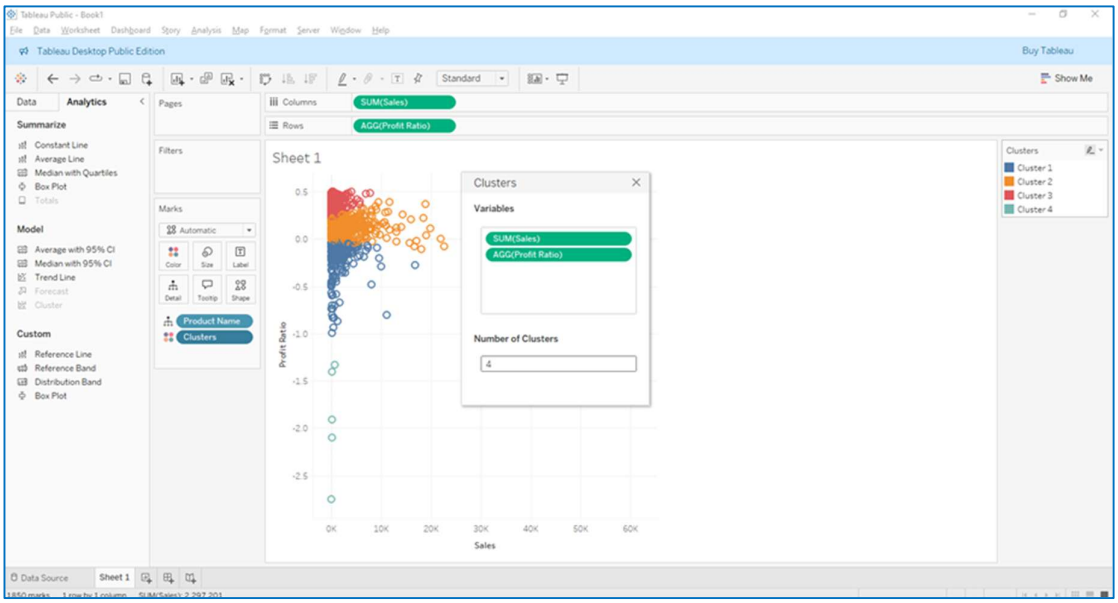
You are ready to initiate the clustering process. To begin, go to the Analytics pane next to the Data pane, as shown below.



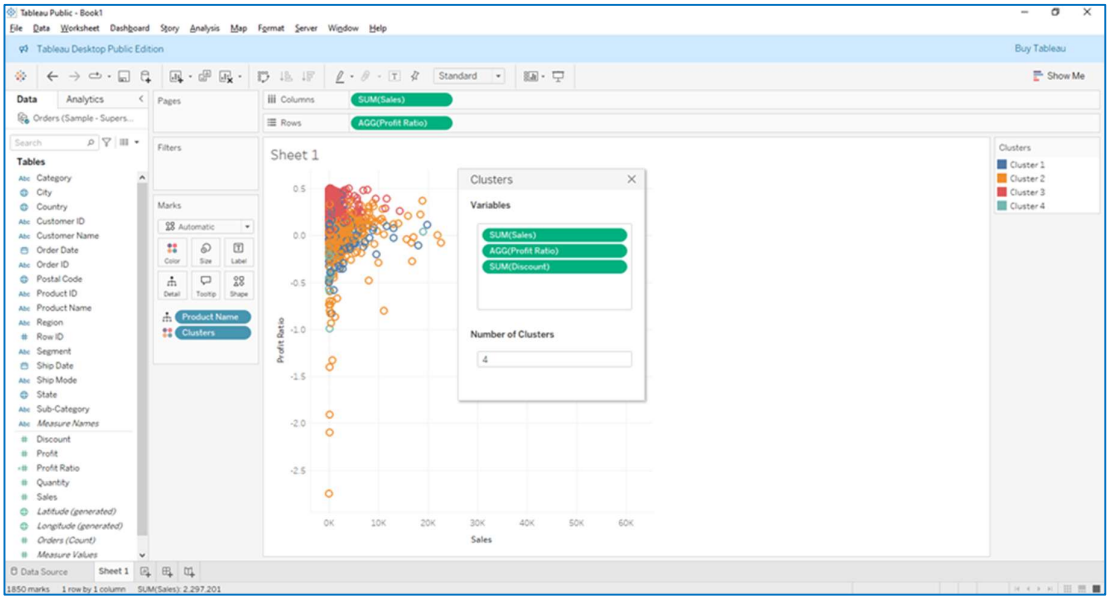
It’s possible to edit the number of clusters. To do this, right click on Clusters and click on Edit Clusters.



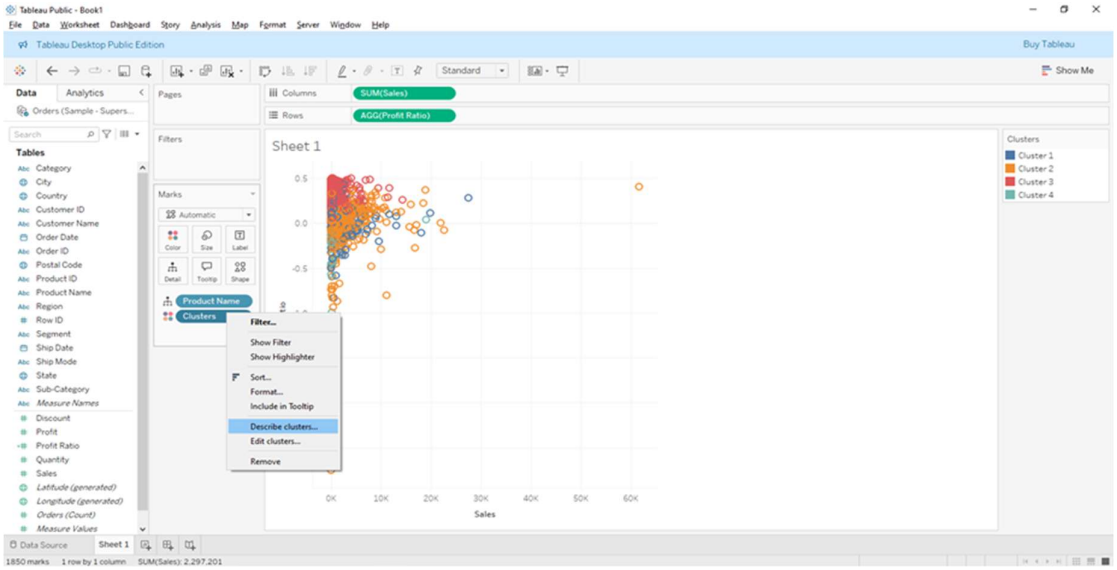
In the pop-up that will be displayed, enter 4 as the Number of Clusters.



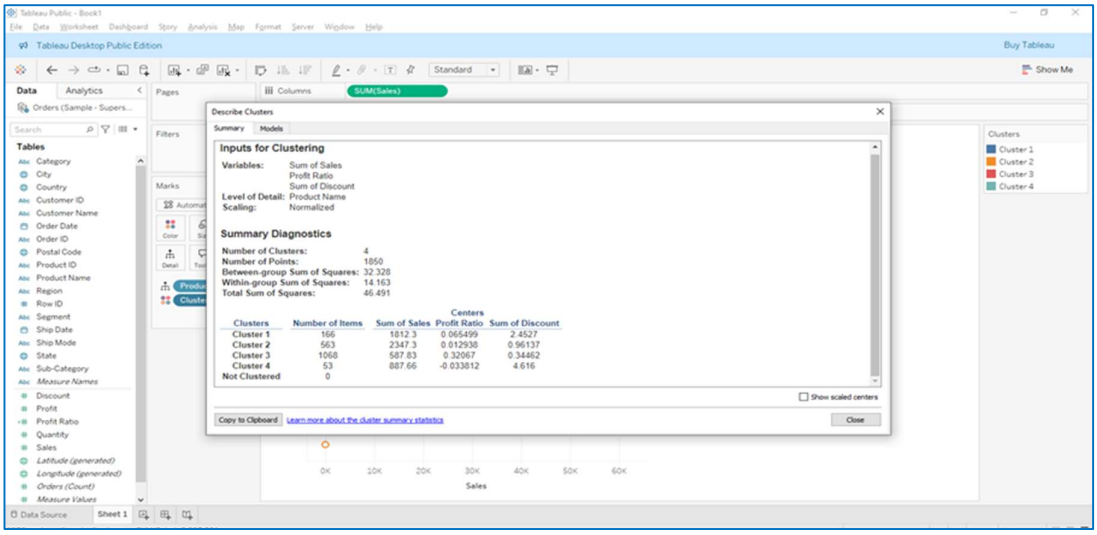
Also, there is an option to add or remove variables from the cluster. In this example, add the Discount variable to the cluster.



This will create the four k-means clusters. In addition, Tableau provides the option to describe the statistical properties of the cluster, which helps to understand the cluster better. To get this information, click on Clusters in the Marks pane and select Describe clusters.



Completing the above steps will display the result in two tabs. The first tab, Summary, displays the inputs used to generate the clusters and provides statistics that characterize the clusters. From the output below, you can observe there are 236 observations grouped in cluster one, 742 observations in cluster two, 867 observations in cluster three, and five in cluster four.



The second tab is the Models tab, which is used to show the statistical information about the model. Analysis of variance, or ANOVA, is used to analyze variation within and between observations that have been grouped into specific clusters. ANOVA is computed per variable, and the result is displayed in the tab, as shown below.

